# TCP

transmission control protocol (TCP)

implements reliable streams of bytes

similar mechanism to what we've described

# TCP extras/differences

bidirectional —

- separate sequence numbers in each direction
- can combine data (from A to B) with acknowledgment (from B to A)

sequence numbers are byte numbers —

- can retransmit data in different sized packets
- sequence numbers = index of *first byte* sent
- acknowledgment numbers = 1 + index of *last byte* acknowledged

dynamic/variable window sizes

- we'll discuss strategies later

offical name for packets = *segments*

# exercise (1)

(assume captured at ← end)

| direction | seq | ack | data len |
|---|---|---|---|
| → | 100 | 200 | 200 |
| → | 300 | 200 | 300 |
| ← | 300 | 600 | 0 |
| ← | 300 | 600 | 100 |
| → | ? | ? | 0 |

# exercise (2a)

(assume captured at $\rightarrow$ end)

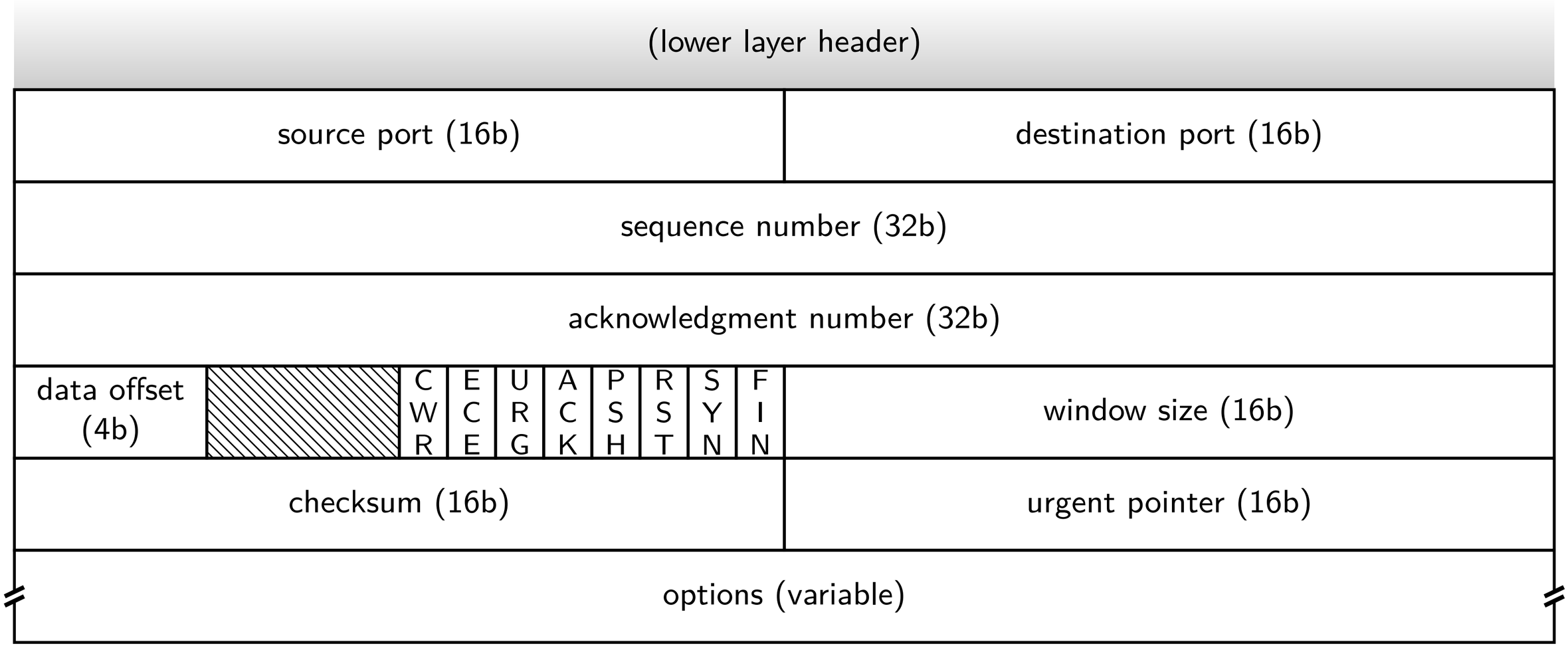
| direction | seq | ack | data len |
|---|---|---|---|
| $\rightarrow$ | 100 | 200 | 200 |
| $\leftarrow$ | ? | 400 | 0 |
| $\rightarrow$ | 300 | 200 | 300 |
| $\leftarrow$ | ? | 600 | 0 |
| $\leftarrow$ | ? | ? | 100 |
| $\rightarrow$ | 600 | 300 | 0 |

# exercise (2b)

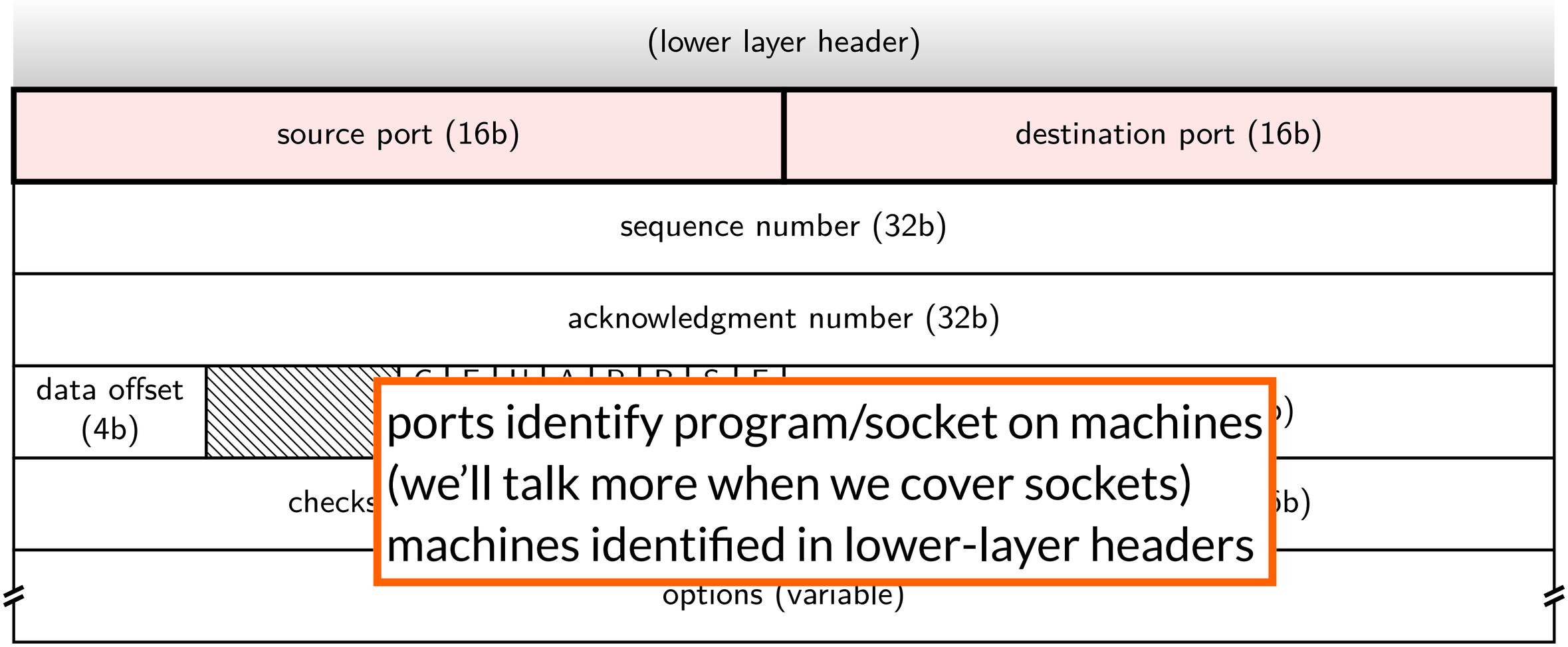
(assume captured at $\rightarrow$ end)

| direction | seq | ack | data len |
|---|---|---|---|
| $\rightarrow$ | 100 | 200 | 200 |
| $\leftarrow$ | 200 | 50 | 0 |
| $\rightarrow$ | 300 | 200 | 300 |
| $\leftarrow$ | 200 | ? | 0 |

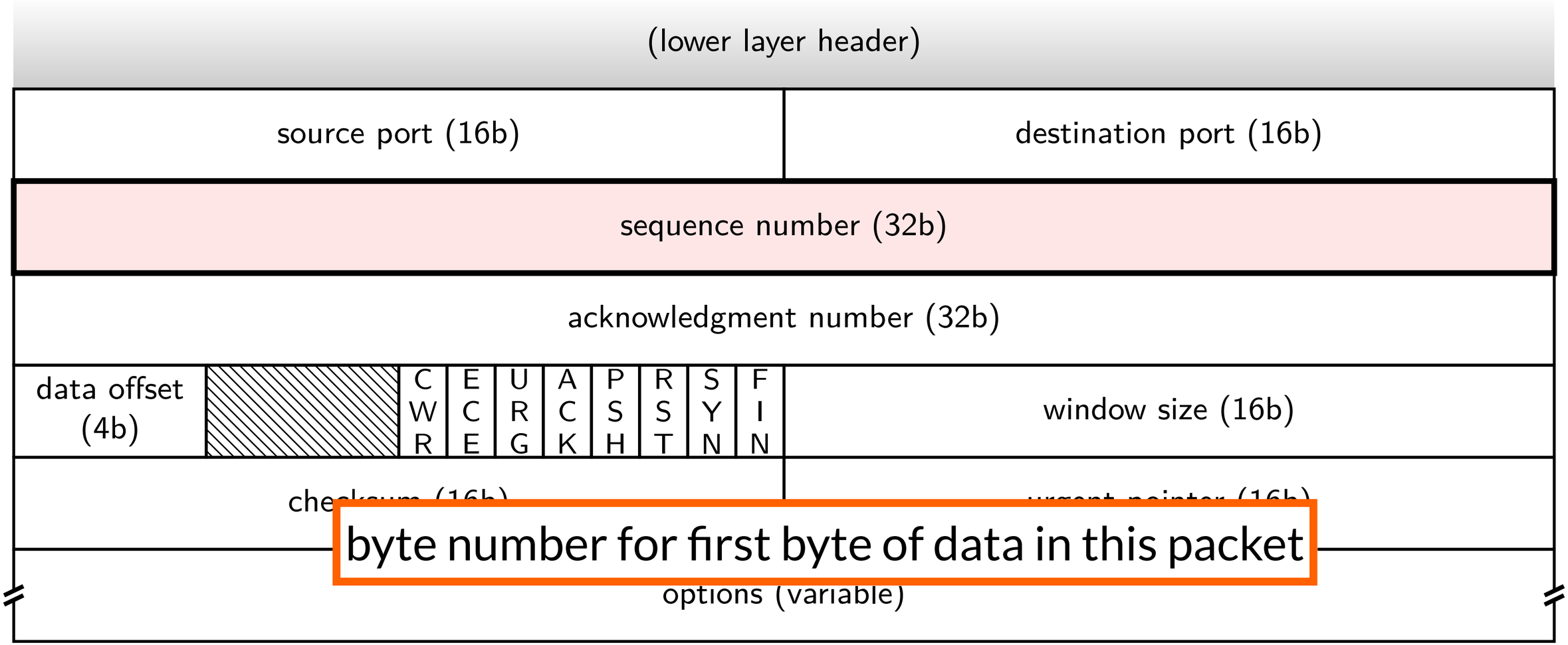
# TCP segment format



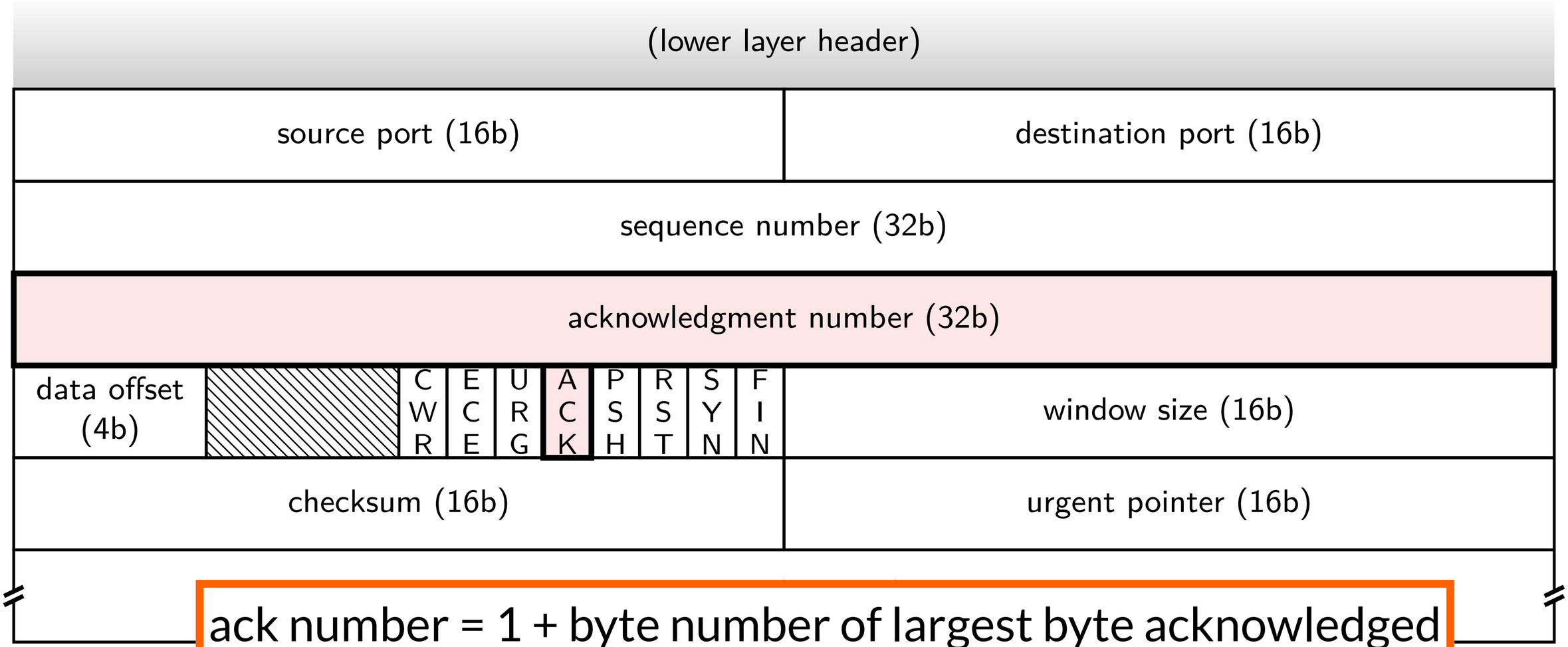
# TCP segment format



# TCP segment format



# TCP segment format



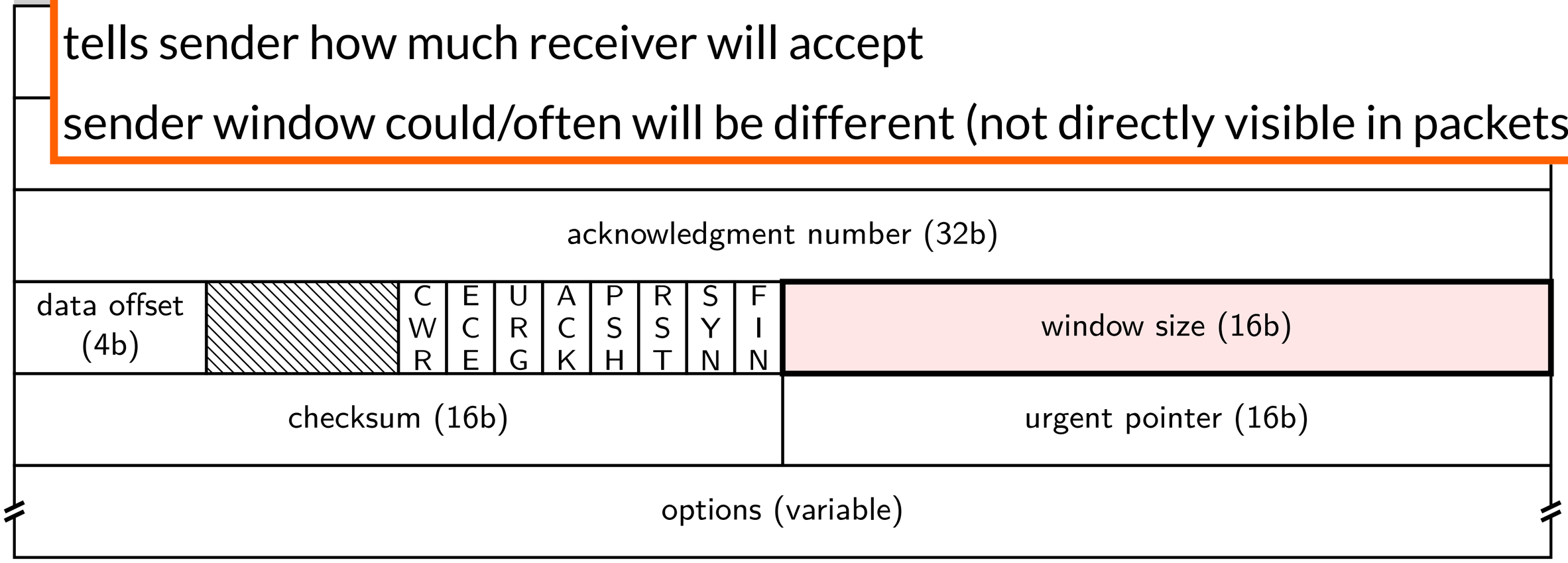

ack number = 1 + byte number of largest byte acknowledged
only meaningful if ACK ‘flag’ is 1

# TCP segment format

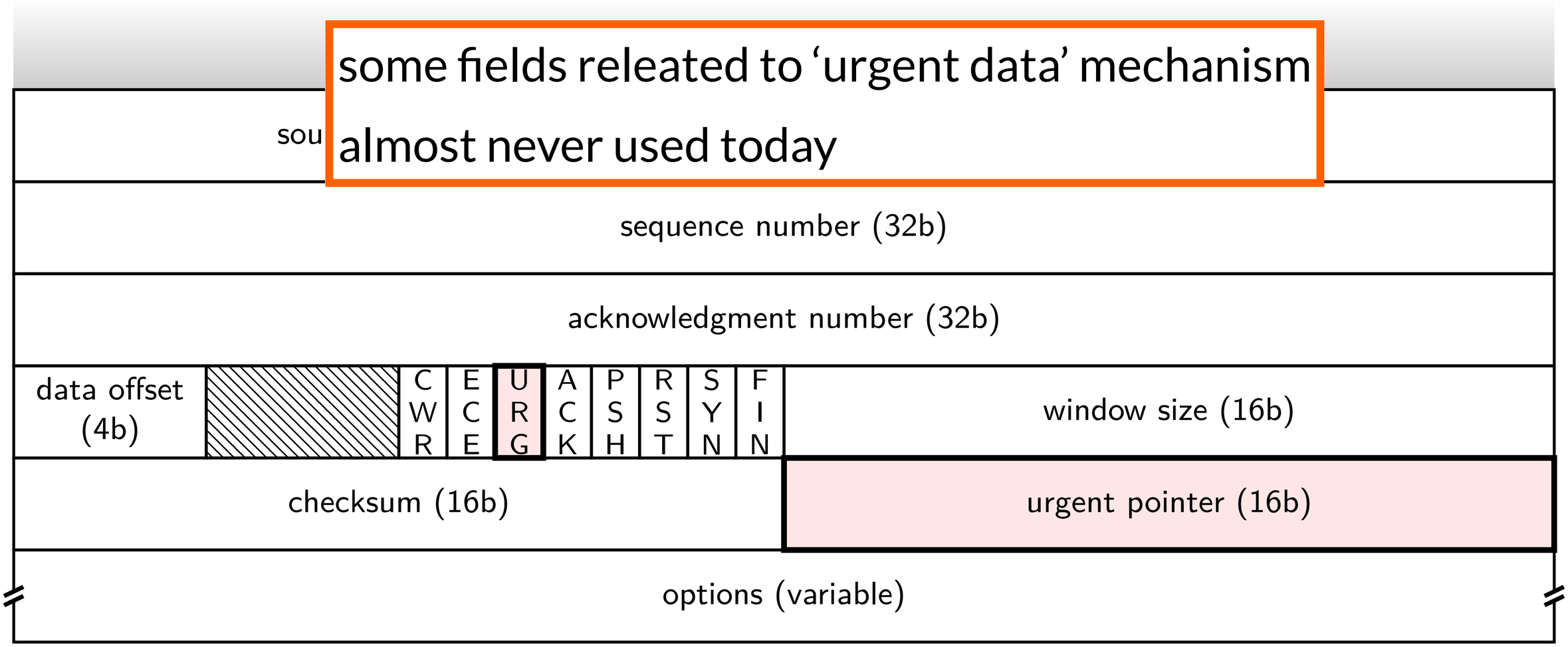
window size is receive window size

tells sender how much receiver will accept

sender window could/often will be different (not directly visible in packets)

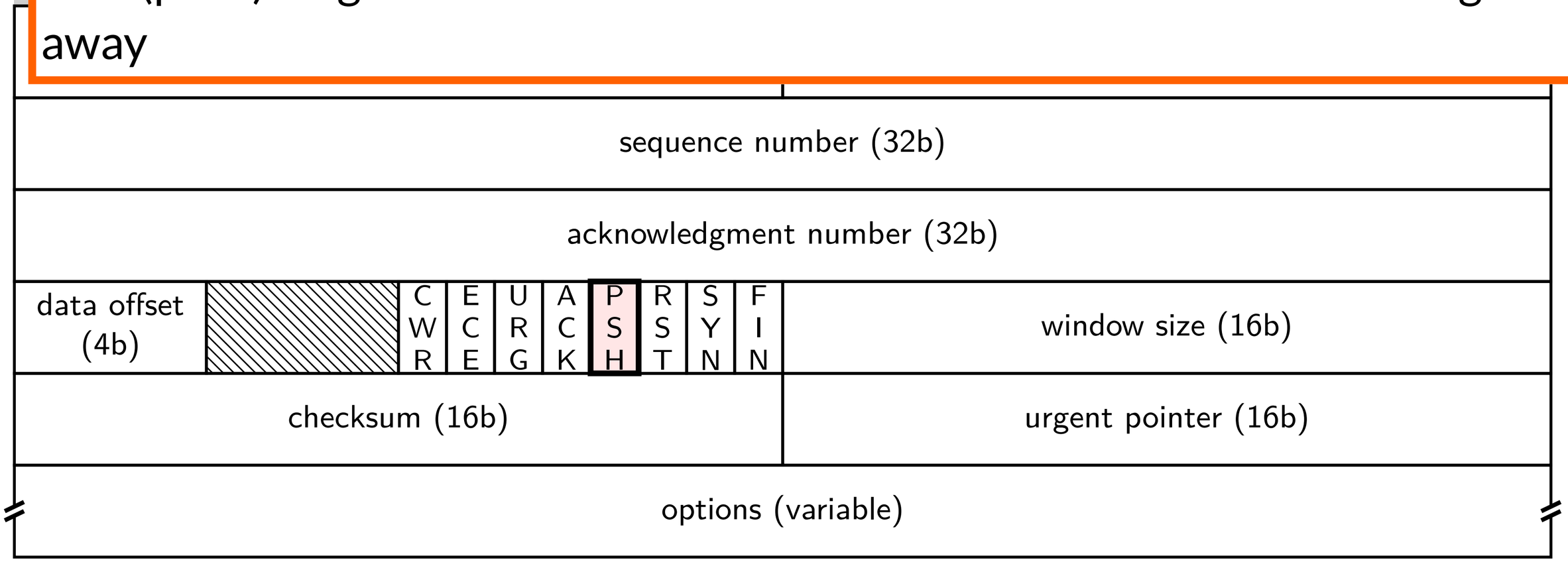


# TCP segment format

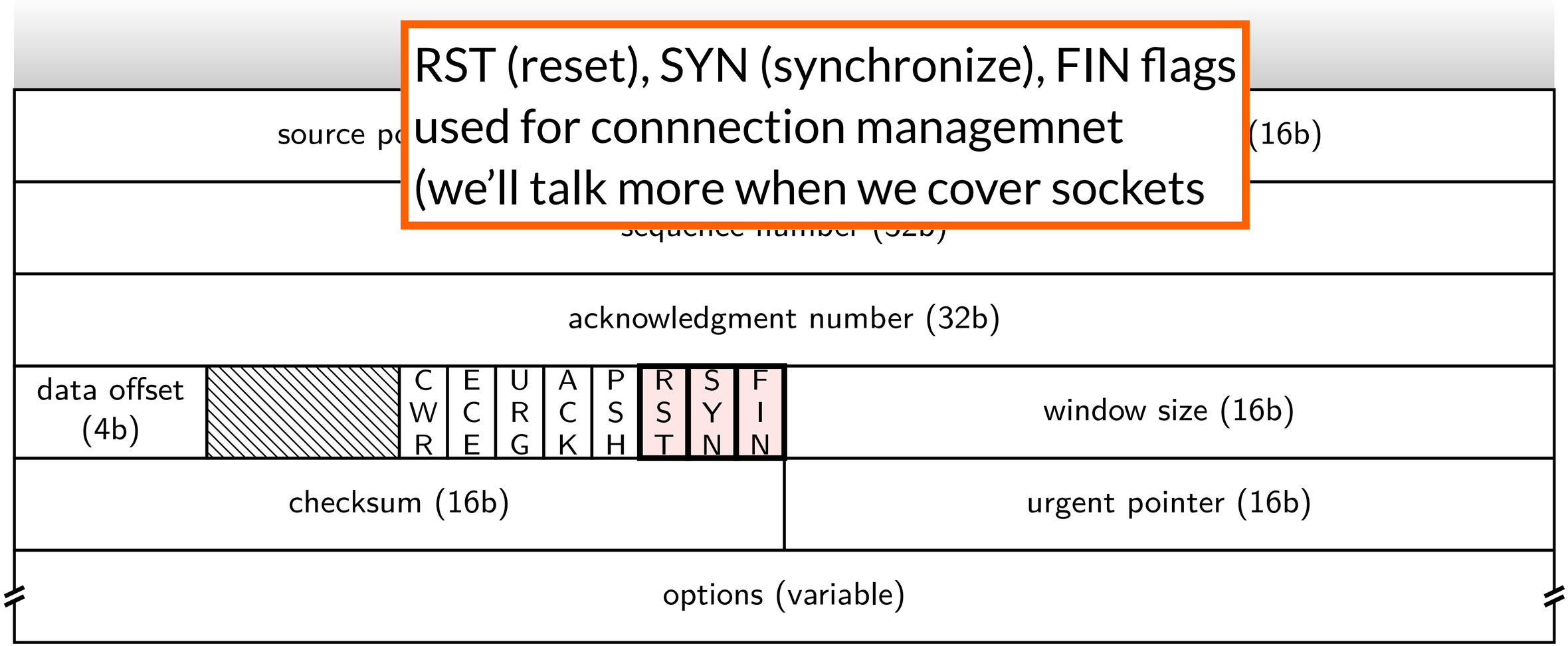


# TCP segment format

PSH (push) ‘flag’ is hint that sender does not have more data to send right away

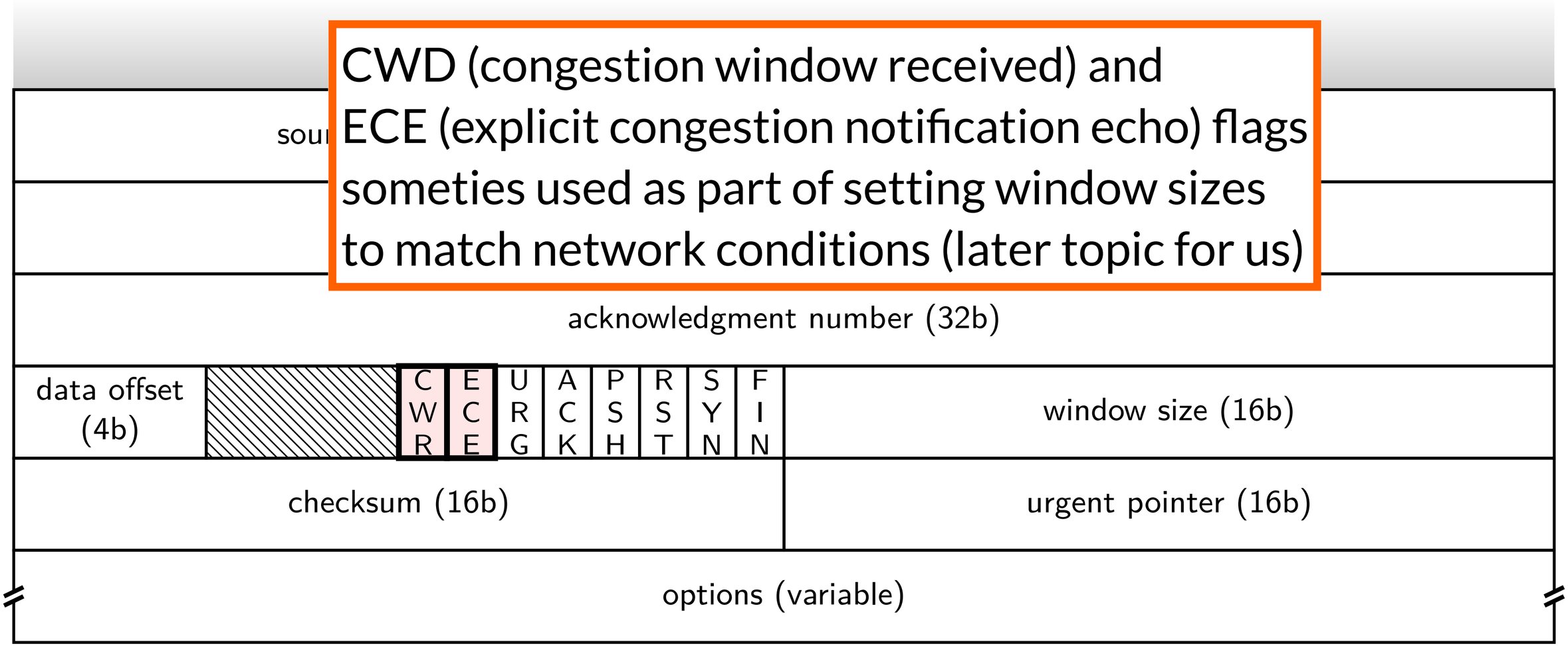


# TCP segment format

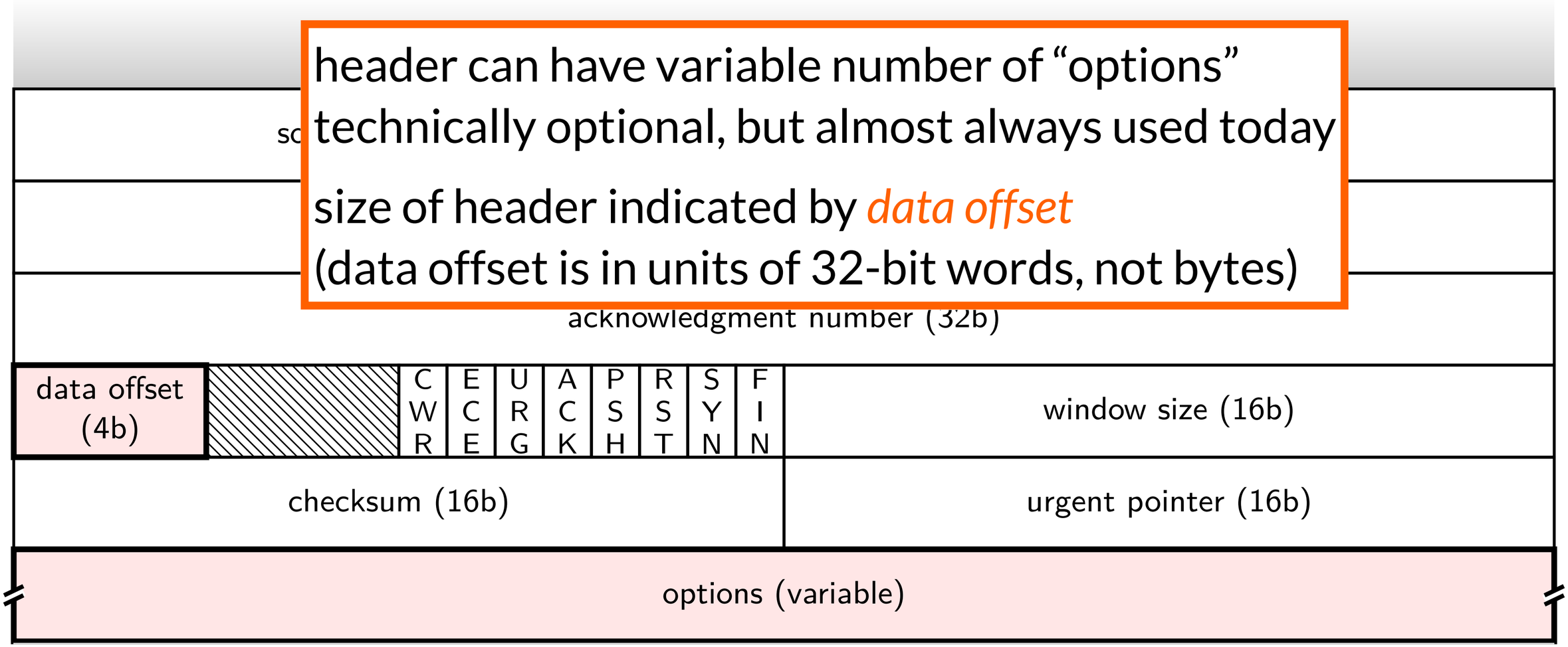



RST (reset), SYN (synchronize), FIN flags used for connnection managemnet (we'll talk more when we cover sockets

# TCP segment format



# TCP segment format



# exercise: maximum throughput

let's say we have a receiver window size of 65535 bytes

receiver window size is 16 bit field

and a round-trip time of 100 ms

if we want to avoid sending data the receiver may reject as outside its window, maximum throughput?

A. around 32kbyte/sec
B. around 64kbyte/sec
C. around 128kbyte/sec
D. around 320kbyte/sec
E. around 640kbyte/sec
F. around 1280kbyte/sec
G. something else

# selected TCP options

window size scale factor

- allow receiver window sizes greater than 64k
- needed to get reasonable bandwidth on modern networks

timestamps

- allow figuring out round trip time to estimate timeout
- extend 32-bit sequence number, which is too small for multi-gigabit networks

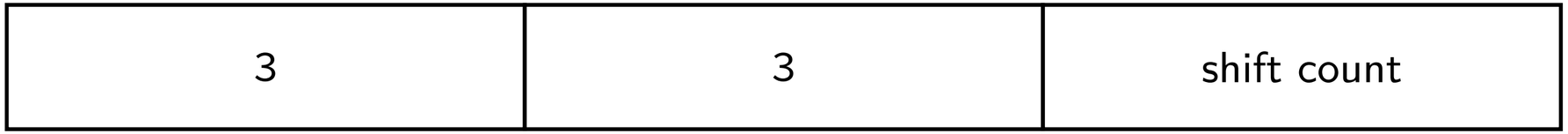
selective acknowledgements

- allow providing information about ‘holes’ in received data
- example: I got bytes 1–5000, 6000–7000, 8000–9000
- without it would only say 5000

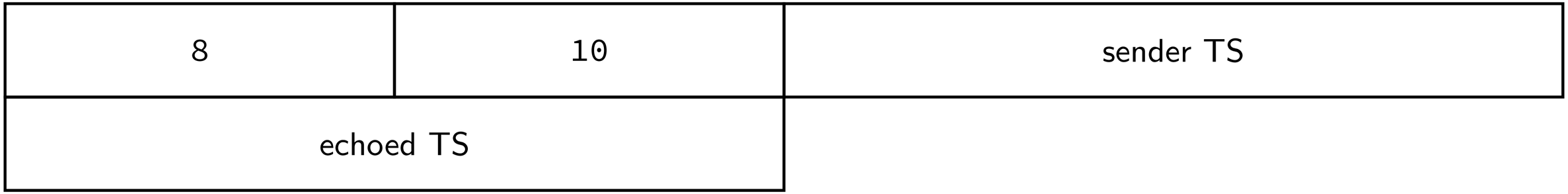
# selected TCP option formats



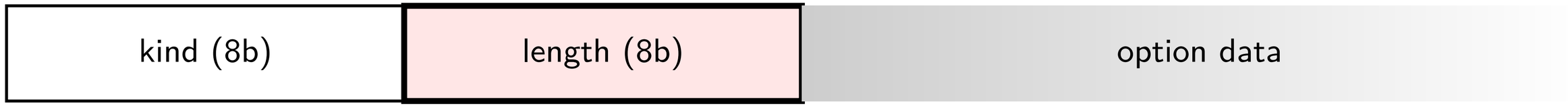

window scale option:



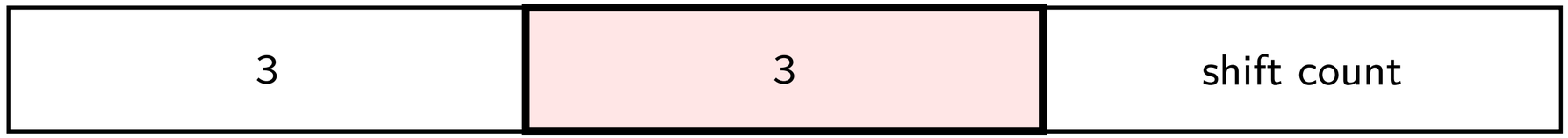

timestamps:



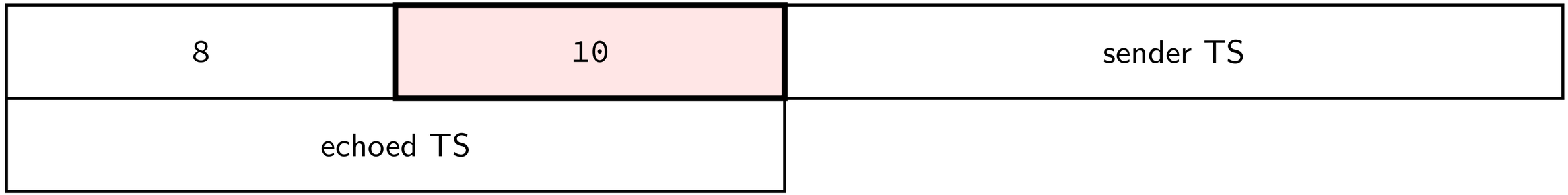
# selected TCP option formats




window scale option:

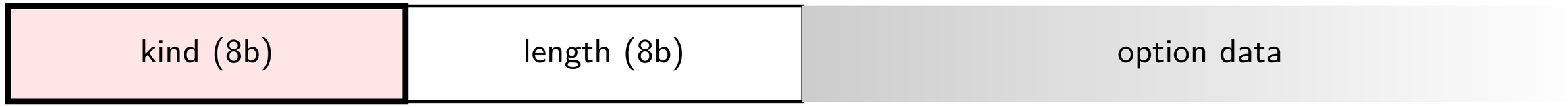



timestamps:

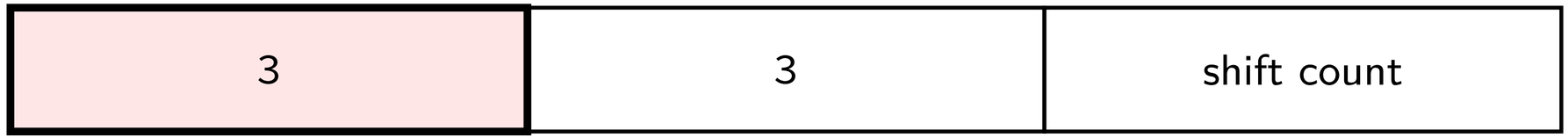



length field permits skipping unrecognized options

# selected TCP option formats




window scale option:




timestamps:

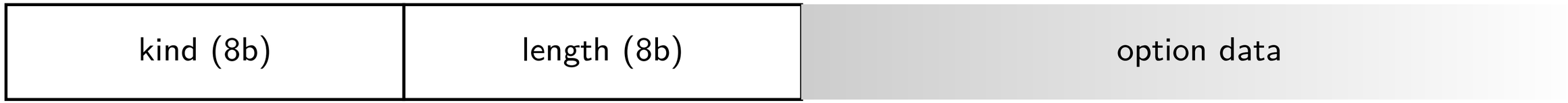




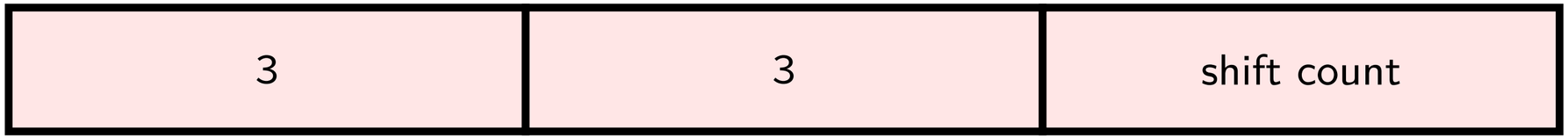


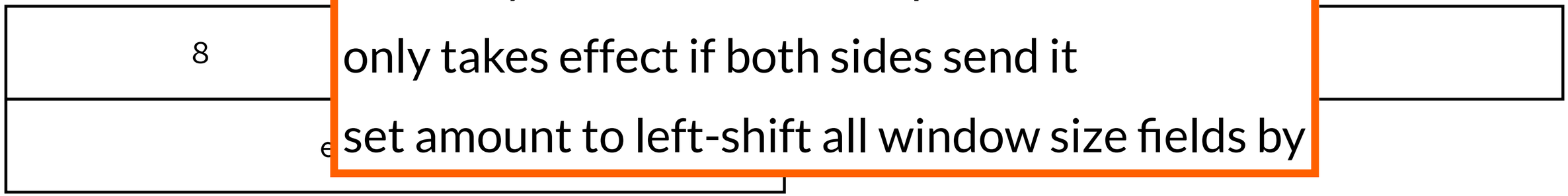
unique kind codes for each option
from list maintained by IANA
(Internet Assigned Numbers Authority)

# selected TCP option formats




*window scale option*:




sent only in connection setup

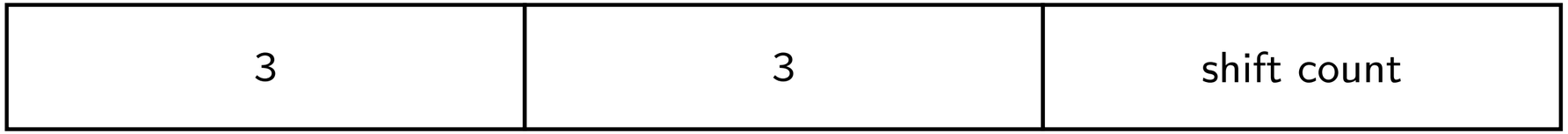
only takes effect if both sides send it

set amount to left-shift all window size fields by

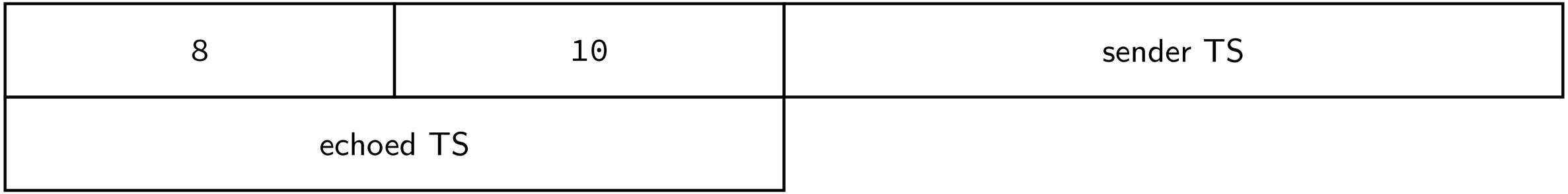
# selected TCP option formats



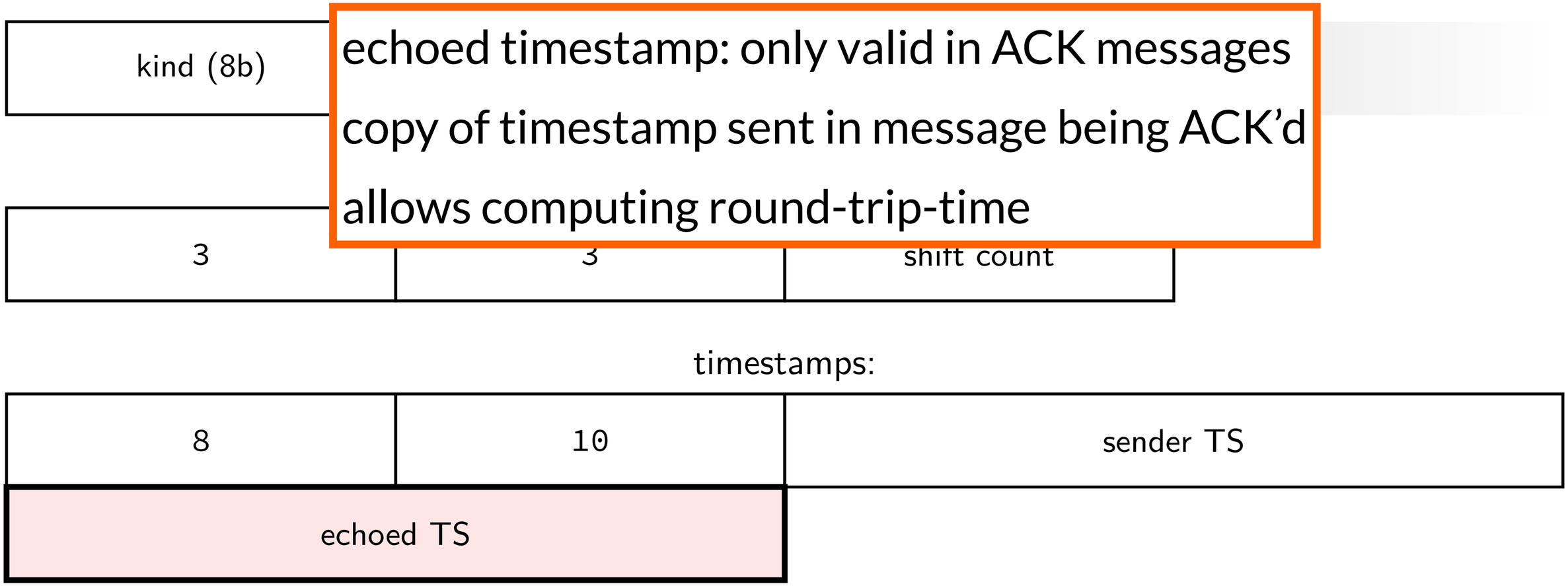

window scale option:




*timestamps*:



# selected TCP option formats



# TCP connection setup preview

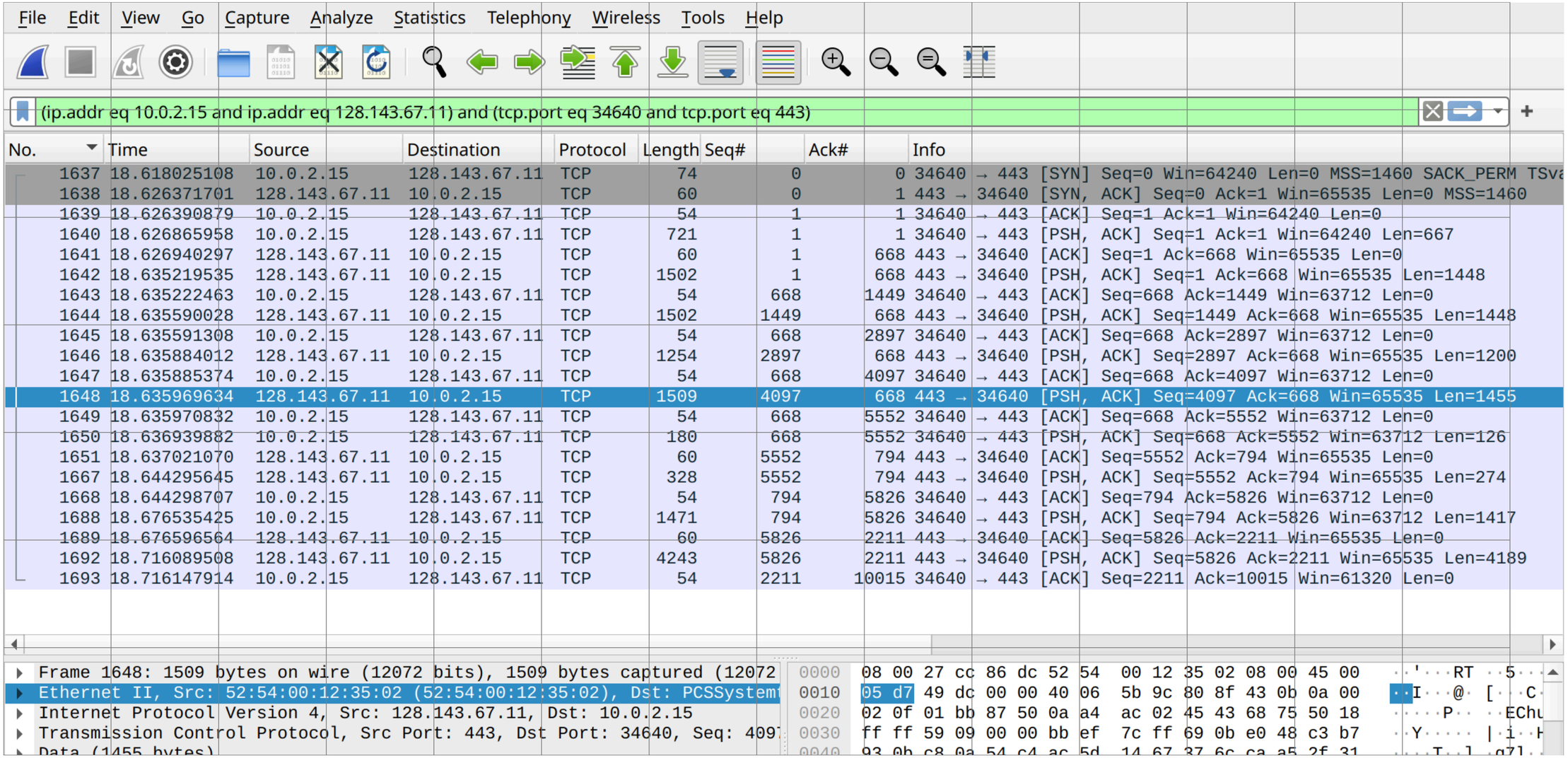
client→server: SYN flag + seq=randomA [no data]

server→client: SYN+ACK flags + seq=randomB ack=randomA+1 [no data]

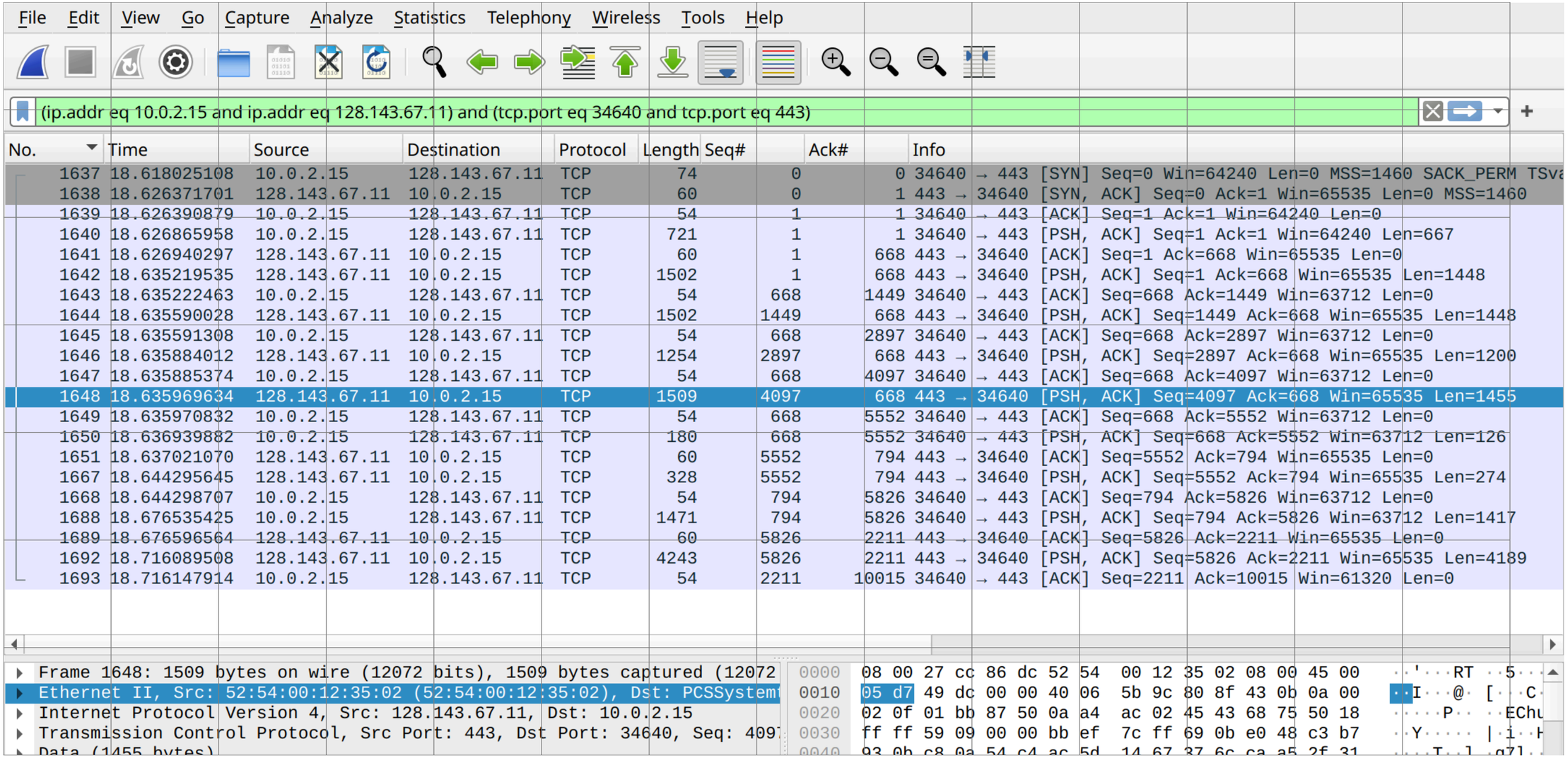
client→server: ACK flags + seq=randomA+1 ack=randomB+1 [no data]

# a TCP connection

# a TCP connection



# a TCP connection



# a TCP connection

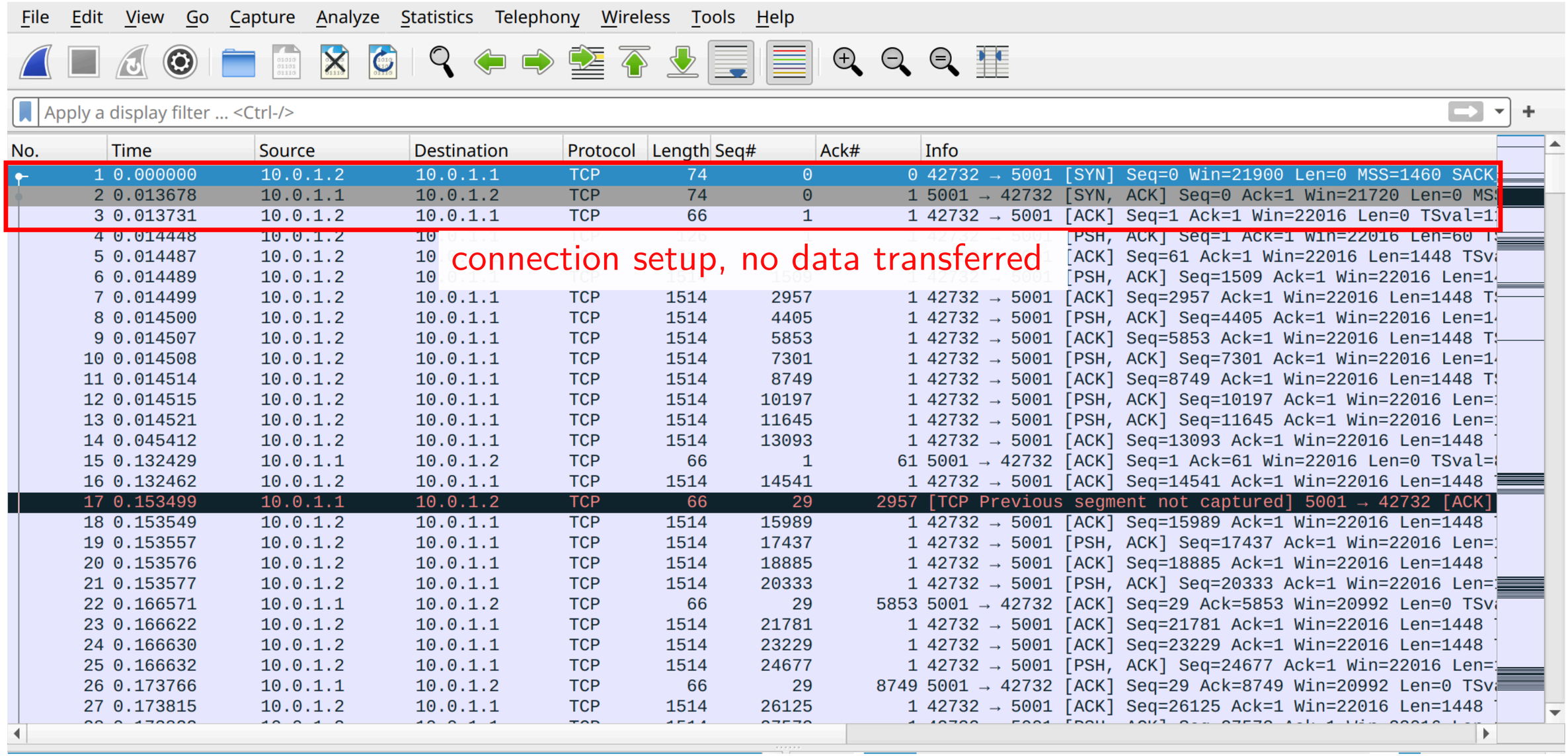
# a TCP connection

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Apply a display filter ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Length | Seq# | Ack# | Info |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.000000 | 10.0.1.2 | 10.0.1.1 | TCP | 74 | 0 | 0 | 42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK |
| 2 | 0.013678 | 10.0.1.1 | 10.0.1.2 | TCP | 74 | 0 | 1 | 5001 → 42732 [SYN, ACK] Seq=0 Ack=1 Win=21720 Len=0 MS |
| 3 | 0.013731 | 10.0.1.2 | 10.0.1.1 | TCP | 66 | 1 | 1 | 42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1 |
| 4 | 0.014448 | 10.0.1.2 | 10.0.1.1 | TCP | 126 | 1 | 1 | 42732 → 5001 [PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T |
| 5 | 0.014487 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 61 | 1 | 42732 → 5001 [ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv |
| 6 | 0.014489 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 1509 | 1 | 42732 → 5001 [PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1 |
| 7 | 0.014499 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 2957 | 1 | 42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T |
| 8 | 0.014500 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 4405 | 1 | 42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1 |
| 9 | 0.014507 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 5853 | 1 | 42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T |
| 10 | 0.014508 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 7301 | 1 | 42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1 |
| 11 | 0.014514 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 8749 | 1 | 42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T |
| 12 | 0.014515 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 10197 | 1 | 42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len= |
| 13 | 0.014521 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 11645 | 1 | 42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len= |
| 14 | 0.045412 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 13093 | 1 | 42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448 |
| 15 | 0.132429 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 1 | 61 | 5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval= |
| 16 | 0.132462 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 14541 | 1 | 42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448 |
| 17 | 0.153499 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 2957 | [TCP Previous segment not captured] 5001 → 42732 [ACK] |
| 18 | 0.153549 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 15989 | 1 | 42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448 |
| 19 | 0.153557 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 17437 | 1 | 42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len= |
| 20 | 0.153576 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 18885 | 1 | 42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448 |
| 21 | 0.153577 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 20333 | 1 | 42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len= |
| 22 | 0.166571 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 5853 | 5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv |
| 23 | 0.166622 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 21781 | 1 | 42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448 |
| 24 | 0.166630 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 23229 | 1 | 42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448 |
| 25 | 0.166632 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 24677 | 1 | 42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len= |
| 26 | 0.173766 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 8749 | 5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv |
| 27 | 0.173815 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 26125 | 1 | 42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448 |

# a TCP connection




| No. | Time | Source | Destination | Protocol | Length | Seq# | Ack# | Info |
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| 4 | 0.014448 | 10.0.1.2 | 10 | | | | | [PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T |
| 5 | 0.014487 | 10.0.1.2 | 10 | | | | | [ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv |
| 6 | 0.014489 | 10.0.1.2 | 10 | | | | | [PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1 |
| 7 | 0.014499 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 2957 | 1 | 42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T |
| 8 | 0.014500 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 4405 | 1 | 42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1 |
| 9 | 0.014507 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 5853 | 1 | 42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T |
| 10 | 0.014508 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 7301 | 1 | 42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1 |
| 11 | 0.014514 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 8749 | 1 | 42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T |
| 12 | 0.014515 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 10197 | 1 | 42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len= |
| 13 | 0.014521 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 11645 | 1 | 42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len= |
| 14 | 0.045412 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 13093 | 1 | 42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448 |
| 15 | 0.132429 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 1 | 61 | 5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval= |
| 16 | 0.132462 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 14541 | 1 | 42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448 |
| 17 | 0.153499 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 2957 | [TCP Previous segment not captured] 5001 → 42732 [ACK] |
| 18 | 0.153549 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 15989 | 1 | 42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448 |
| 19 | 0.153557 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 17437 | 1 | 42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len= |
| 20 | 0.153576 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 18885 | 1 | 42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448 |
| 21 | 0.153577 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 20333 | 1 | 42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len= |
| 22 | 0.166571 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 5853 | 5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv |
| 23 | 0.166622 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 21781 | 1 | 42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448 |
| 24 | 0.166630 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 23229 | 1 | 42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448 |
| 25 | 0.166632 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 24677 | 1 | 42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len= |
| 26 | 0.173766 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 8749 | 5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv |
| 27 | 0.173815 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 26125 | 1 | 42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448 |

# a TCP connection

server+client sequence numbers advance by 1 to indicate where in setup

| ngth | Seq# | Ack# | In |
|---|---|---|---|
| 74 | 0 | 0 | 42 |
| 74 | 0 | 1 | 50 |
| 66 | 1 | 1 | 42 |

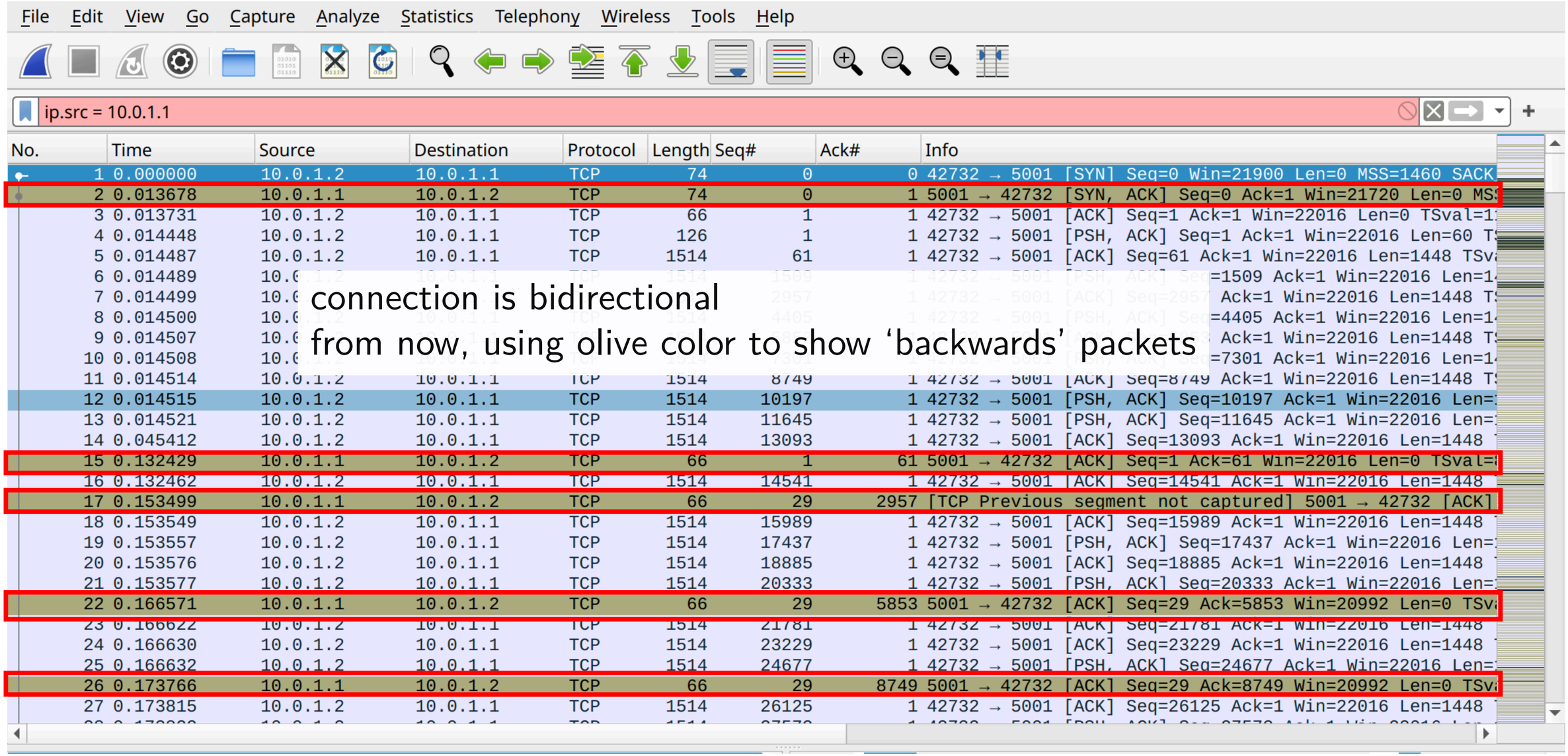
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src = 10.0.1.1

| No. | Time | Source | Destination | Protocol | Length | Seq# | Ack# | Info |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.000000 | 10.0.1.2 | 10.0.1.1 | TCP | 74 | 0 | 0 | 42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK |
| 2 | 0.013678 | 10.0.1.1 | 10.0.1.2 | TCP | 74 | 0 | 1 | 5001 → 42732 [SYN, ACK] Seq=0 Ack=1 Win=21720 Len=0 MS |
| 3 | 0.013731 | 10.0.1.2 | 10.0.1.1 | TCP | 66 | 1 | 1 | 42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1 |
| 4 | 0.014448 | 10.0.1.2 | 10 | | | | | [PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T |
| 5 | 0.014487 | 10.0.1.2 | 10 | | | | | [ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv |
| 6 | 0.014489 | 10.0.1.2 | 10 | | | | | [PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1 |
| 7 | 0.014499 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 2957 | 1 | 42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T |
| 8 | 0.014500 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 4405 | 1 | 42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1 |
| 9 | 0.014507 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 5853 | 1 | 42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T |
| 10 | 0.014508 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 7301 | 1 | 42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1 |
| 11 | 0.014514 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 8749 | 1 | 42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T |
| 12 | 0.014515 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 10197 | 1 | 42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len= |
| 13 | 0.014521 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 11645 | 1 | 42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len= |
| 14 | 0.045412 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 13093 | 1 | 42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448 |
| 15 | 0.132429 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 1 | 61 | 5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval= |
| 16 | 0.132462 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 14541 | 1 | 42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448 |
| 17 | 0.153499 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 2957 | [TCP Previous segment not captured] 5001 → 42732 [ACK] |
| 18 | 0.153549 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 15989 | 1 | 42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448 |
| 19 | 0.153557 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 17437 | 1 | 42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len= |
| 20 | 0.153576 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 18885 | 1 | 42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448 |
| 21 | 0.153577 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 20333 | 1 | 42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len= |
| 22 | 0.166571 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 5853 | 5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv |
| 23 | 0.166622 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 21781 | 1 | 42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448 |
| 24 | 0.166630 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 23229 | 1 | 42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448 |
| 25 | 0.166632 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 24677 | 1 | 42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len= |
| 26 | 0.173766 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 8749 | 5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv |
| 27 | 0.173815 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 26125 | 1 | 42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448 |

connection setup, no data transferred

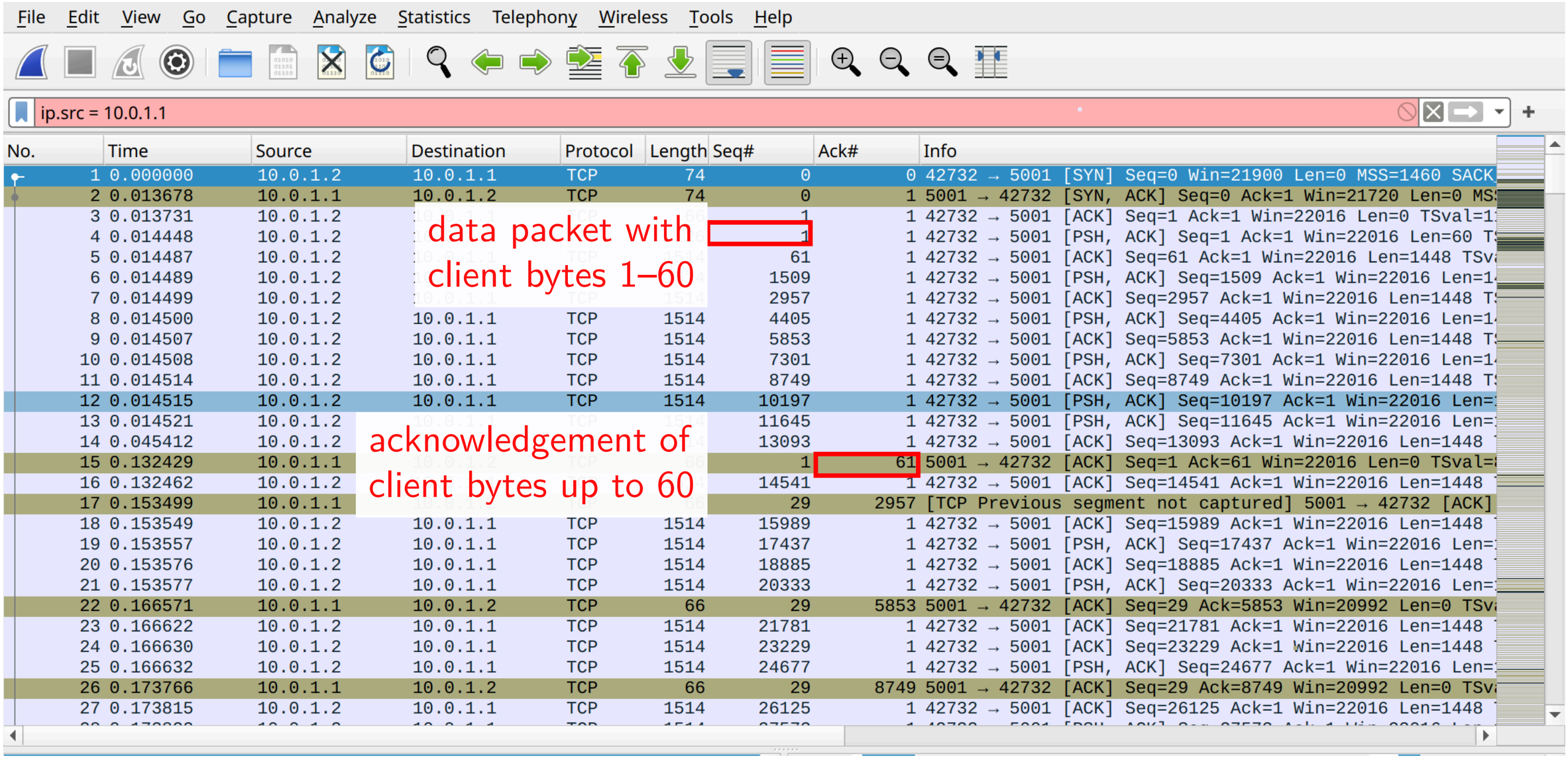
# a TCP connection



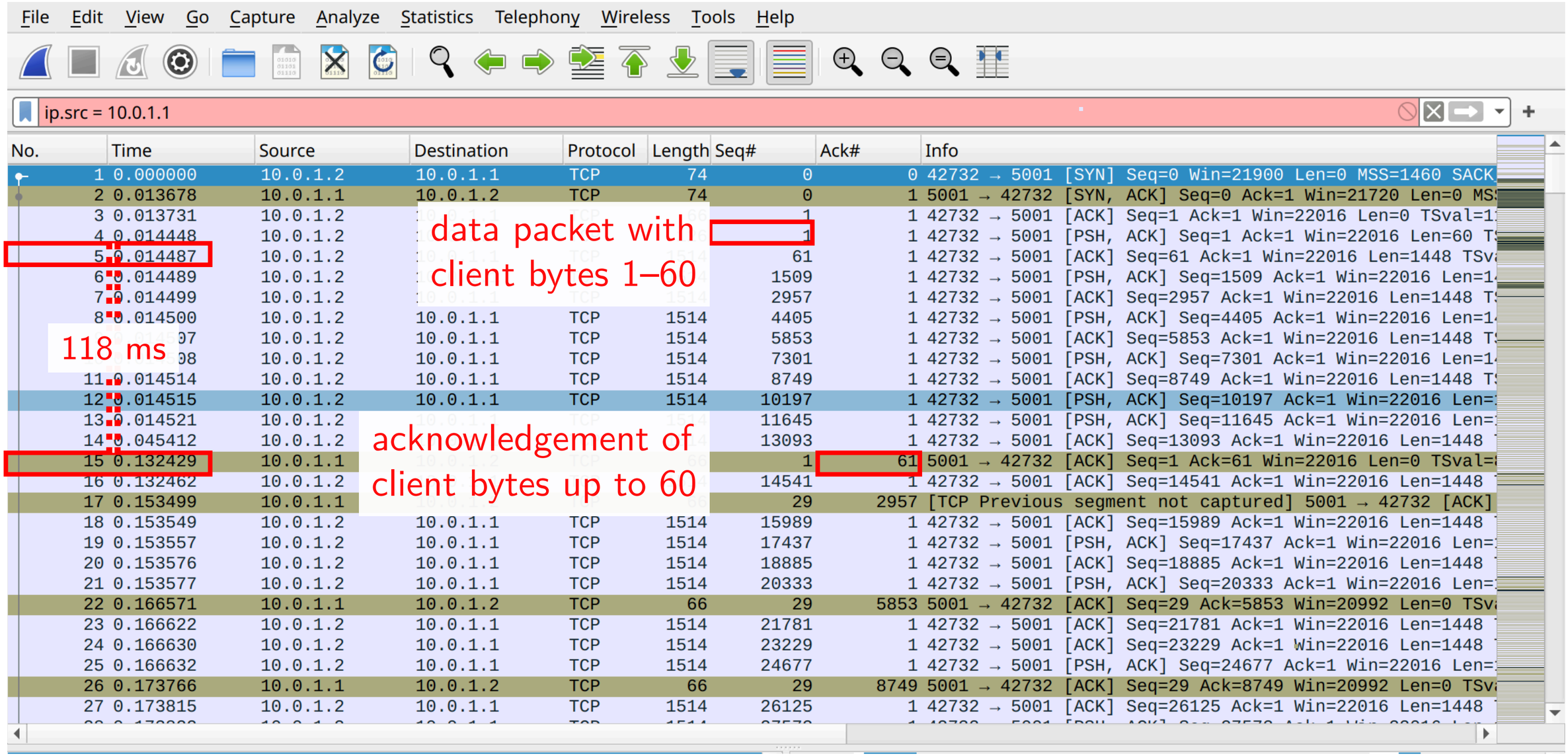

connection is bidirectional

from now, using olive color to show ‘backwards’ packets

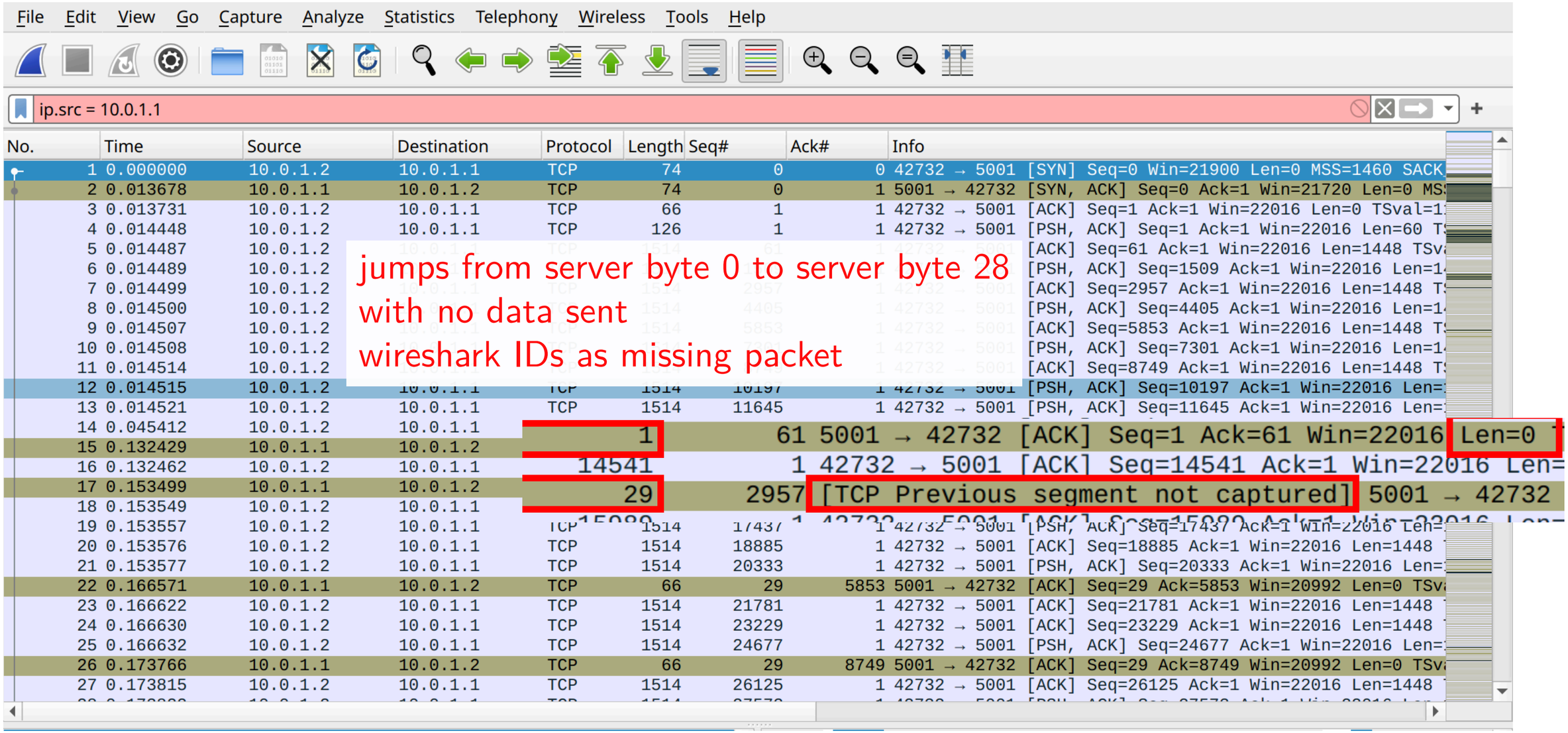
# a TCP connection



# a TCP connection

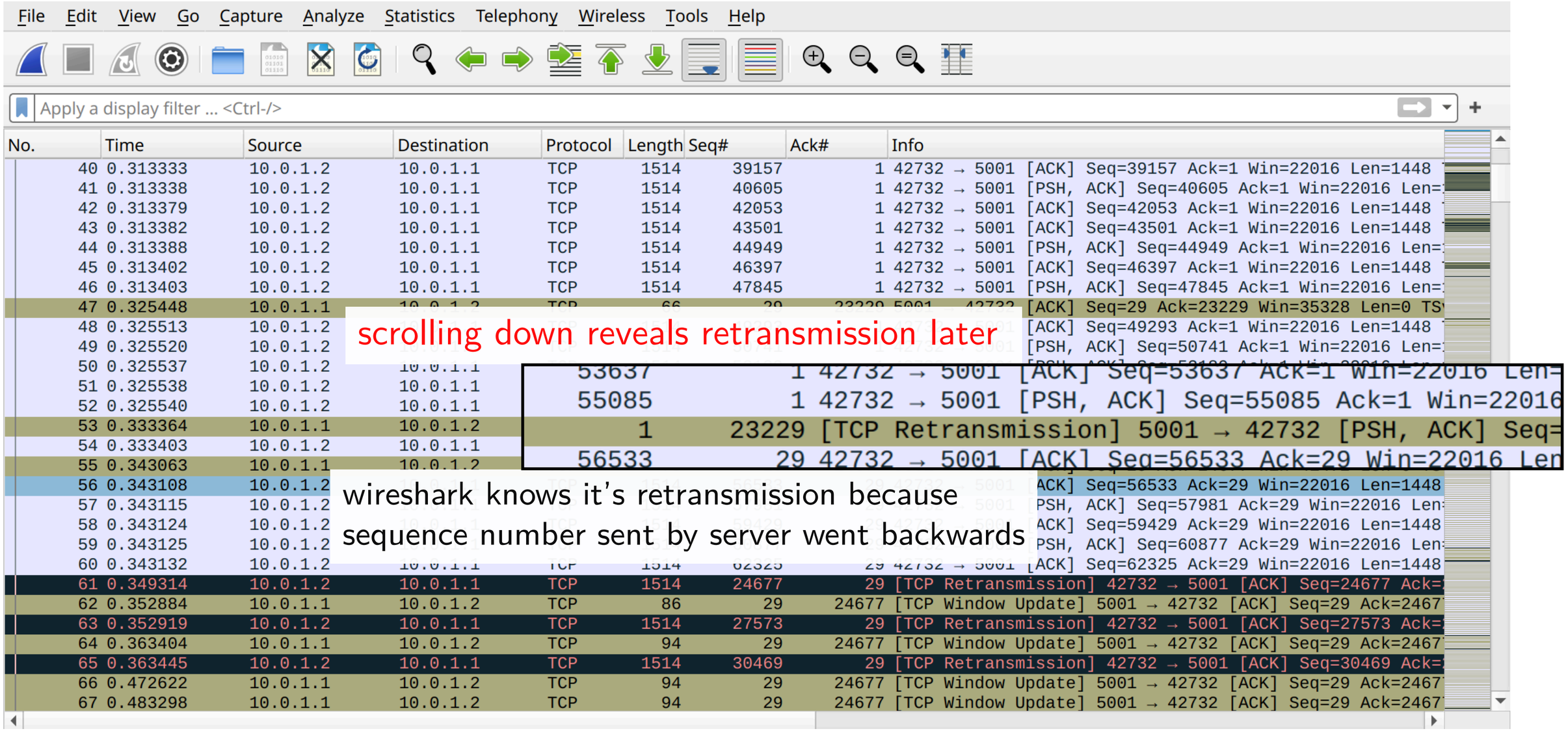


# a TCP connection



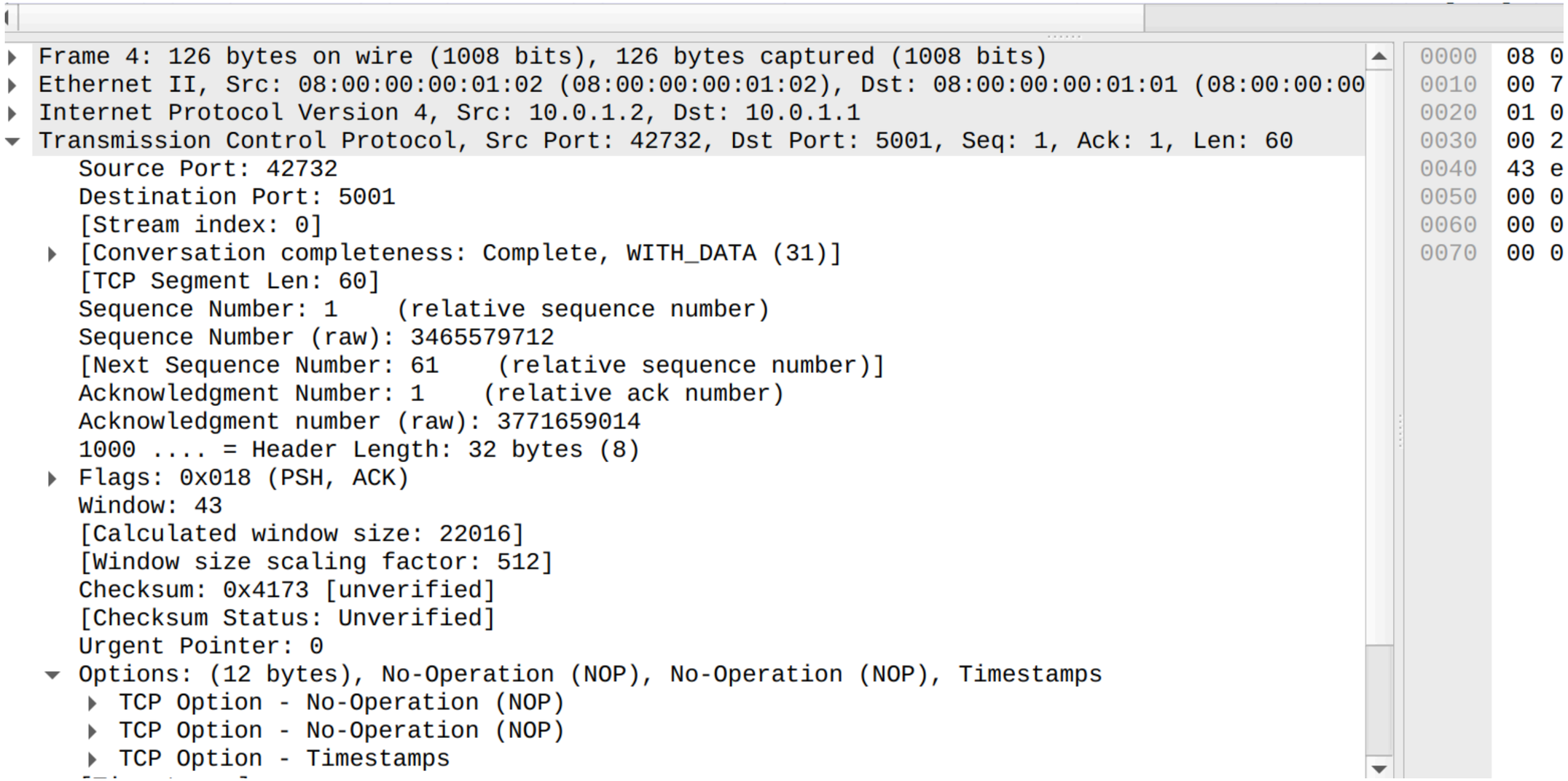

jumps from server byte 0 to server byte 28
with no data sent
wireshark IDs as missing packet

# a TCP connection

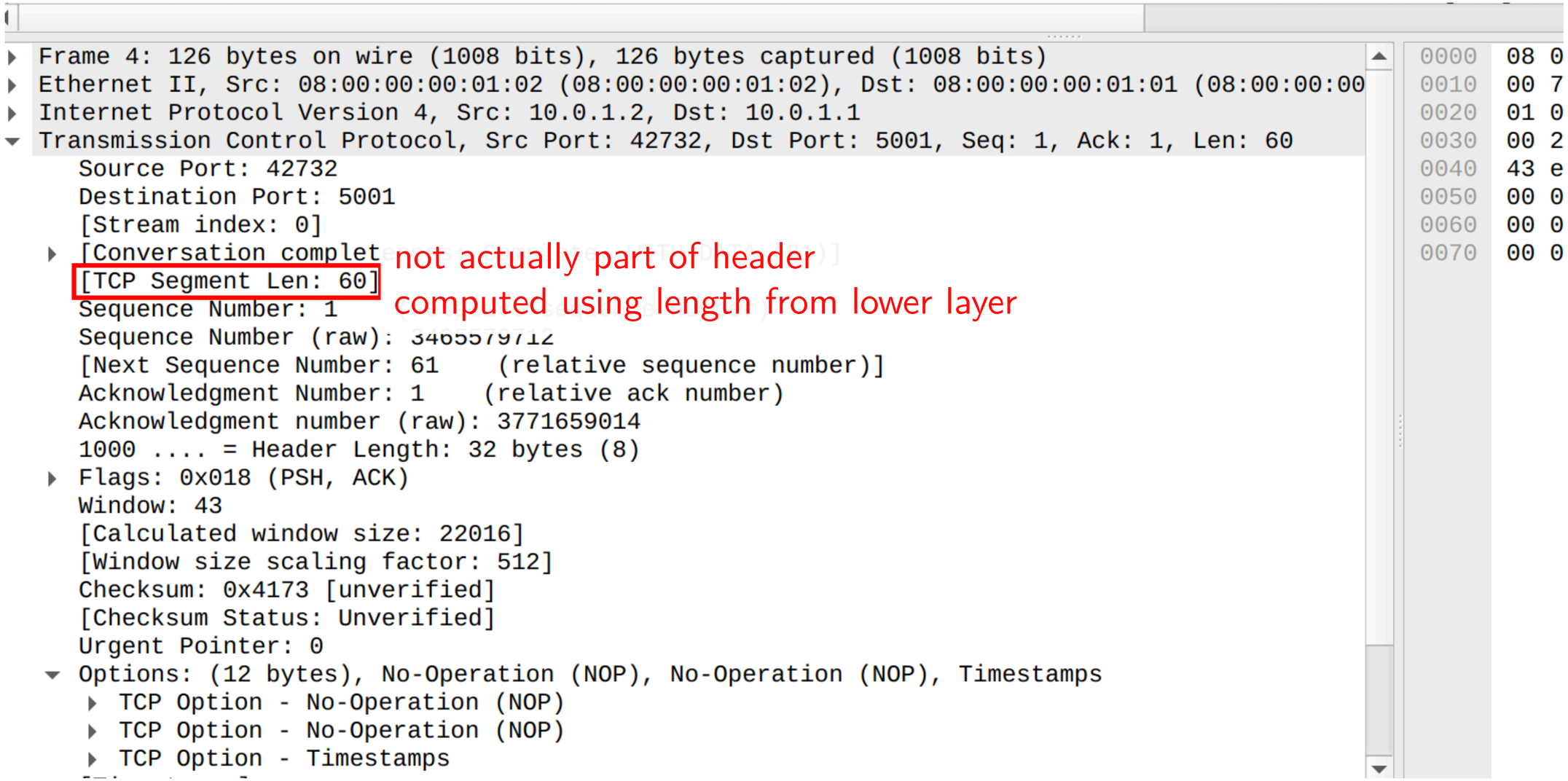


# first data packet

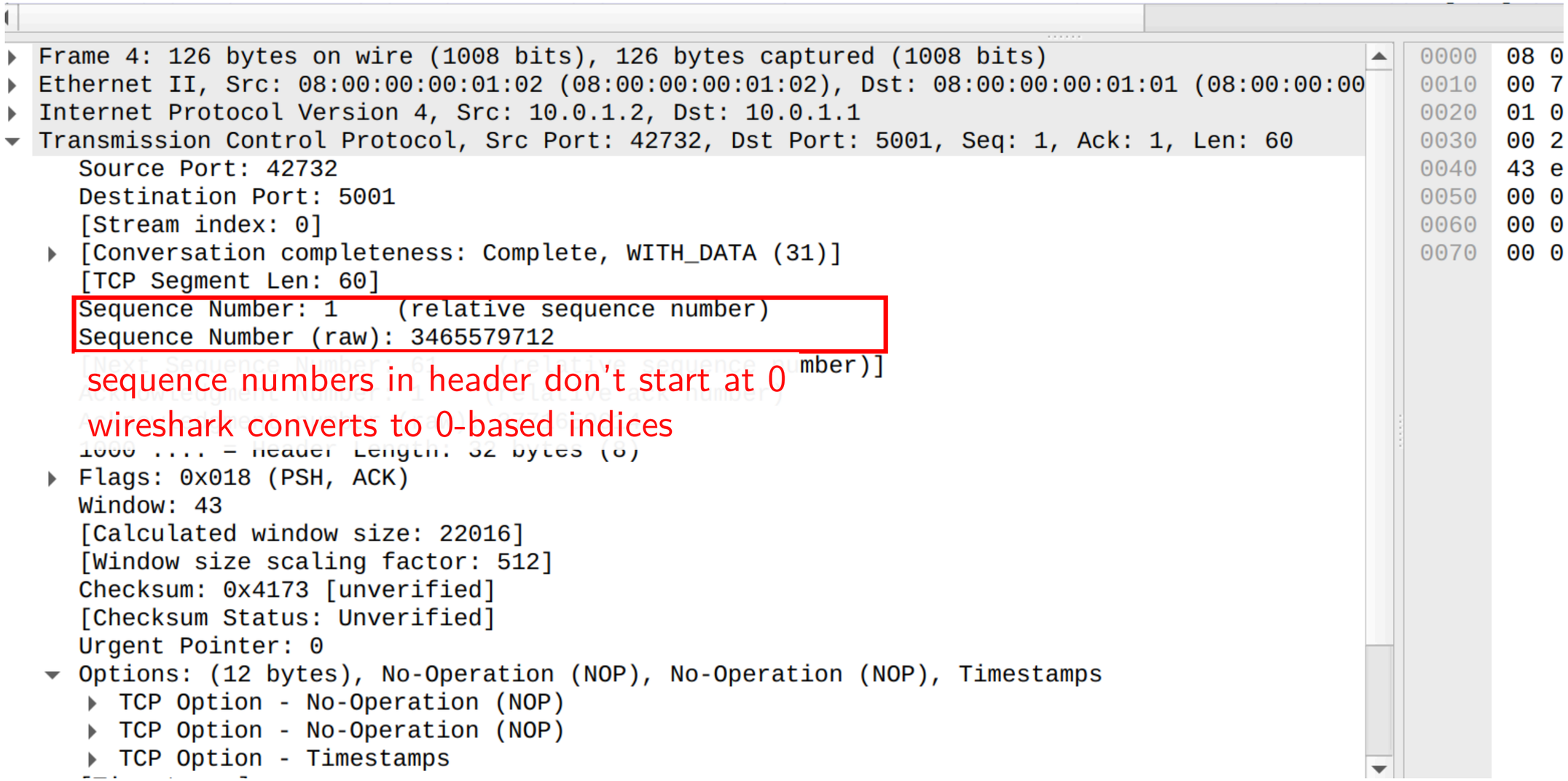
# first data packet



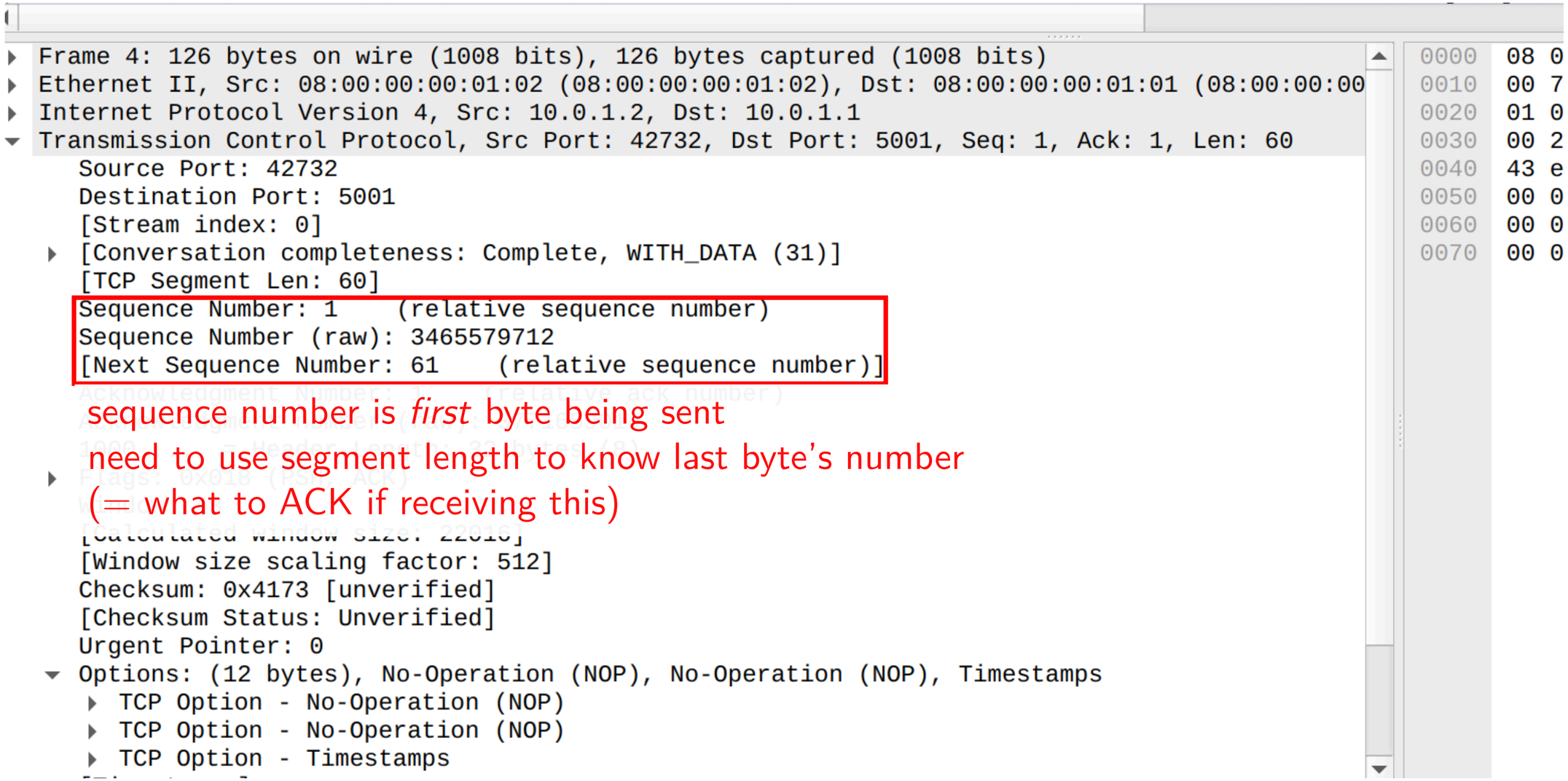
# first data packet



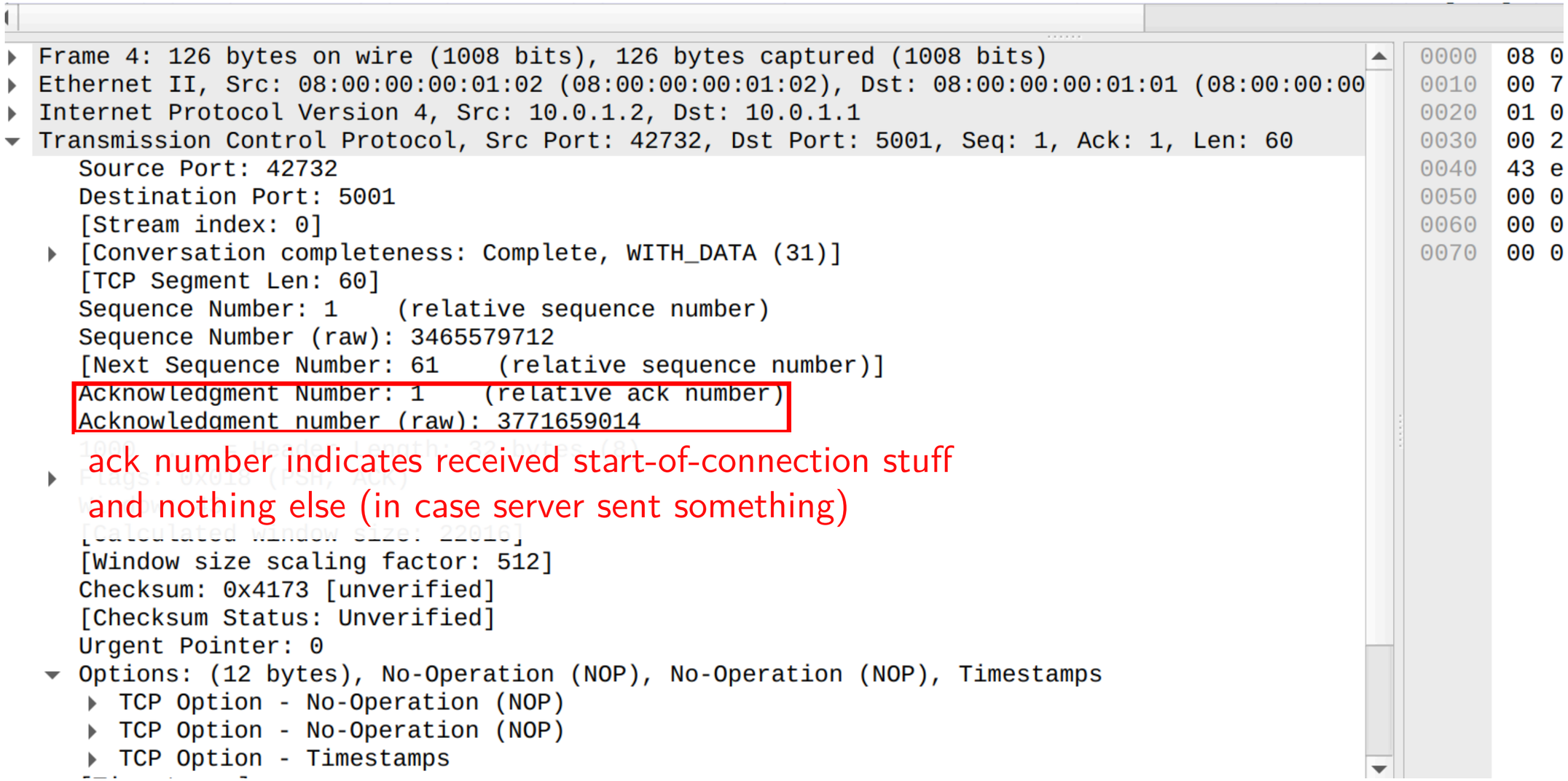
# first data packet



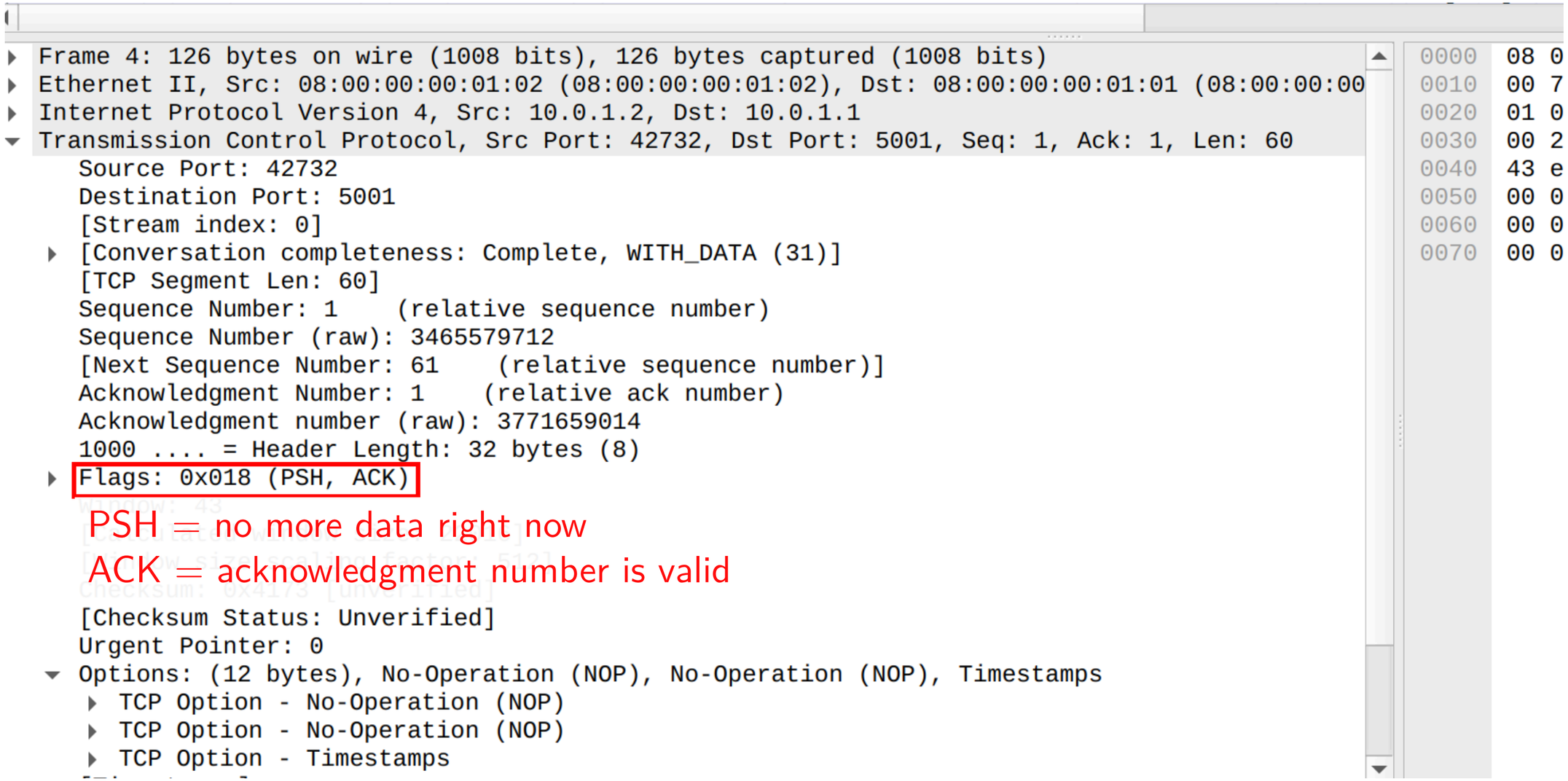
# first data packet



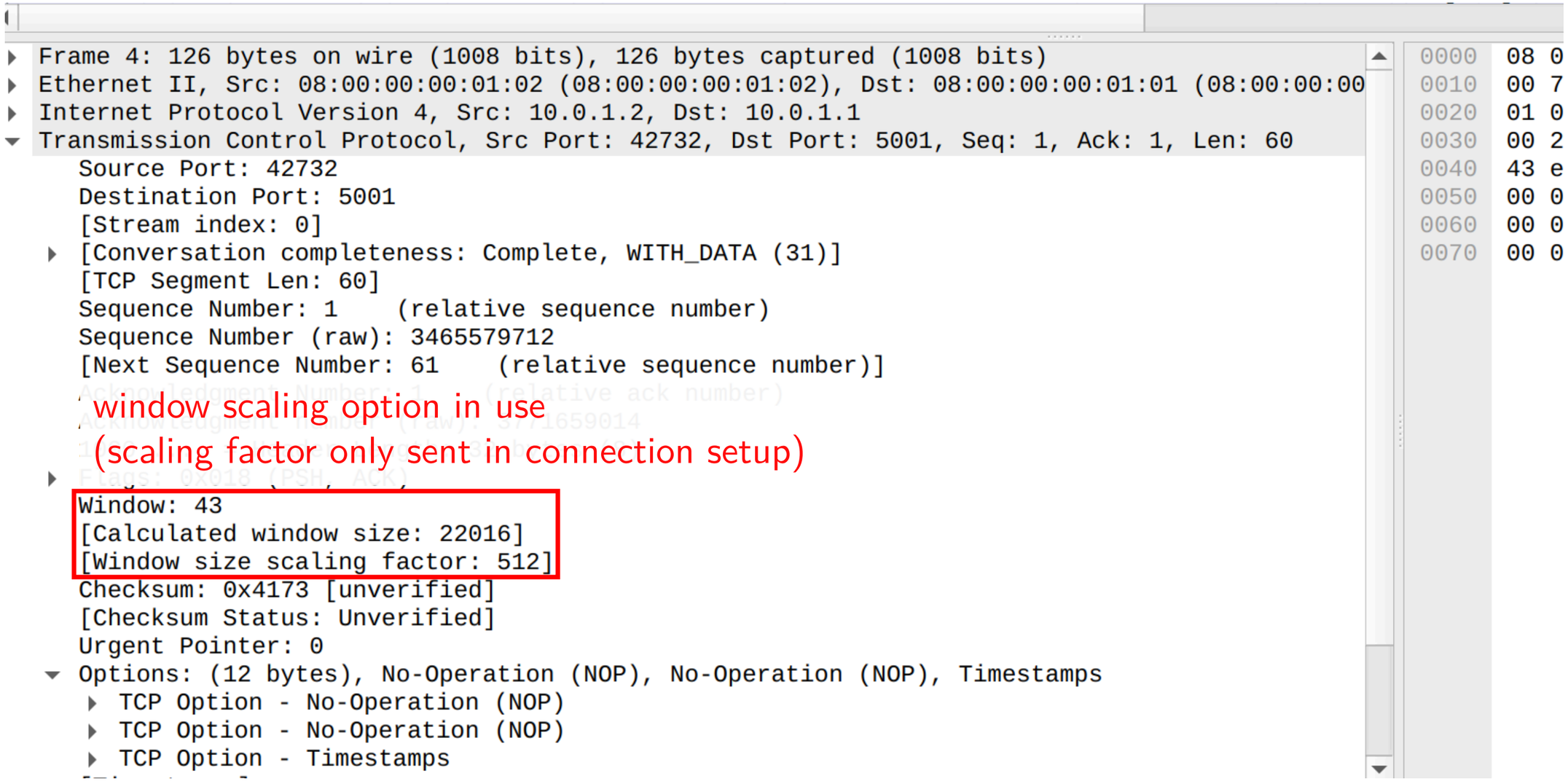
# first data packet



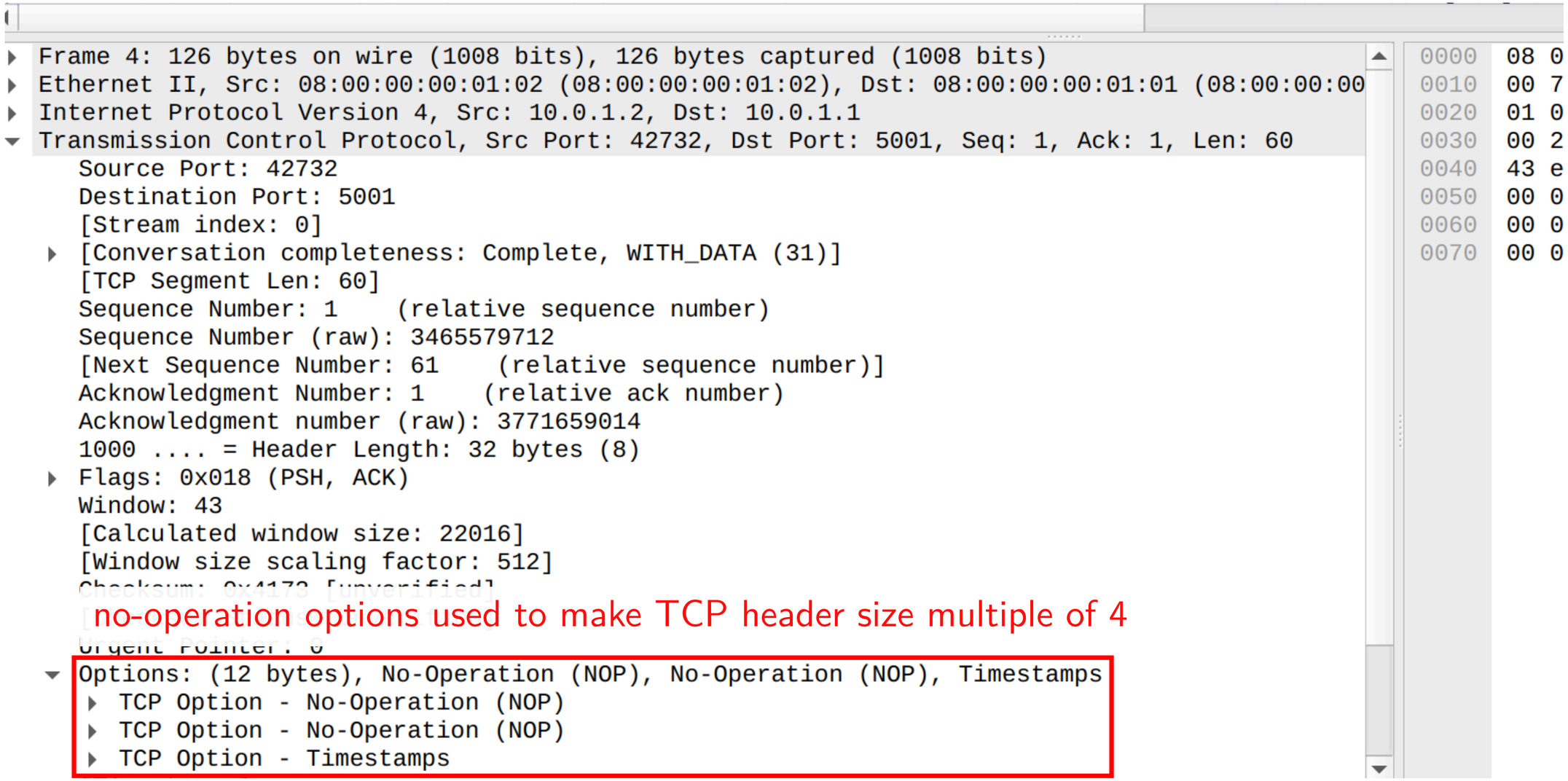
# first data packet



# first data packet

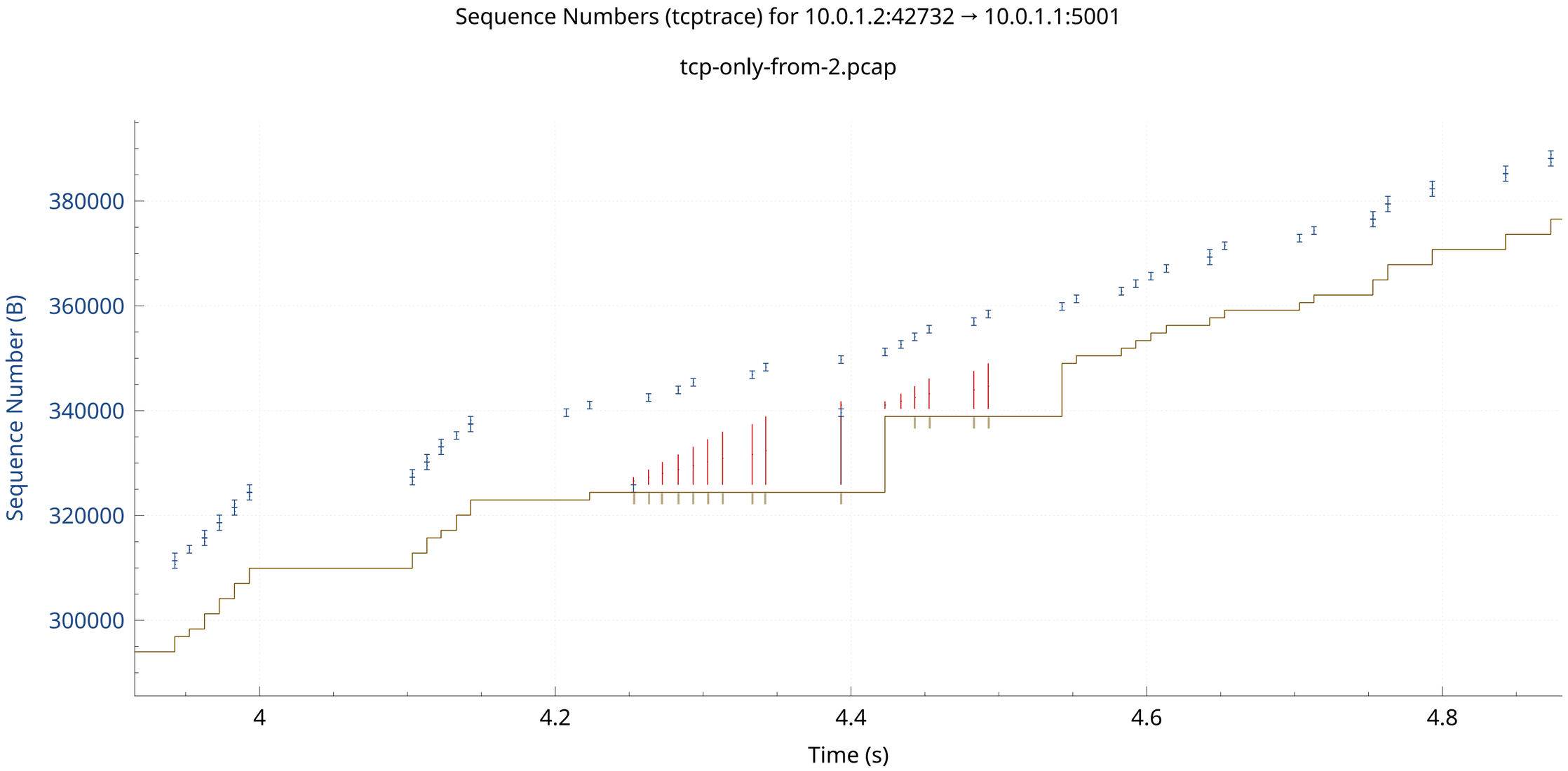


# first data packet




no-operation options used to make TCP header size multiple of 4

# sequence numbers graph



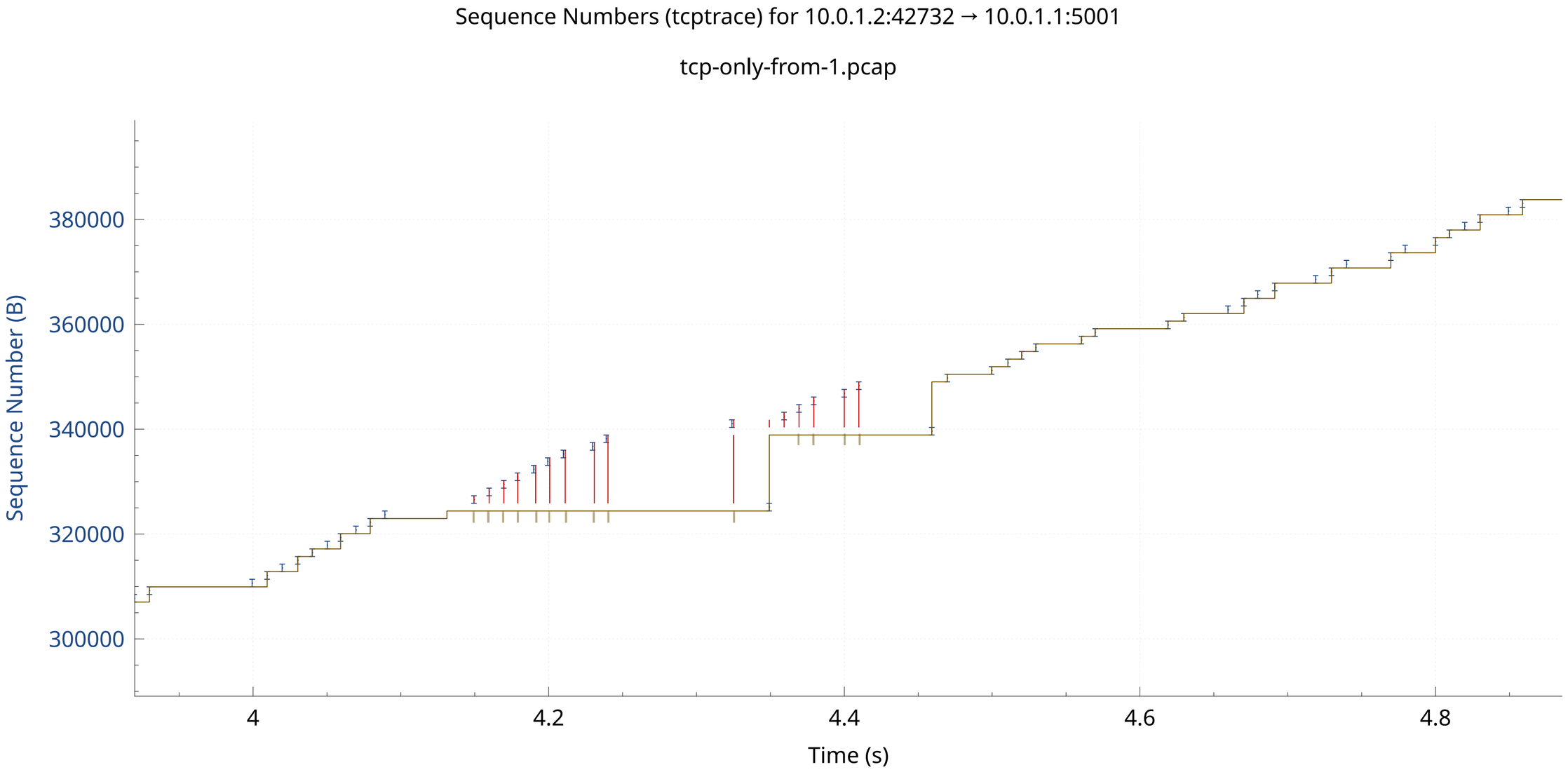
# reading thigs graph

bottom line = last ack number

notches on bottom line = duplicate acks

red lines = selective ACK info

# diff. timing in opposite direction



# backup slides

# TCP

transmission control protocol (TCP)

implements reliable streams of bytes

similar mechanism to what we've described

# TCP extras/differences

bidirectional —

- separate sequence numbers in each direction
- can combine data (from A to B) with acknowledgment (from B to A)

sequence numbers are byte numbers —

- can retransmit data in different sized packets
- sequence numbers = index of *first byte* sent
- acknowledgment numbers = 1 + index of *last byte* acknowledged

dynamic/variable window sizes

- we'll discuss strategies later

offical name for packets = *segments*

# exercise (1)

(assume captured at ← end)

| direction | seq | ack | data len |
|---|---|---|---|
| → | 100 | 200 | 200 |
| → | 300 | 200 | 300 |
| ← | 300 | 600 | 0 |
| ← | 300 | 600 | 100 |
| → | ? | ? | 0 |

# exercise (2a)

(assume captured at $\rightarrow$ end)

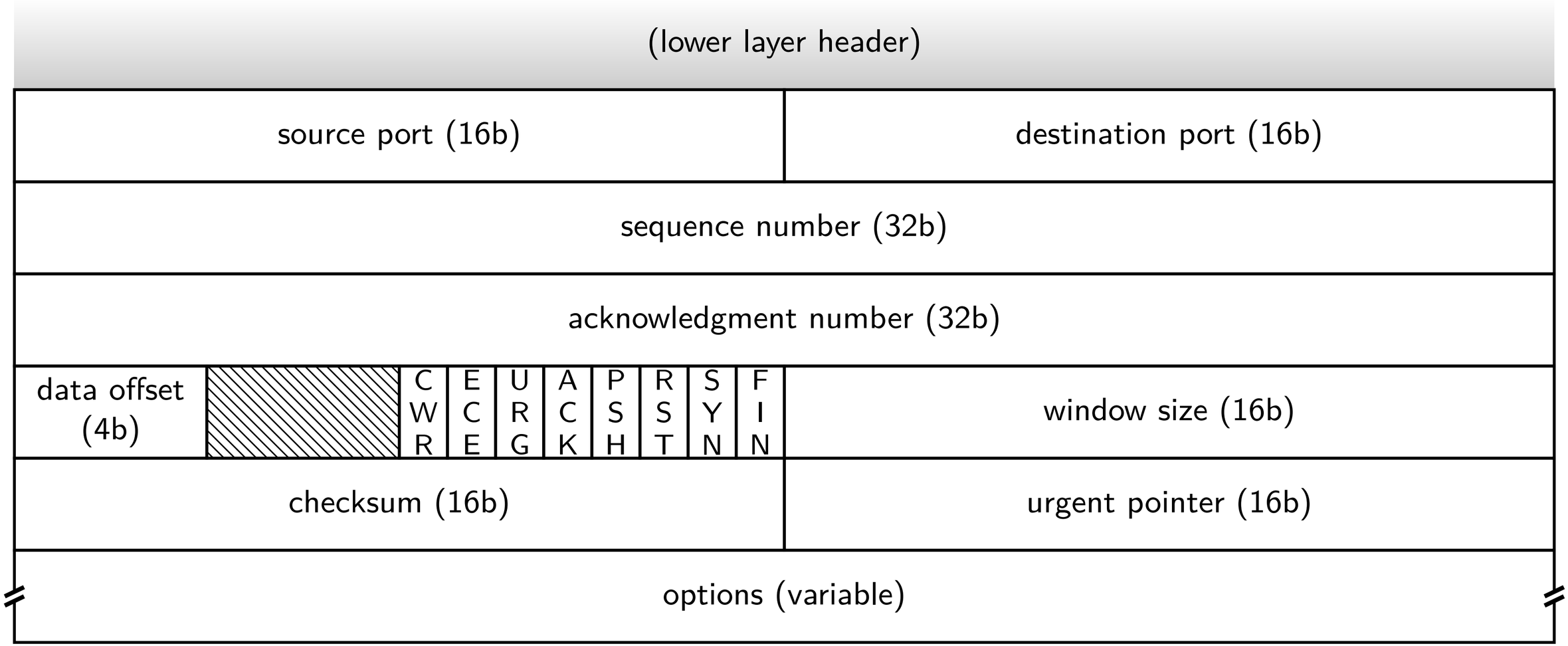
| direction | seq | ack | data len |
|---|---|---|---|
| $\rightarrow$ | 100 | 200 | 200 |
| $\leftarrow$ | ? | 400 | 0 |
| $\rightarrow$ | 300 | 200 | 300 |
| $\leftarrow$ | ? | 600 | 0 |
| $\leftarrow$ | ? | ? | 100 |
| $\rightarrow$ | 600 | 300 | 0 |

# exercise (2b)

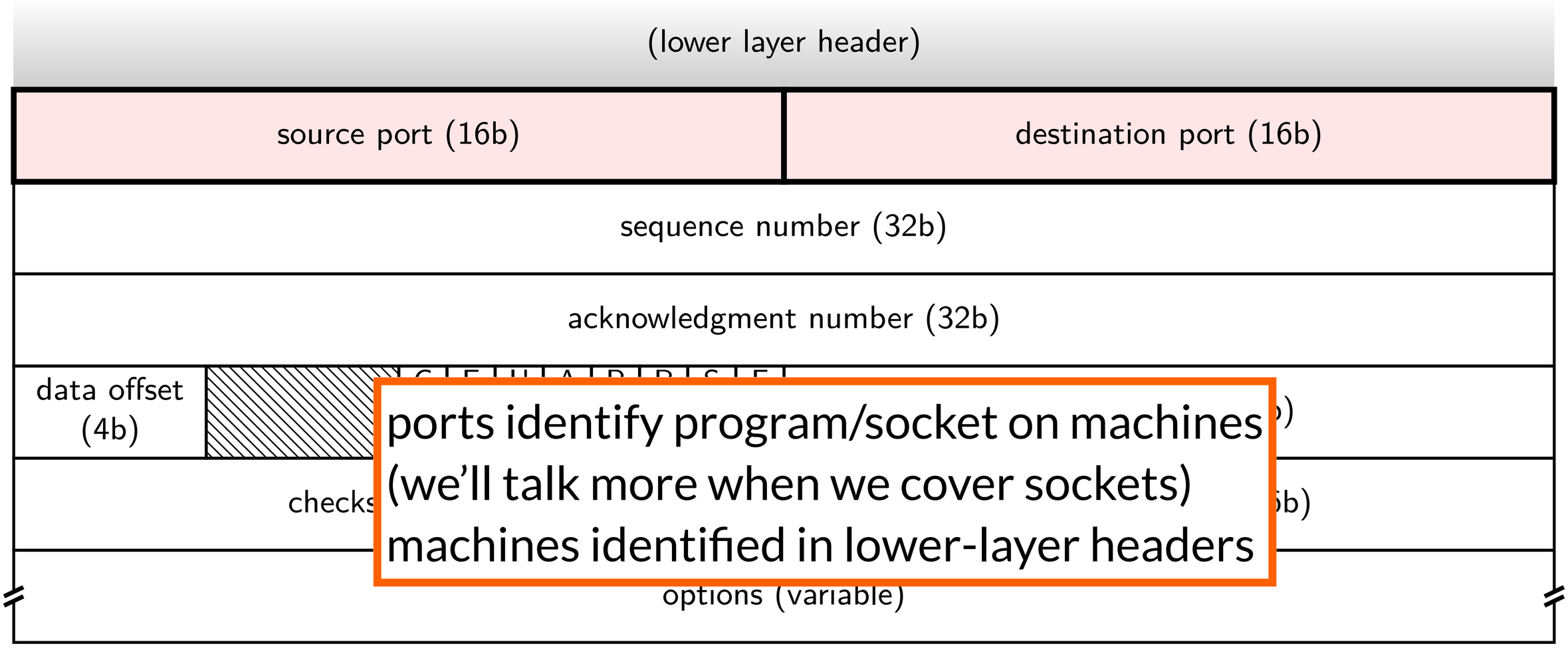
(assume captured at $\rightarrow$ end)

| direction | seq | ack | data len |
|---|---|---|---|
| $\rightarrow$ | 100 | 200 | 200 |
| $\leftarrow$ | 200 | 50 | 0 |
| $\rightarrow$ | 300 | 200 | 300 |
| $\leftarrow$ | 200 | ? | 0 |

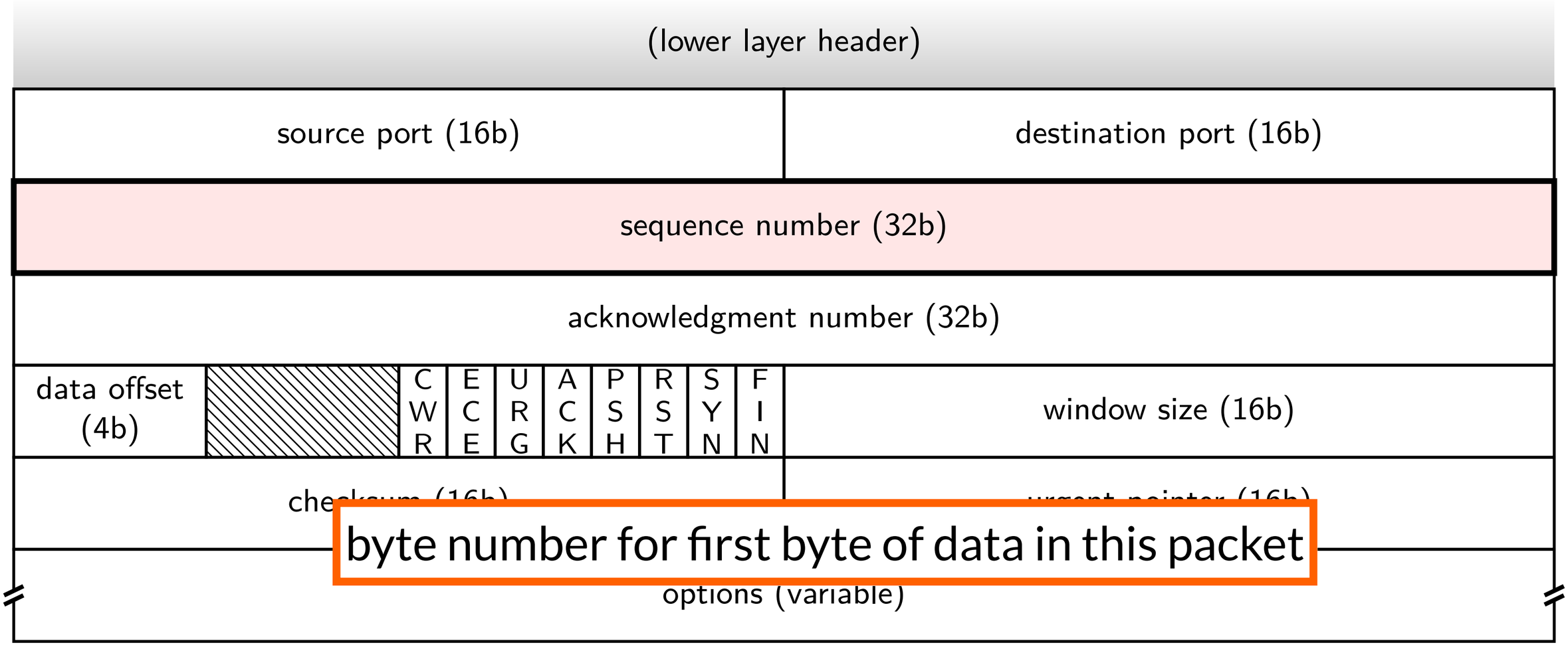
# TCP segment format



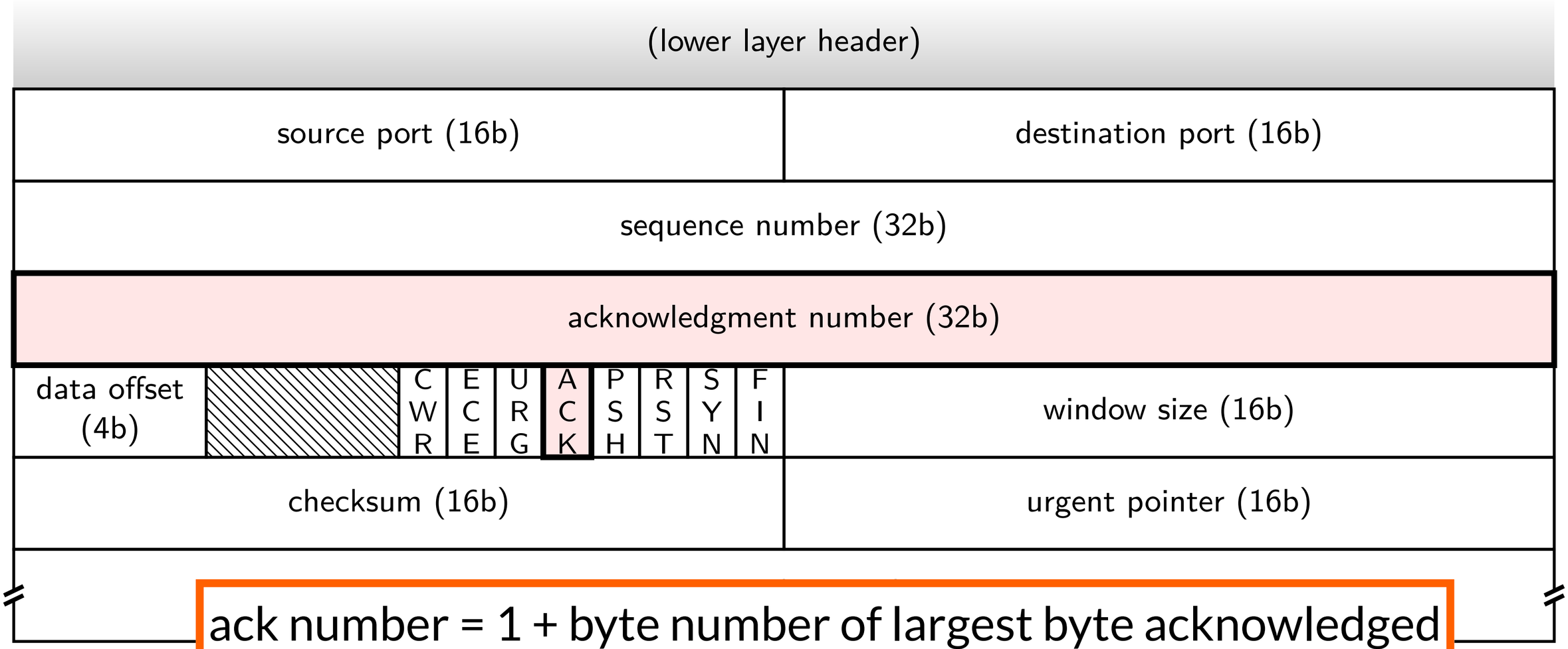
# TCP segment format



# TCP segment format



# TCP segment format




ack number = 1 + byte number of largest byte acknowledged

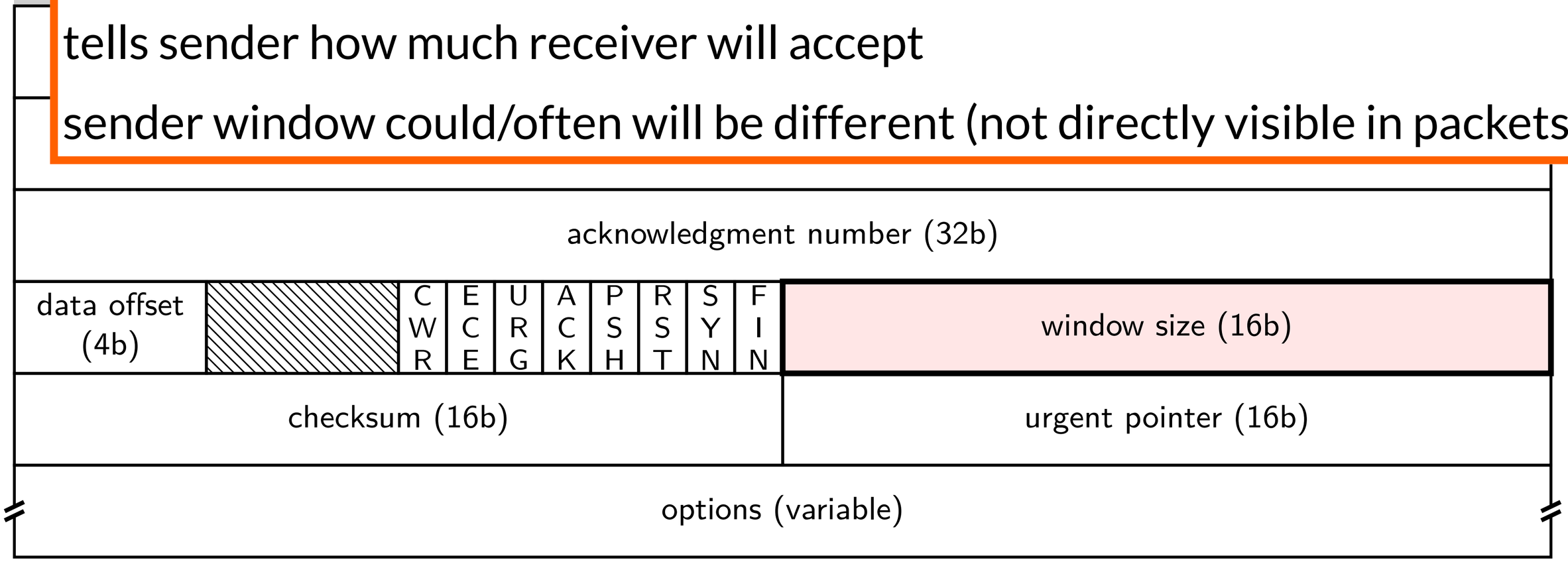
only meaningful if ACK ‘flag’ is 1

# TCP segment format

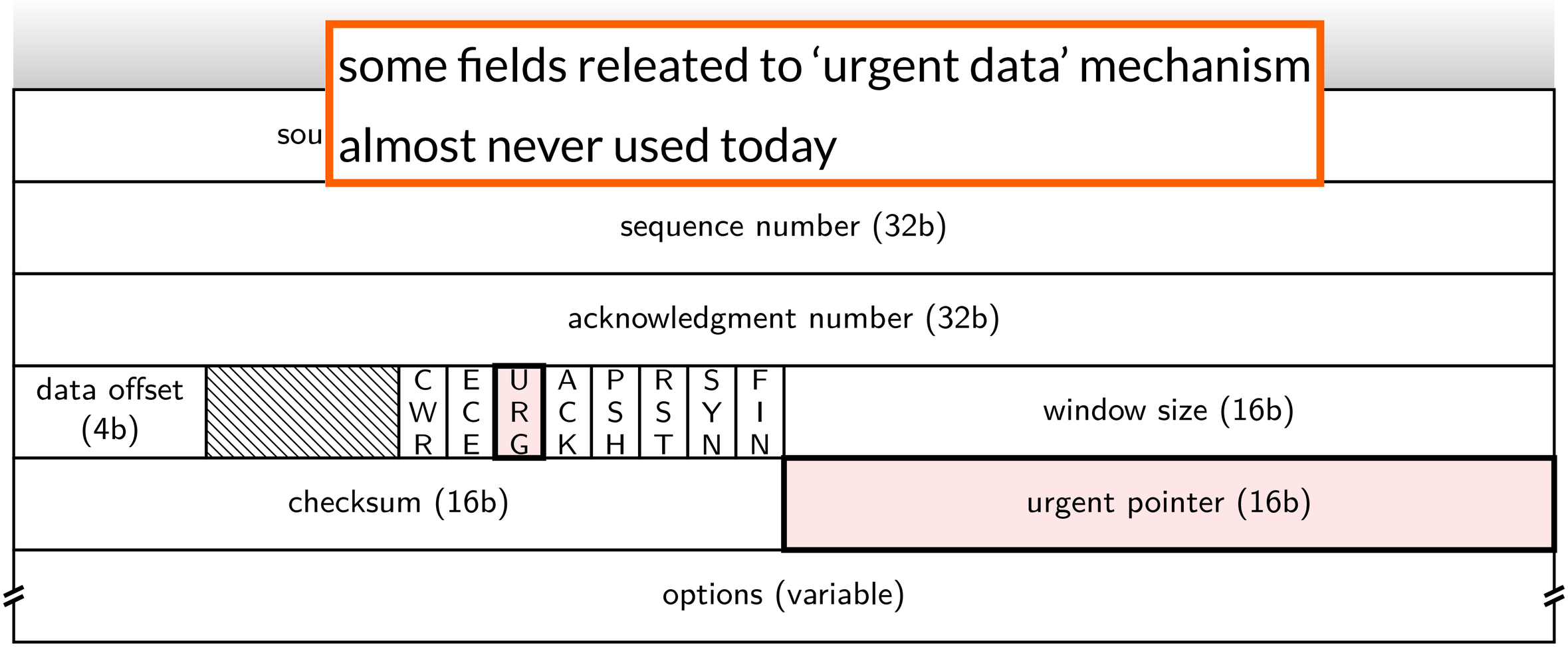
window size is receive window size

tells sender how much receiver will accept

sender window could/often will be different (not directly visible in packets)

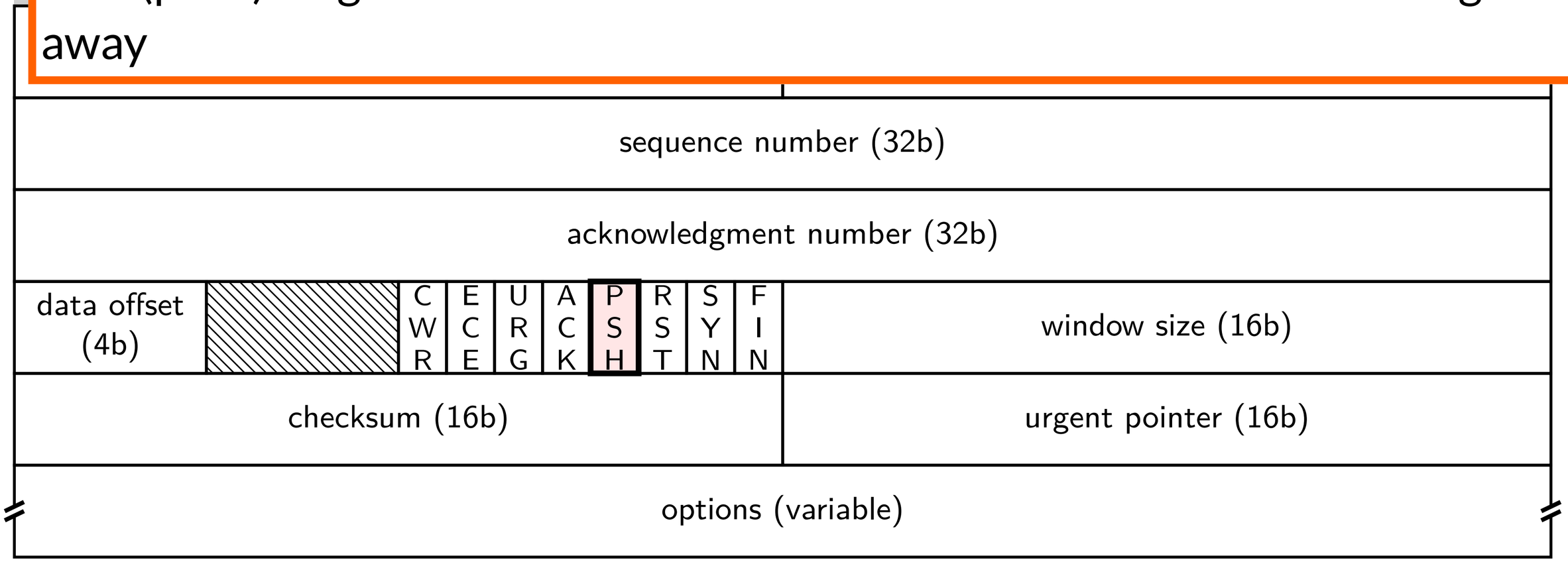


# TCP segment format

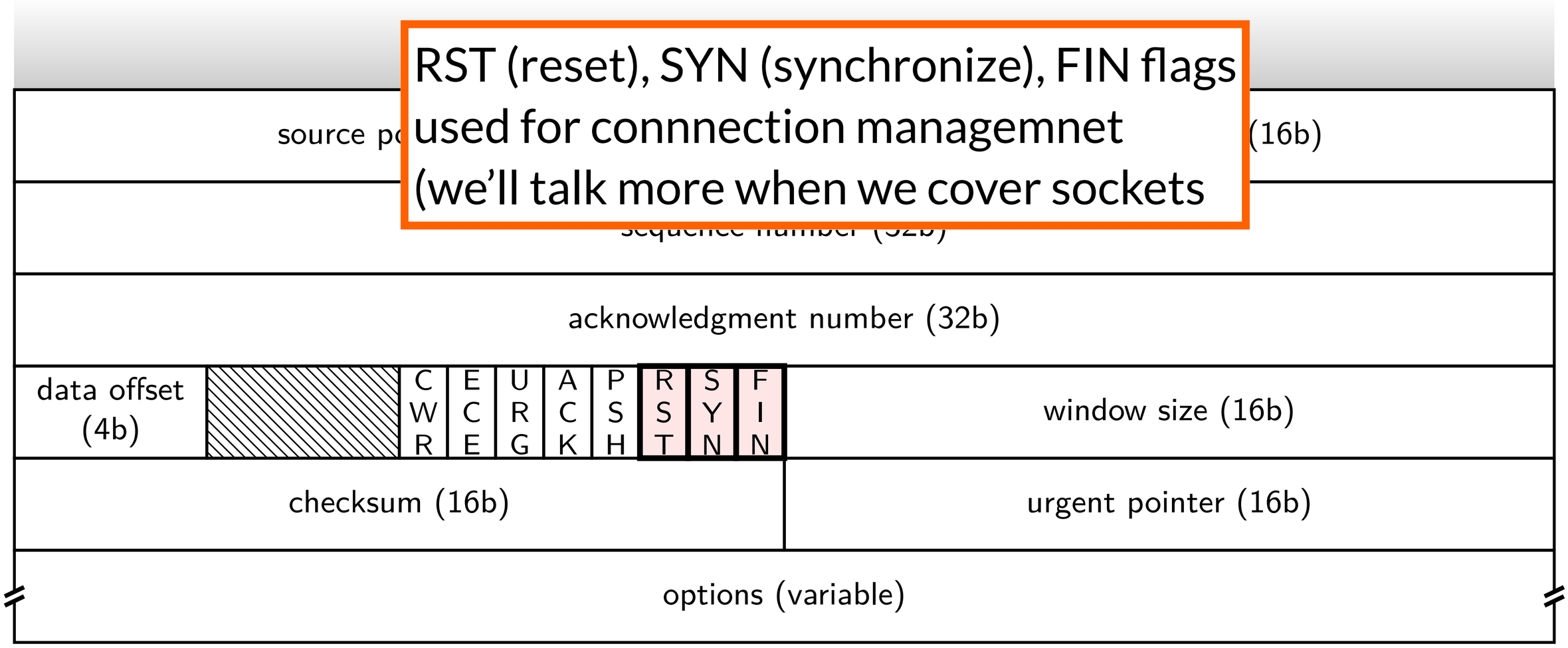


# TCP segment format

PSH (push) ‘flag’ is hint that sender does not have more data to send right away

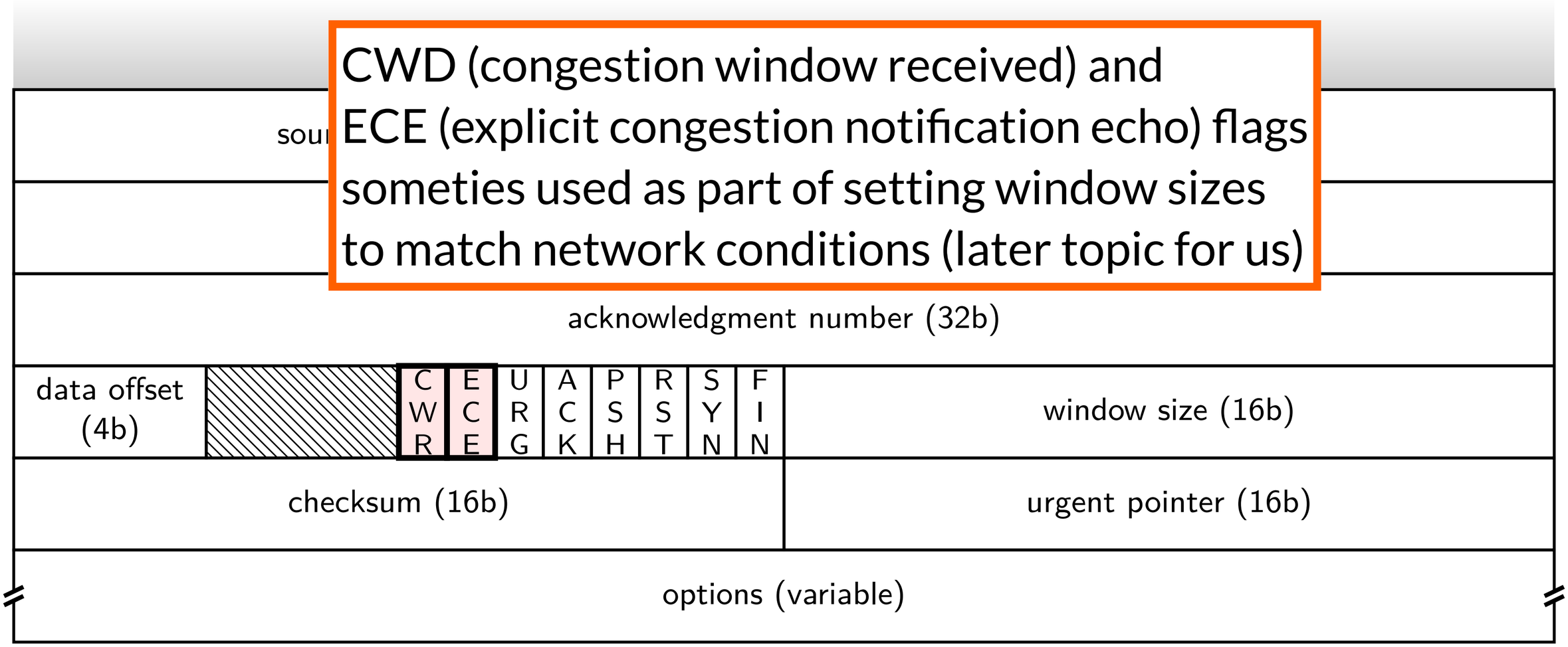


# TCP segment format




RST (reset), SYN (synchronize), FIN flags used for connnection managemnet (we'll talk more when we cover sockets

# TCP segment format

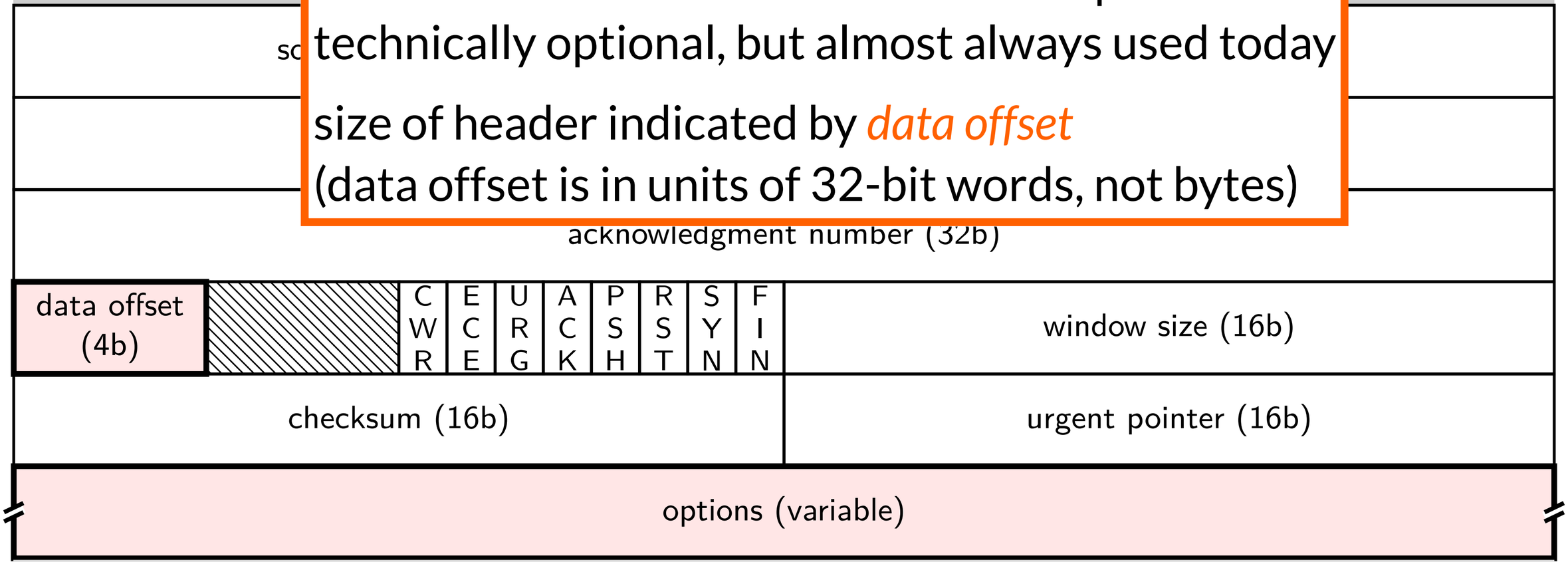



CWD (congestion window received) and
ECE (explicit congestion notification echo) flags
someties used as part of setting window sizes
to match network conditions (later topic for us)

# TCP segment format

header can have variable number of “options”
technically optional, but almost always used today

size of header indicated by *data offset*
(data offset is in units of 32-bit words, not bytes)



# exercise: maximum throughput

let's say we have a receiver window size of 65535 bytes

receiver window size is 16 bit field

and a round-trip time of 100 ms

if we want to avoid sending data the receiver may reject as outside its window, maximum throughput?

A. around 32kbyte/sec
B. around 64kbyte/sec
C. around 128kbyte/sec
D. around 320kbyte/sec
E. around 640kbyte/sec
F. around 1280kbyte/sec
G. something else

# selected TCP options

window size scale factor

- allow receiver window sizes greater than 64k
- needed to get reasonable bandwidth on modern networks

timestamps

- allow figuring out round trip time to estimate timeout
- extend 32-bit sequence number, which is too small for multi-gigabit networks

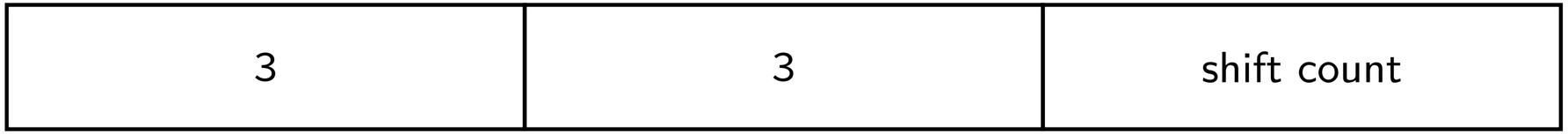
selective acknowledgements

- allow providing information about ‘holes’ in received data
- example: I got bytes 1–5000, 6000–7000, 8000–9000
- without it would only say 5000

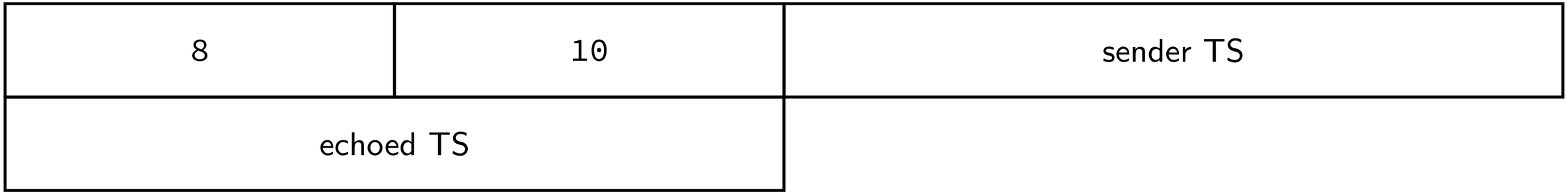
# selected TCP option formats



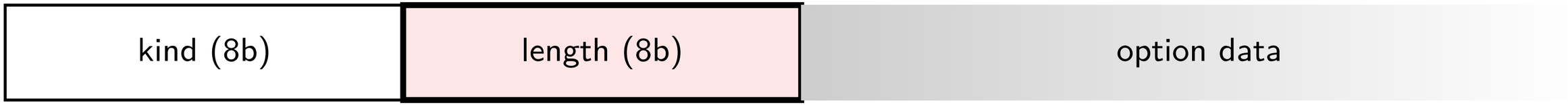

window scale option:



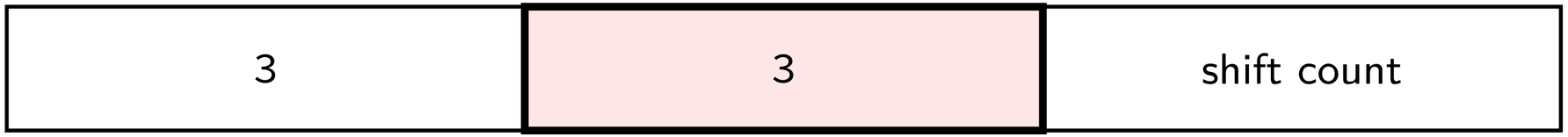

timestamps:



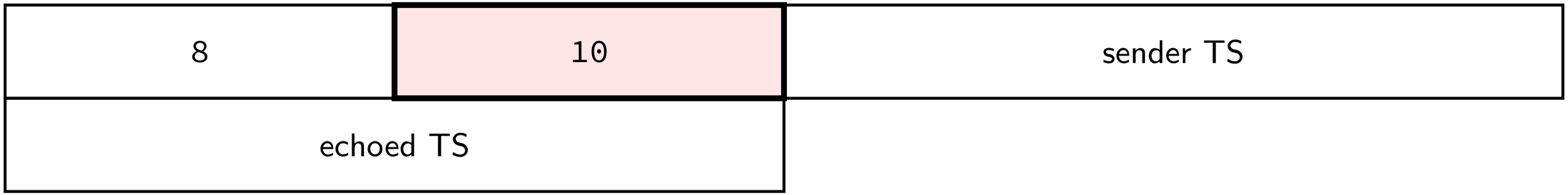
# selected TCP option formats




window scale option:

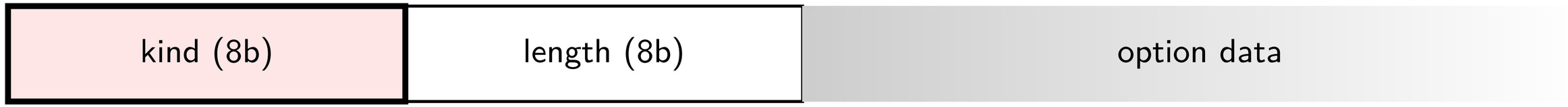



timestamps:

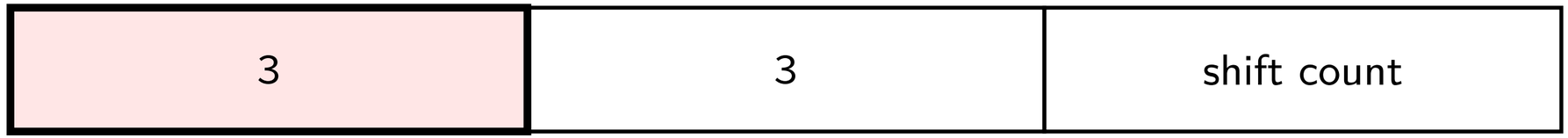



length field permits skipping unrecognized options

# selected TCP option formats




window scale option:




timestamps:

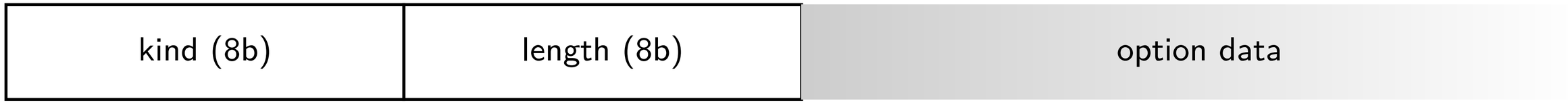




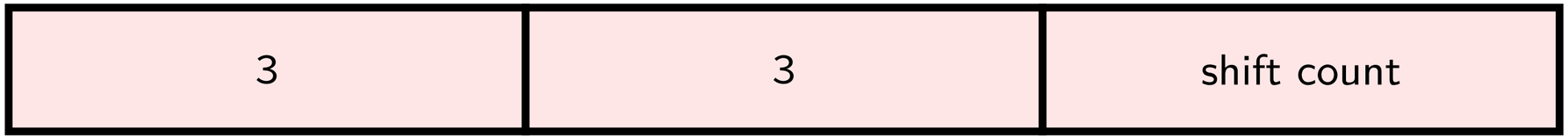


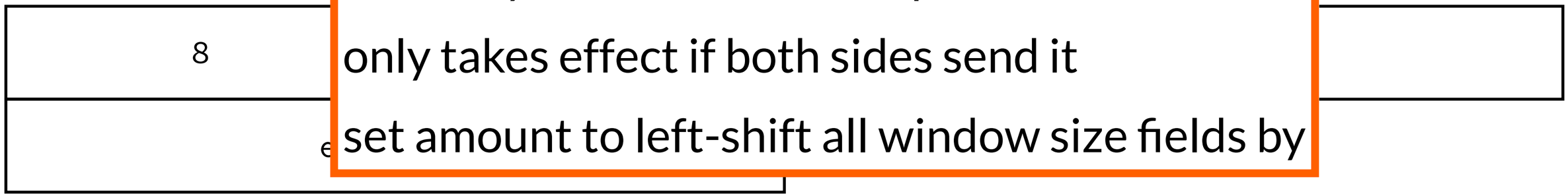
unique kind codes for each option
from list maintained by IANA
(Internet Assigned Numbers Authority)

# selected TCP option formats




*window scale option*:




sent only in connection setup

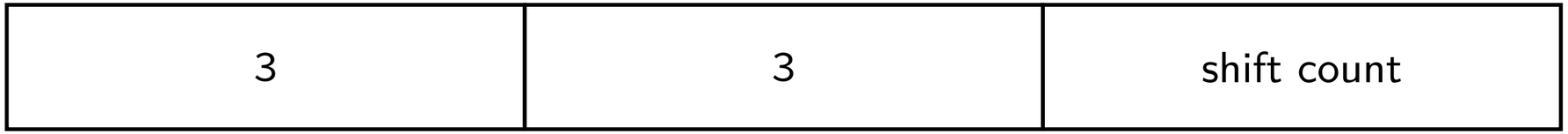
only takes effect if both sides send it

set amount to left-shift all window size fields by

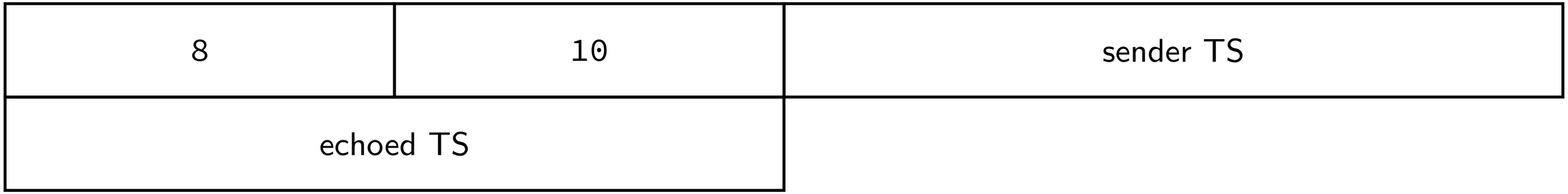
# selected TCP option formats



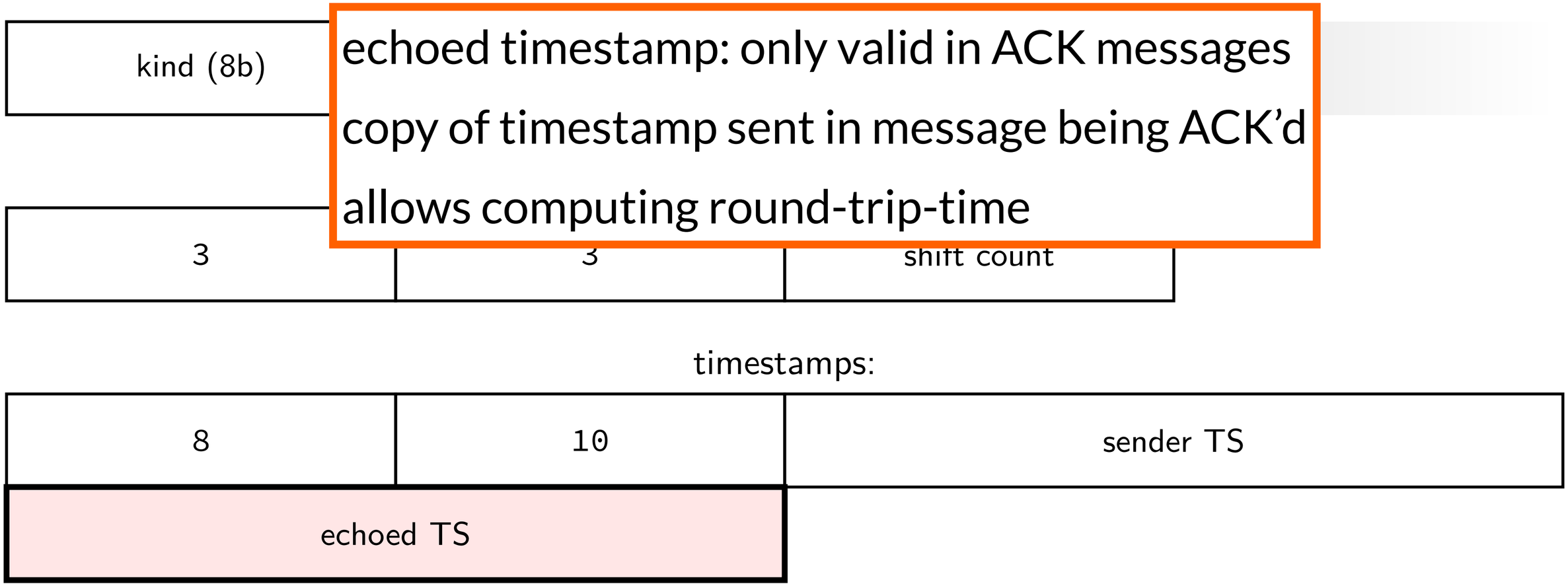

window scale option:




*timestamps*:



# selected TCP option formats




echoed timestamp: only valid in ACK messages

copy of timestamp sent in message being ACK'd

allows computing round-trip-time

# TCP connection setup preview

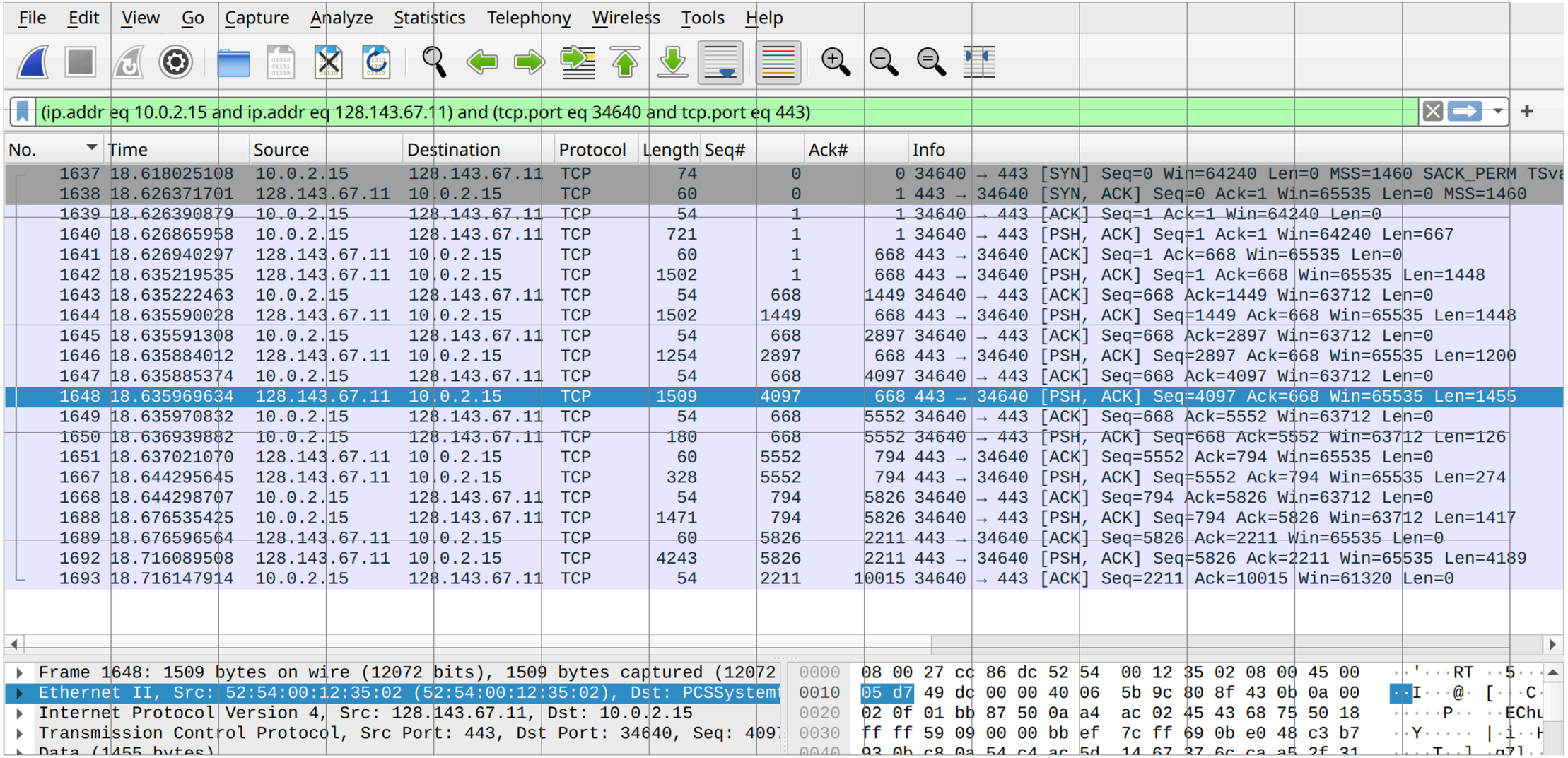
client→server: SYN flag + seq=randomA [no data]

server→client: SYN+ACK flags + seq=randomB ack=randomA+1 [no data]

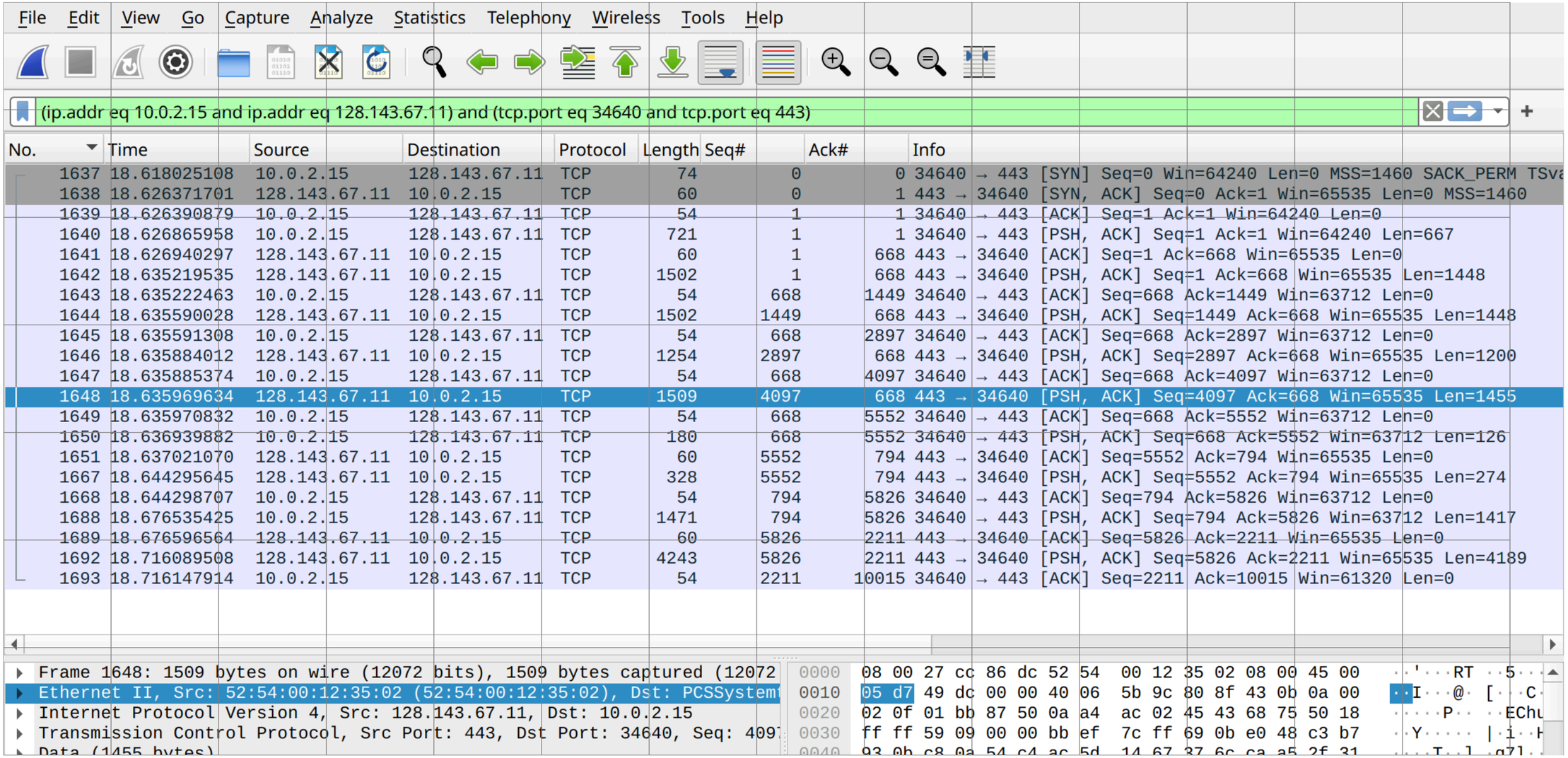
client→server: ACK flags + seq=randomA+1 ack=randomB+1 [no data]

# a TCP connection

# a TCP connection



# a TCP connection



# a TCP connection

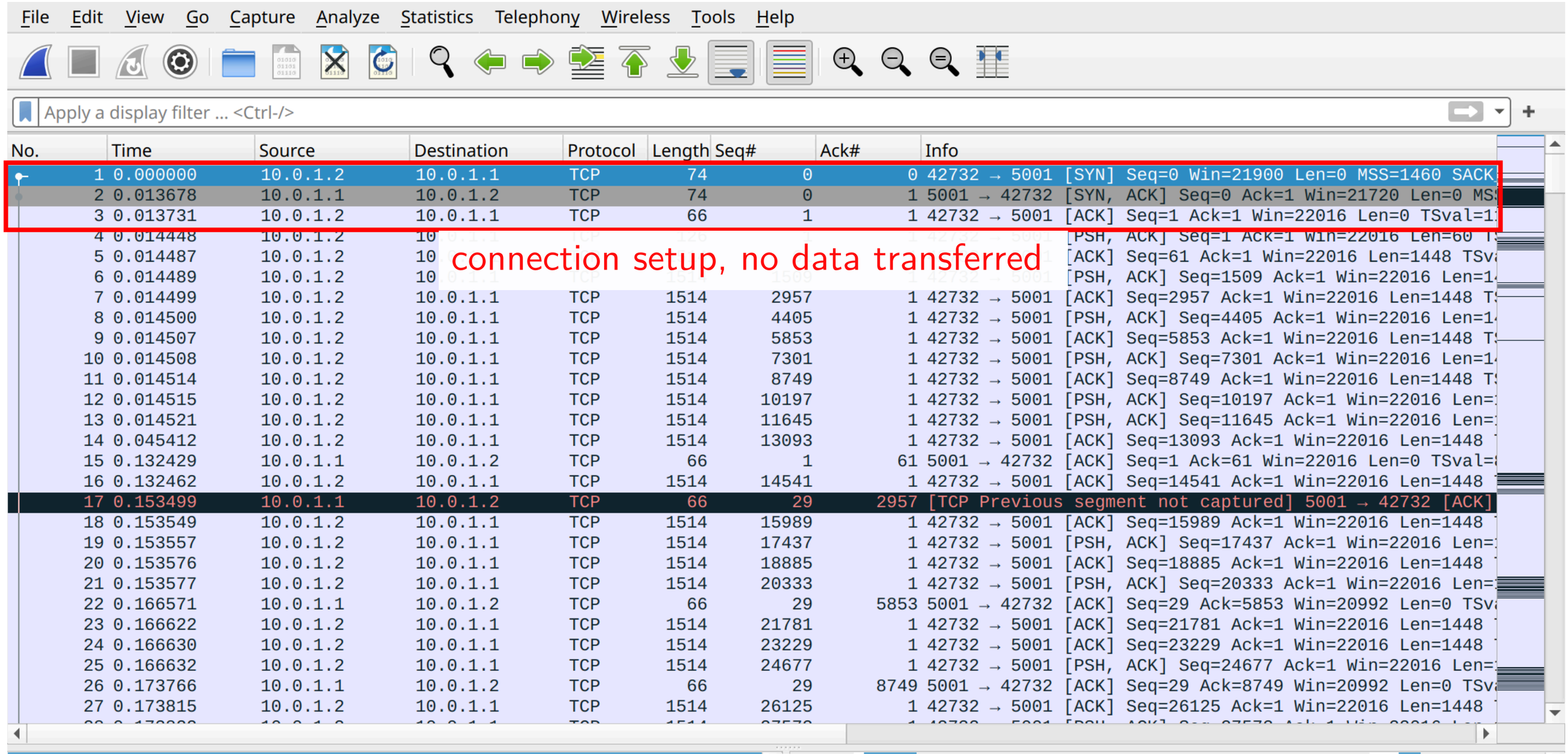
# a TCP connection

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Length | Seq# | Ack# | Info |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.000000 | 10.0.1.2 | 10.0.1.1 | TCP | 74 | 0 | 0 | 42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK |
| 2 | 0.013678 | 10.0.1.1 | 10.0.1.2 | TCP | 74 | 0 | 1 | 5001 → 42732 [SYN, ACK] Seq=0 Ack=1 Win=21720 Len=0 MS |
| 3 | 0.013731 | 10.0.1.2 | 10.0.1.1 | TCP | 66 | 1 | 1 | 42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1 |
| 4 | 0.014448 | 10.0.1.2 | 10.0.1.1 | TCP | 126 | 1 | 1 | 42732 → 5001 [PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T |
| 5 | 0.014487 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 61 | 1 | 42732 → 5001 [ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv |
| 6 | 0.014489 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 1509 | 1 | 42732 → 5001 [PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1 |
| 7 | 0.014499 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 2957 | 1 | 42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T |
| 8 | 0.014500 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 4405 | 1 | 42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1 |
| 9 | 0.014507 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 5853 | 1 | 42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T |
| 10 | 0.014508 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 7301 | 1 | 42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1 |
| 11 | 0.014514 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 8749 | 1 | 42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T |
| 12 | 0.014515 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 10197 | 1 | 42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len= |
| 13 | 0.014521 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 11645 | 1 | 42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len= |
| 14 | 0.045412 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 13093 | 1 | 42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448 |
| 15 | 0.132429 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 1 | 61 | 5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval= |
| 16 | 0.132462 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 14541 | 1 | 42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448 |
| 17 | 0.153499 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 2957 | [TCP Previous segment not captured] 5001 → 42732 [ACK] |
| 18 | 0.153549 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 15989 | 1 | 42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448 |
| 19 | 0.153557 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 17437 | 1 | 42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len= |
| 20 | 0.153576 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 18885 | 1 | 42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448 |
| 21 | 0.153577 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 20333 | 1 | 42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len= |
| 22 | 0.166571 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 5853 | 5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv |
| 23 | 0.166622 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 21781 | 1 | 42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448 |
| 24 | 0.166630 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 23229 | 1 | 42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448 |
| 25 | 0.166632 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 24677 | 1 | 42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len= |
| 26 | 0.173766 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 8749 | 5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv |
| 27 | 0.173815 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 26125 | 1 | 42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448 |

# a TCP connection



# a TCP connection

server+client sequence numbers advance by 1 to indicate where in setup

| ngth | Seq# | Ack# | In |
|---|---|---|---|
| 74 | 0 | 0 | 42 |
| 74 | 0 | 1 | 50 |
| 66 | 1 | 1 | 42 |

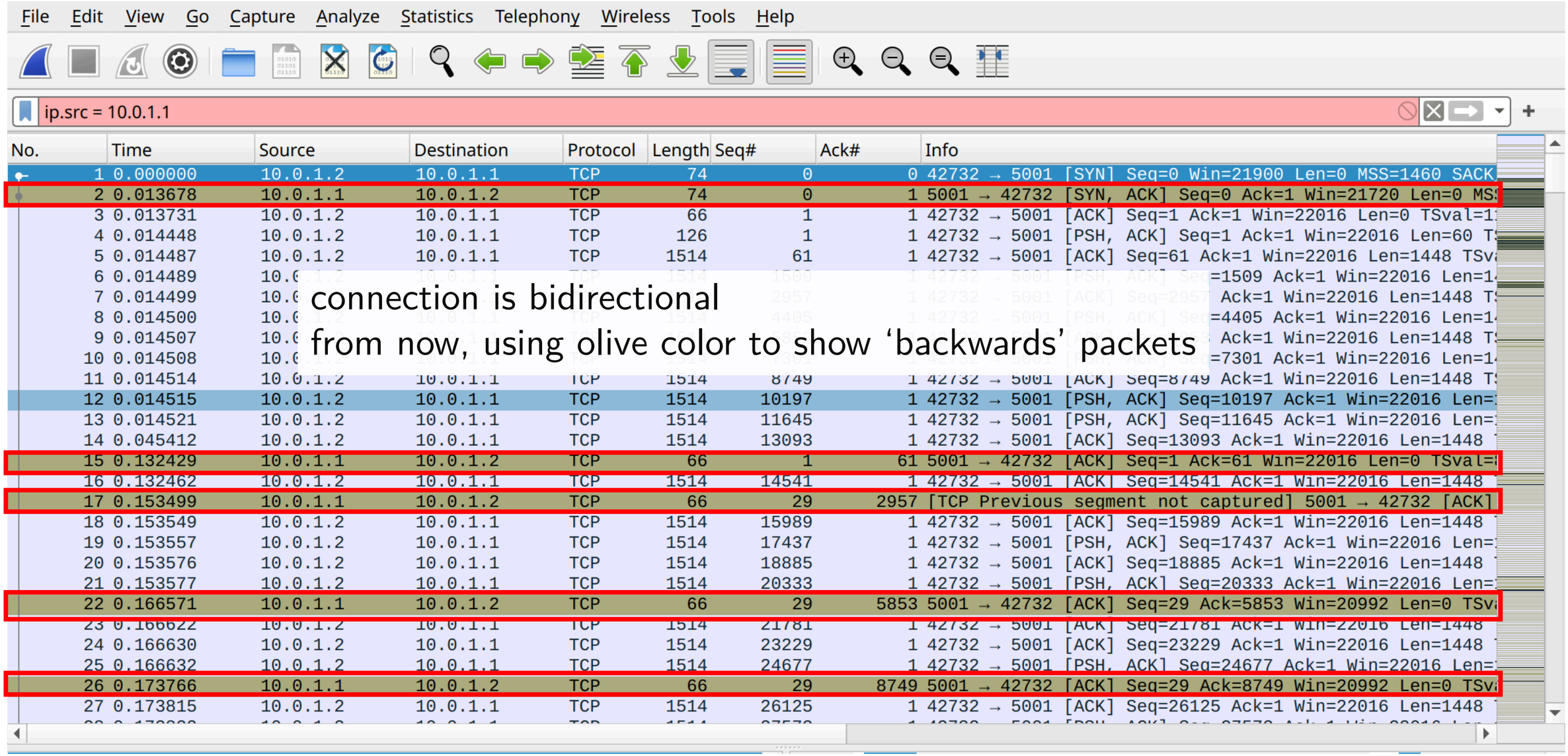
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src = 10.0.1.1

| No. | Time | Source | Destination | Protocol | Length | Seq# | Ack# | Info |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.000000 | 10.0.1.2 | 10.0.1.1 | TCP | 74 | 0 | 0 | 42732 → 5001 [SYN] Seq=0 Win=21900 Len=0 MSS=1460 SACK |
| 2 | 0.013678 | 10.0.1.1 | 10.0.1.2 | TCP | 74 | 0 | 1 | 5001 → 42732 [SYN, ACK] Seq=0 Ack=1 Win=21720 Len=0 MS |
| 3 | 0.013731 | 10.0.1.2 | 10.0.1.1 | TCP | 66 | 1 | 1 | 42732 → 5001 [ACK] Seq=1 Ack=1 Win=22016 Len=0 TSval=1 |
| 4 | 0.014448 | 10.0.1.2 | 10 | | | | | [PSH, ACK] Seq=1 Ack=1 Win=22016 Len=60 T |
| 5 | 0.014487 | 10.0.1.2 | 10 | | | | | [ACK] Seq=61 Ack=1 Win=22016 Len=1448 TSv |
| 6 | 0.014489 | 10.0.1.2 | 10 | | | | | [PSH, ACK] Seq=1509 Ack=1 Win=22016 Len=1 |
| 7 | 0.014499 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 2957 | 1 | 42732 → 5001 [ACK] Seq=2957 Ack=1 Win=22016 Len=1448 T |
| 8 | 0.014500 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 4405 | 1 | 42732 → 5001 [PSH, ACK] Seq=4405 Ack=1 Win=22016 Len=1 |
| 9 | 0.014507 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 5853 | 1 | 42732 → 5001 [ACK] Seq=5853 Ack=1 Win=22016 Len=1448 T |
| 10 | 0.014508 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 7301 | 1 | 42732 → 5001 [PSH, ACK] Seq=7301 Ack=1 Win=22016 Len=1 |
| 11 | 0.014514 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 8749 | 1 | 42732 → 5001 [ACK] Seq=8749 Ack=1 Win=22016 Len=1448 T |
| 12 | 0.014515 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 10197 | 1 | 42732 → 5001 [PSH, ACK] Seq=10197 Ack=1 Win=22016 Len= |
| 13 | 0.014521 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 11645 | 1 | 42732 → 5001 [PSH, ACK] Seq=11645 Ack=1 Win=22016 Len= |
| 14 | 0.045412 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 13093 | 1 | 42732 → 5001 [ACK] Seq=13093 Ack=1 Win=22016 Len=1448 |
| 15 | 0.132429 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 1 | 61 | 5001 → 42732 [ACK] Seq=1 Ack=61 Win=22016 Len=0 TSval= |
| 16 | 0.132462 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 14541 | 1 | 42732 → 5001 [ACK] Seq=14541 Ack=1 Win=22016 Len=1448 |
| 17 | 0.153499 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 2957 | [TCP Previous segment not captured] 5001 → 42732 [ACK] |
| 18 | 0.153549 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 15989 | 1 | 42732 → 5001 [ACK] Seq=15989 Ack=1 Win=22016 Len=1448 |
| 19 | 0.153557 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 17437 | 1 | 42732 → 5001 [PSH, ACK] Seq=17437 Ack=1 Win=22016 Len= |
| 20 | 0.153576 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 18885 | 1 | 42732 → 5001 [ACK] Seq=18885 Ack=1 Win=22016 Len=1448 |
| 21 | 0.153577 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 20333 | 1 | 42732 → 5001 [PSH, ACK] Seq=20333 Ack=1 Win=22016 Len= |
| 22 | 0.166571 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 5853 | 5001 → 42732 [ACK] Seq=29 Ack=5853 Win=20992 Len=0 TSv |
| 23 | 0.166622 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 21781 | 1 | 42732 → 5001 [ACK] Seq=21781 Ack=1 Win=22016 Len=1448 |
| 24 | 0.166630 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 23229 | 1 | 42732 → 5001 [ACK] Seq=23229 Ack=1 Win=22016 Len=1448 |
| 25 | 0.166632 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 24677 | 1 | 42732 → 5001 [PSH, ACK] Seq=24677 Ack=1 Win=22016 Len= |
| 26 | 0.173766 | 10.0.1.1 | 10.0.1.2 | TCP | 66 | 29 | 8749 | 5001 → 42732 [ACK] Seq=29 Ack=8749 Win=20992 Len=0 TSv |
| 27 | 0.173815 | 10.0.1.2 | 10.0.1.1 | TCP | 1514 | 26125 | 1 | 42732 → 5001 [ACK] Seq=26125 Ack=1 Win=22016 Len=1448 |

connection setup, no data transferred

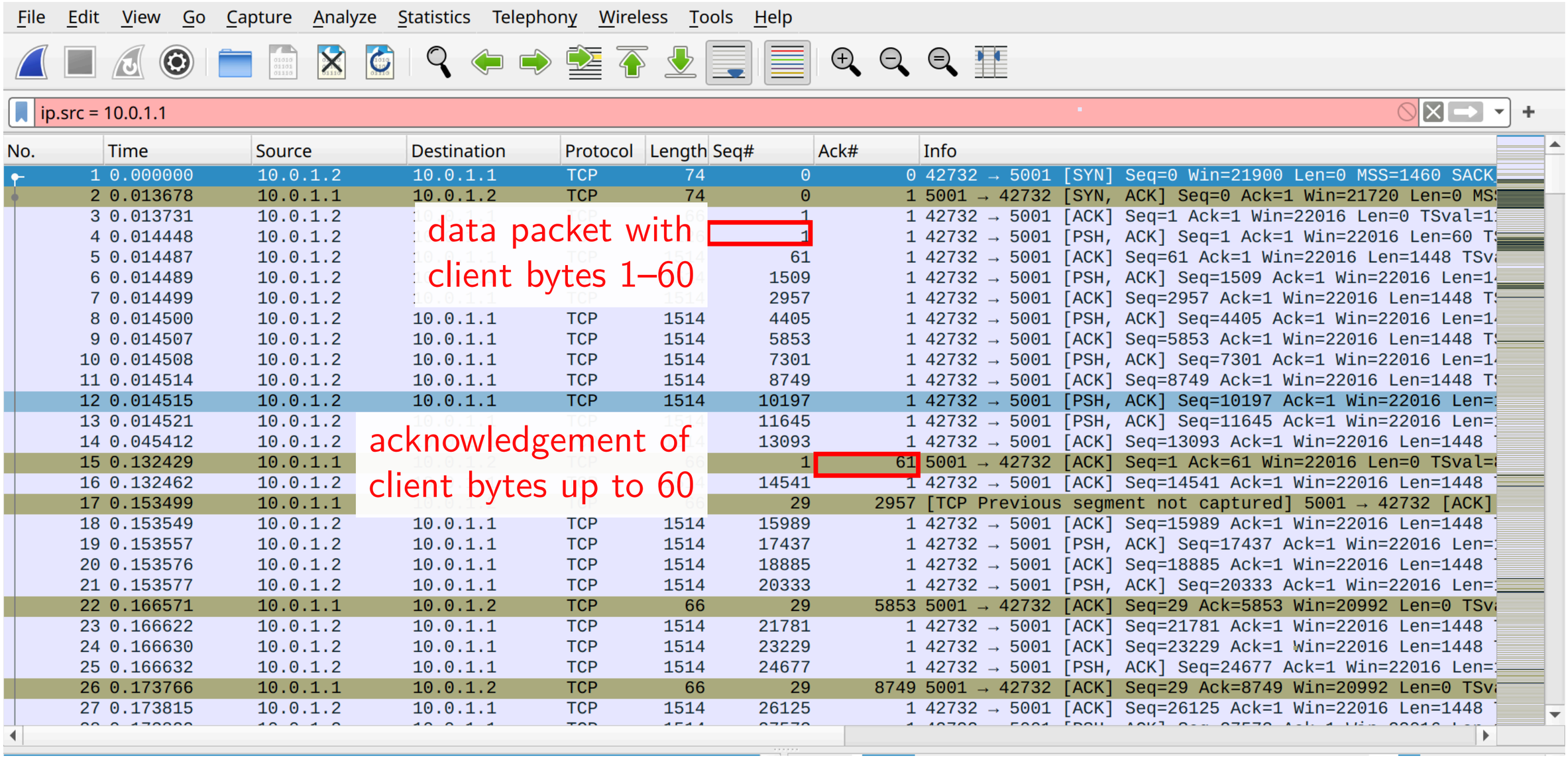
# a TCP connection



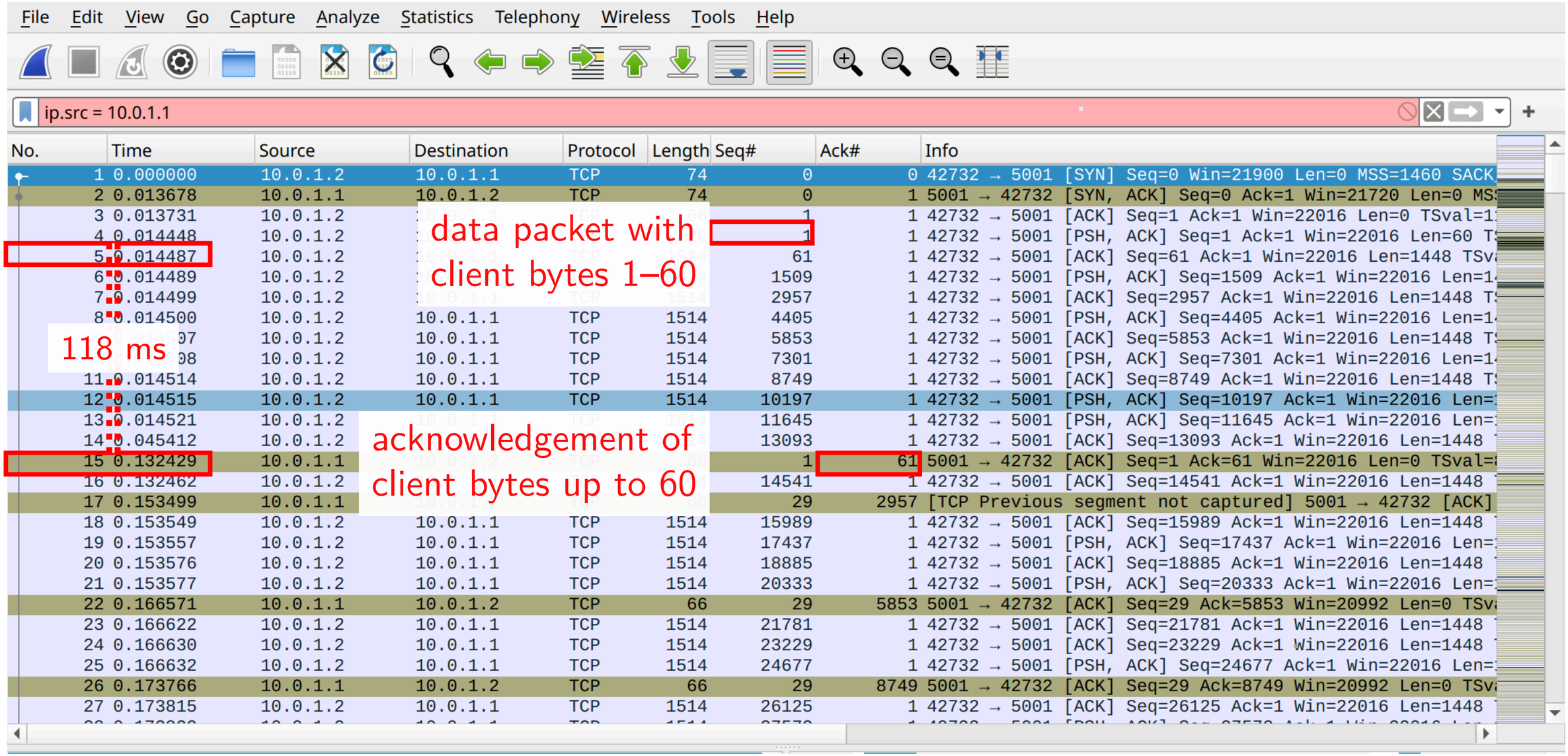

connection is bidirectional

from now, using olive color to show ‘backwards’ packets

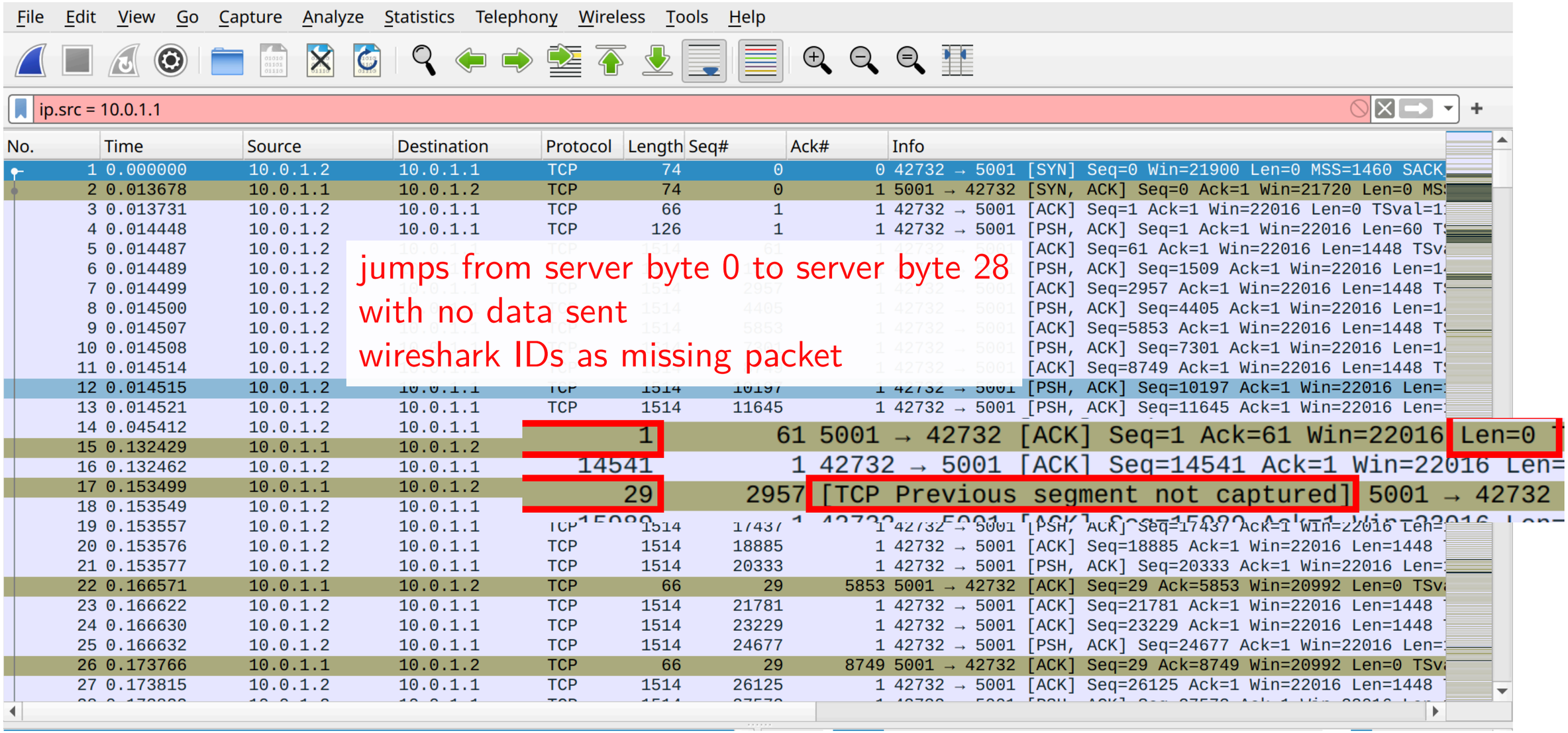
# a TCP connection



# a TCP connection

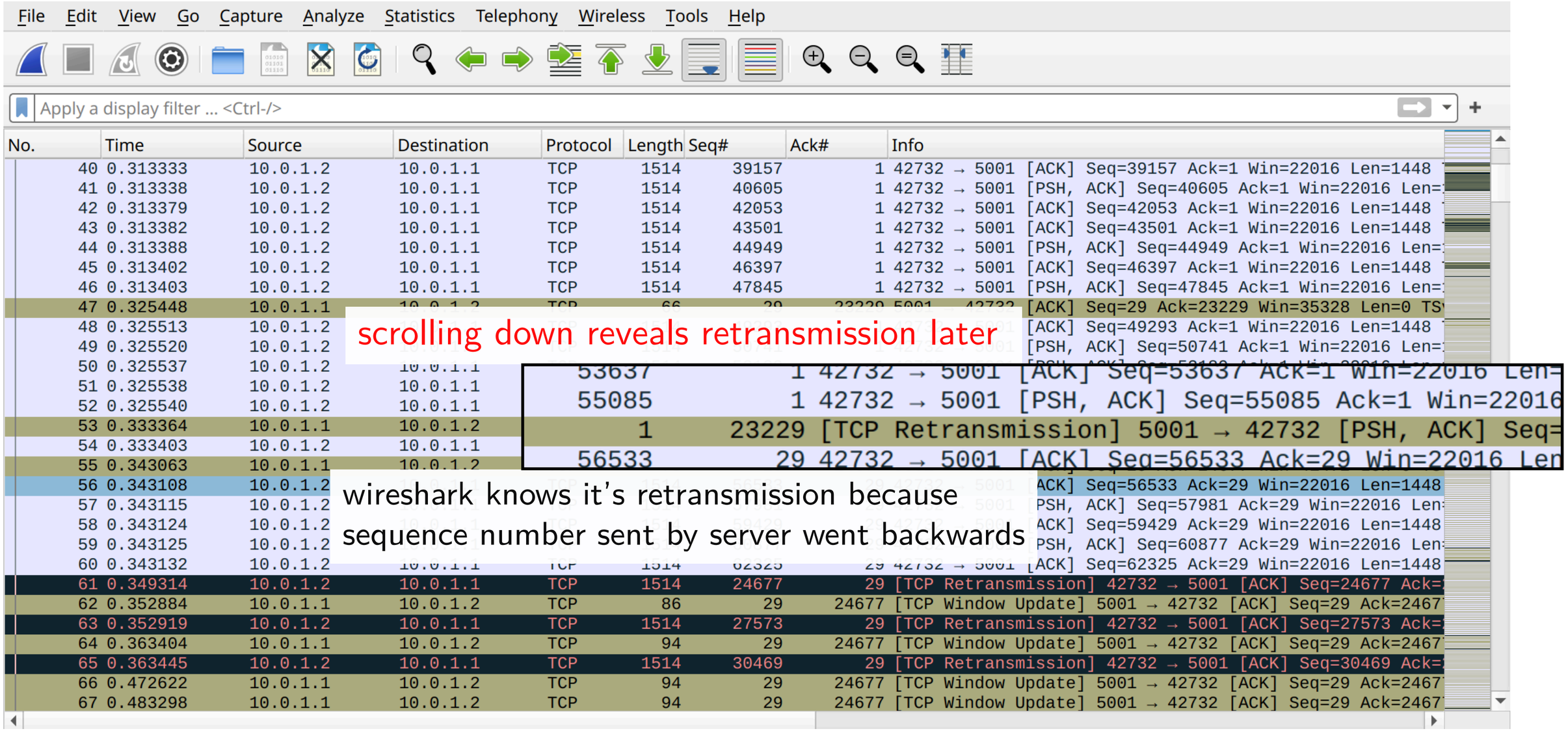


# a TCP connection



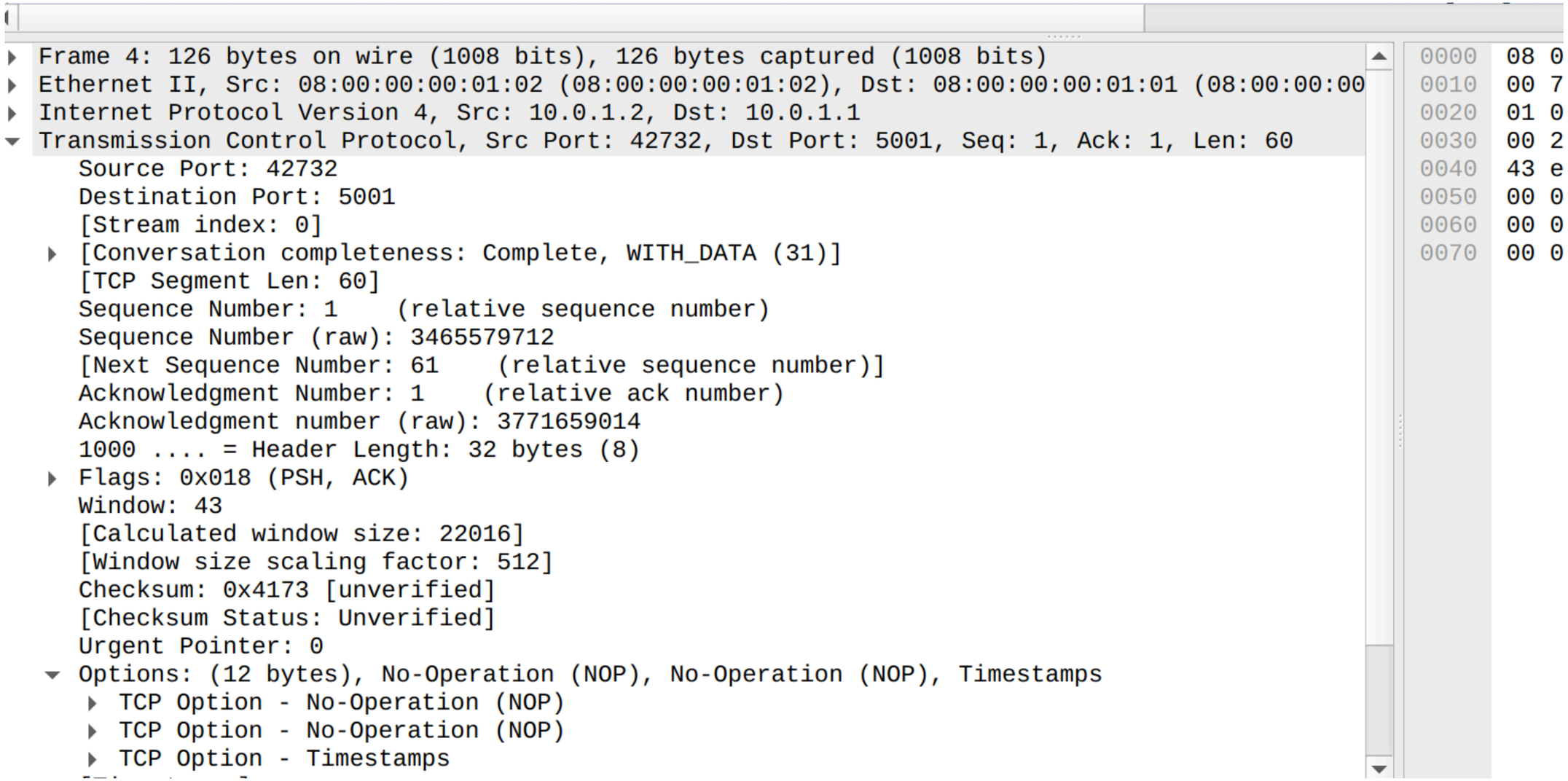

jumps from server byte 0 to server byte 28
with no data sent
wireshark IDs as missing packet

# a TCP connection

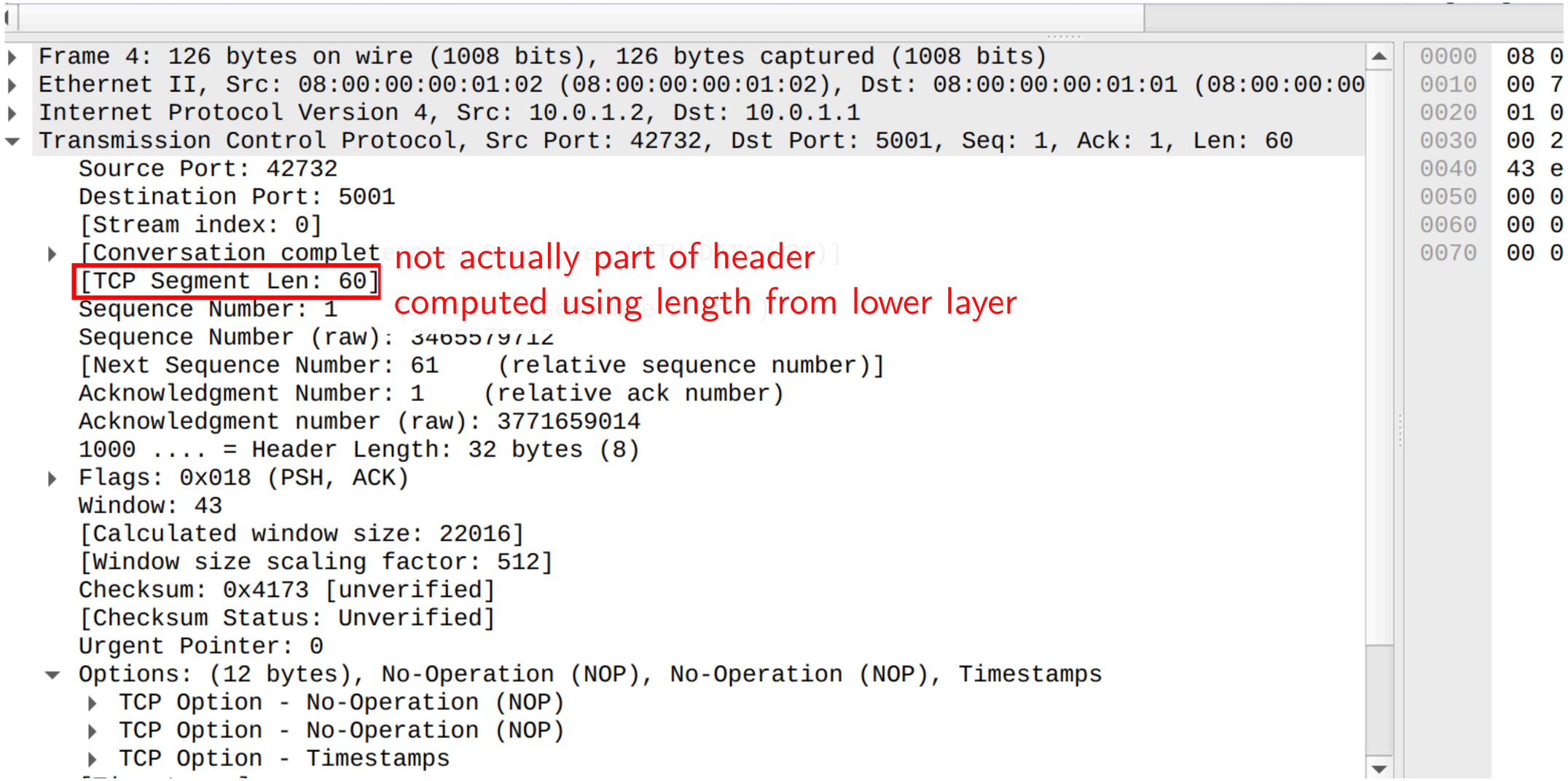


# first data packet

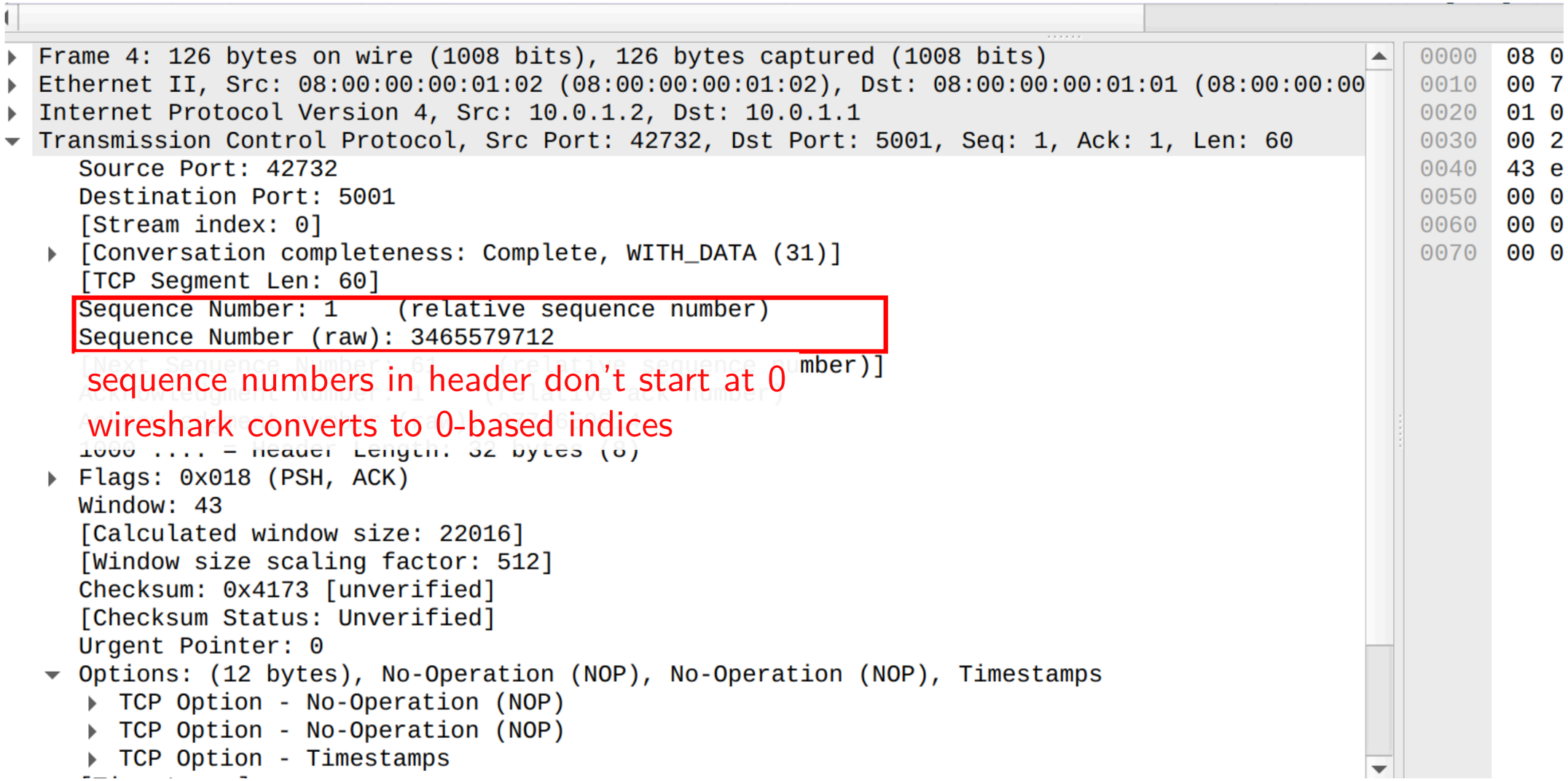
# first data packet



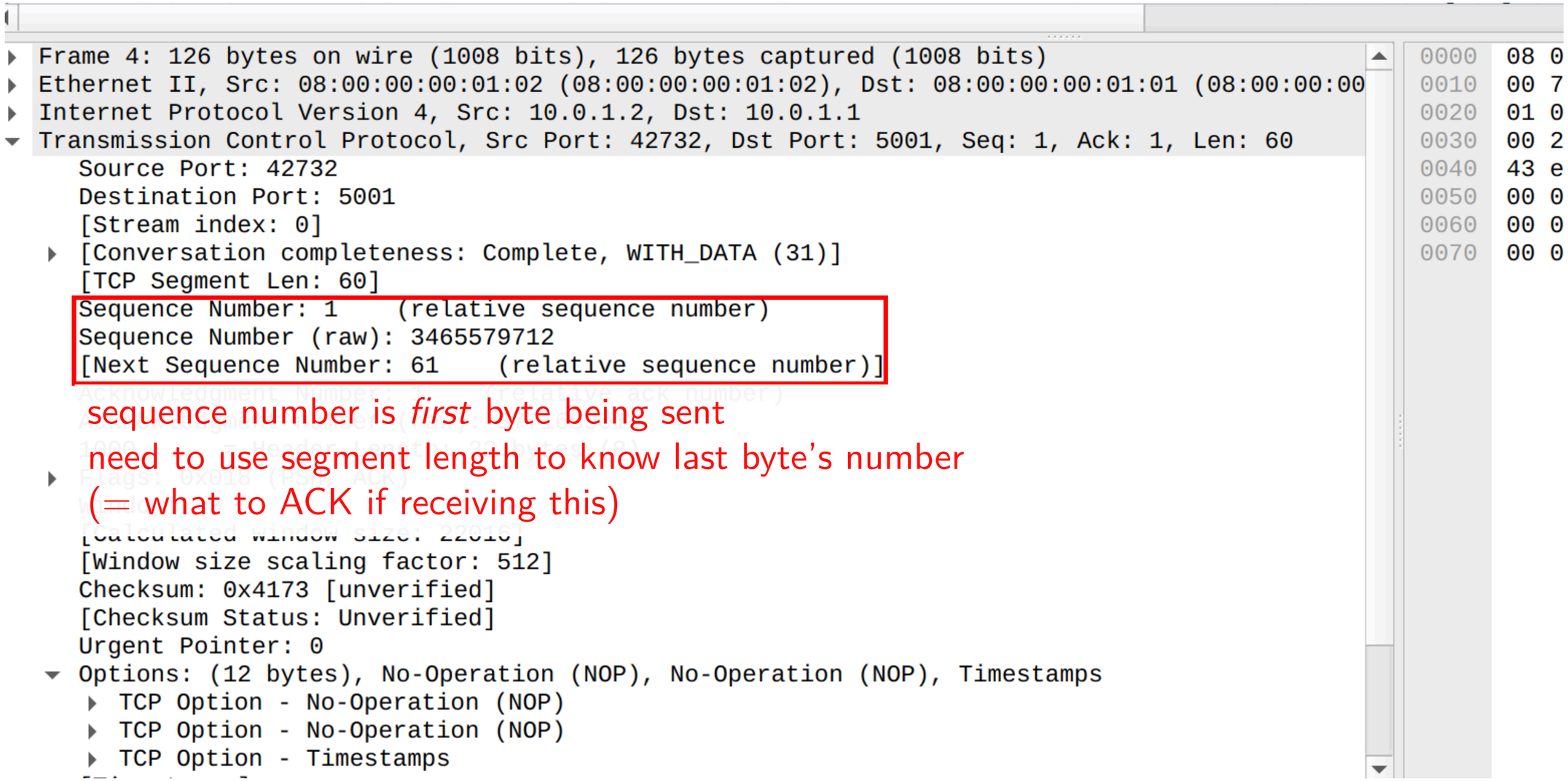
# first data packet



# first data packet



# first data packet

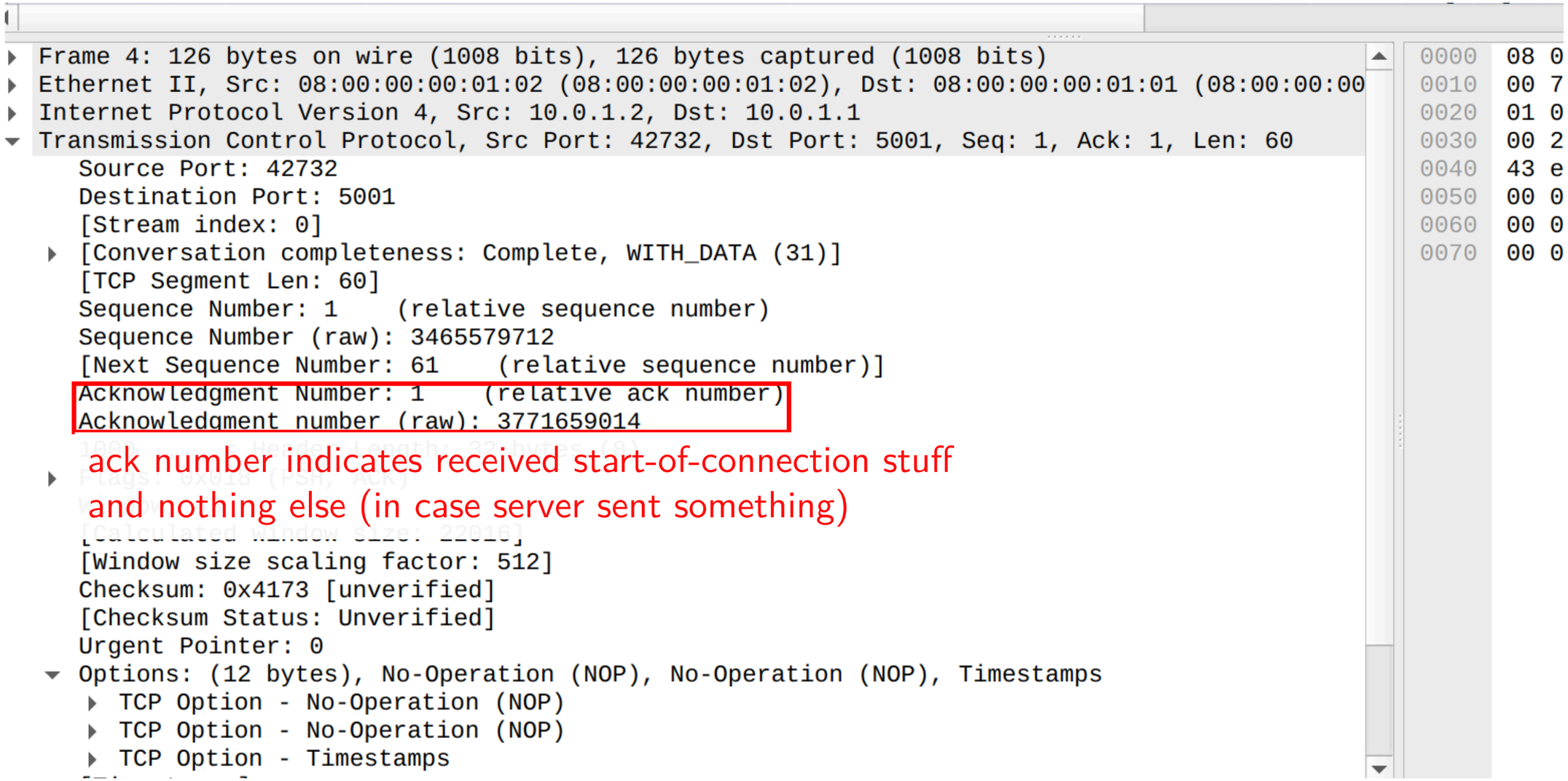



sequence number is *first* byte being sent

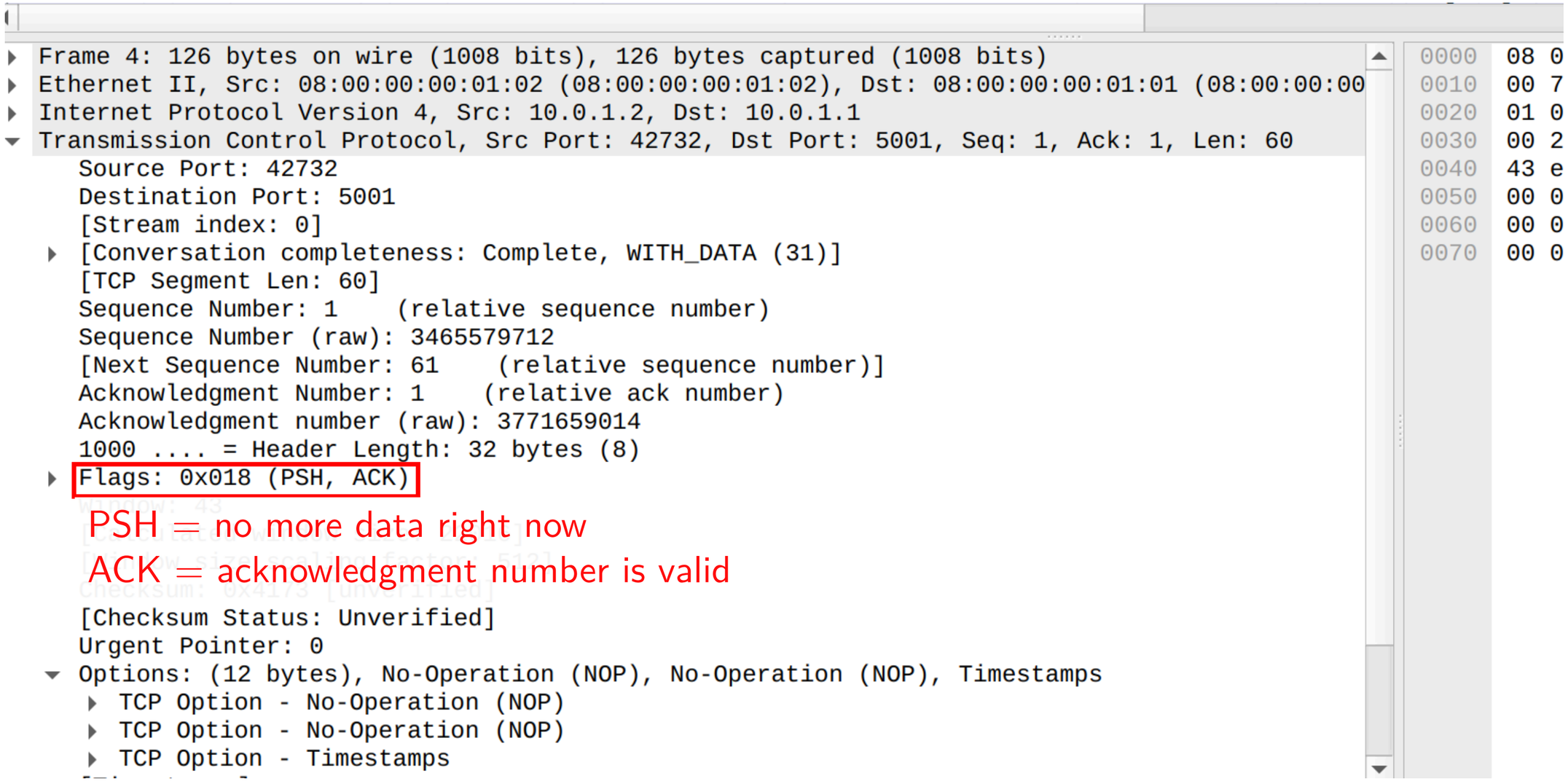
need to use segment length to know last byte's number

(= what to ACK if receiving this)

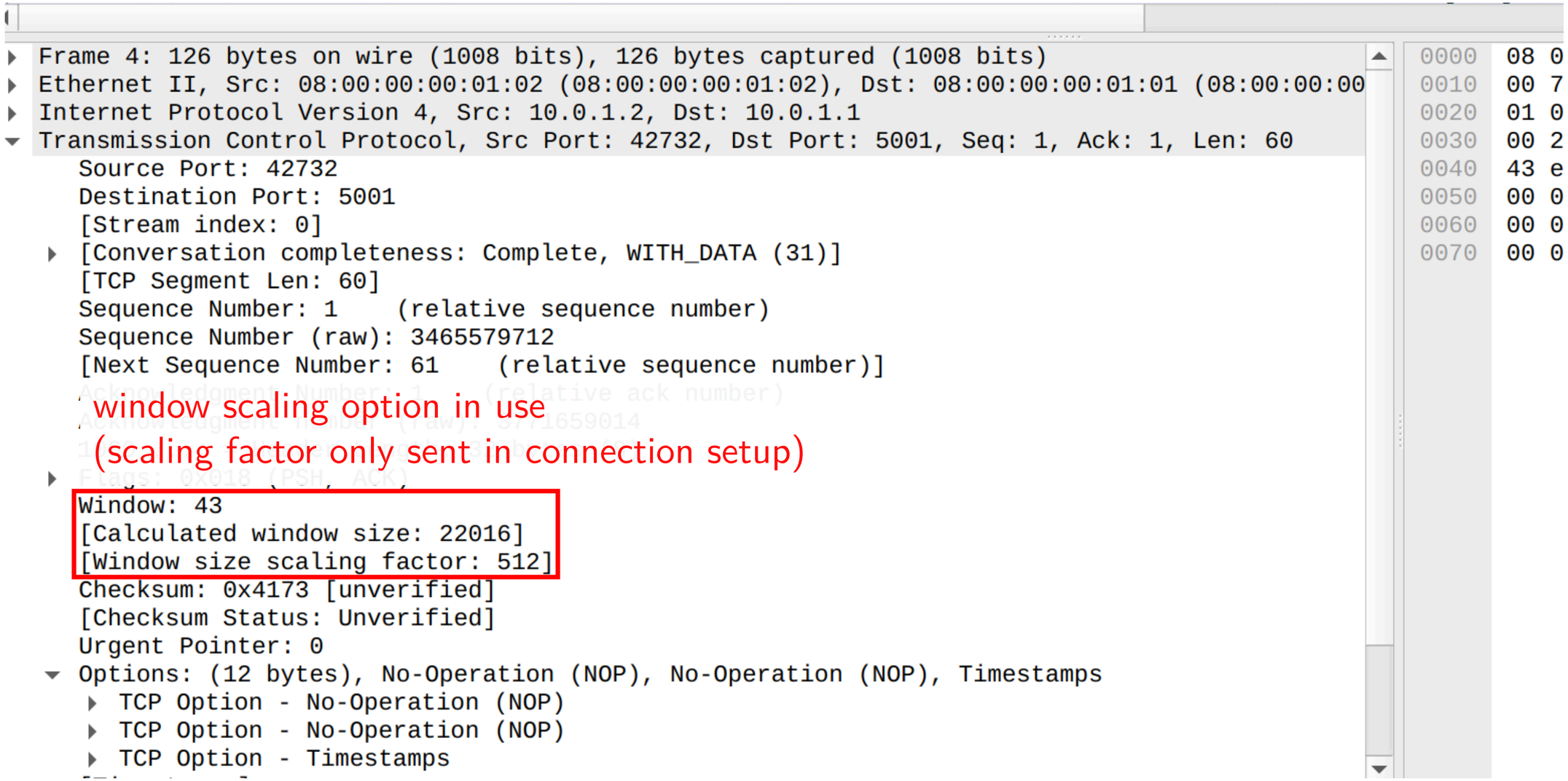
# first data packet



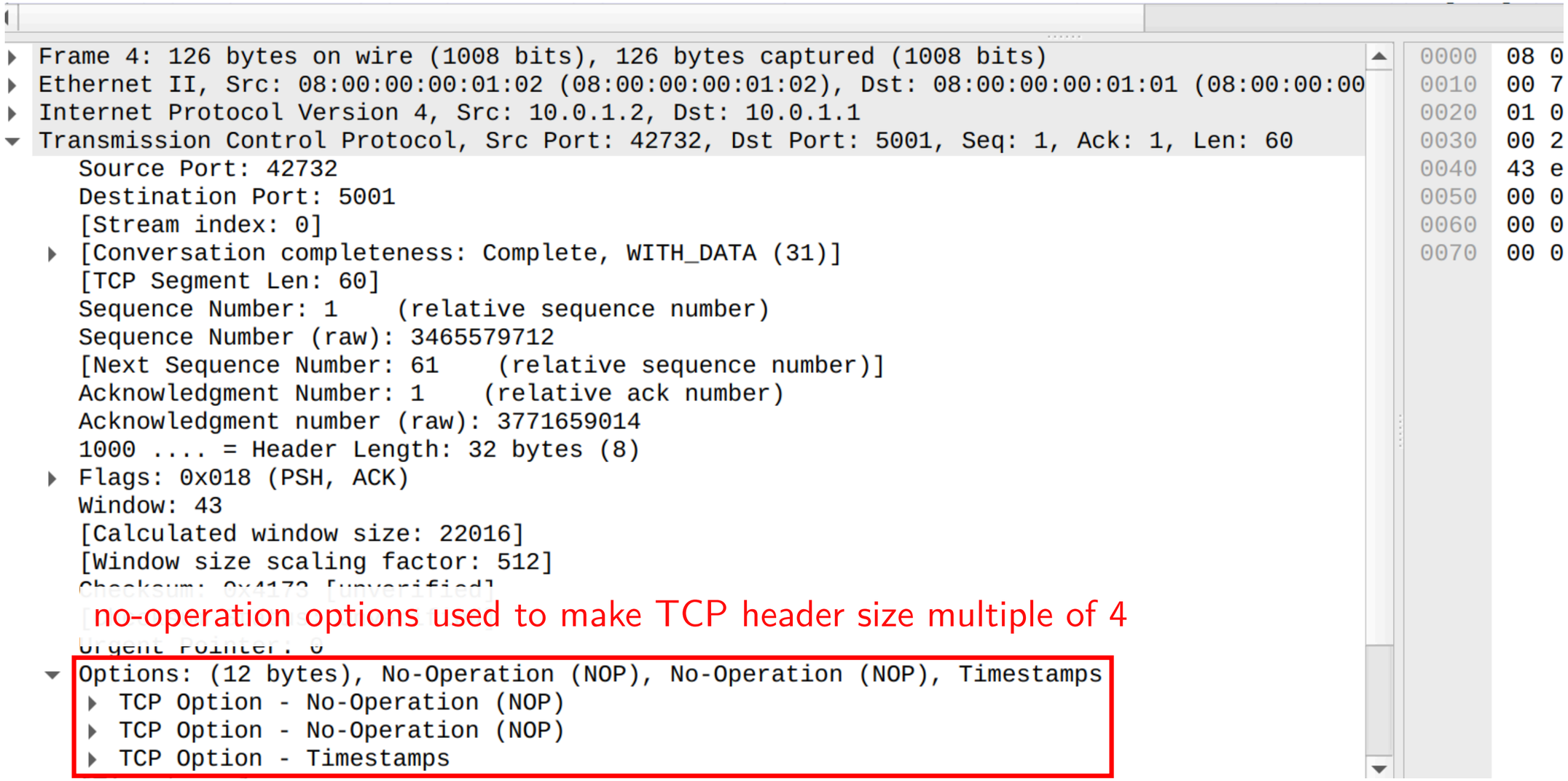
# first data packet



# first data packet

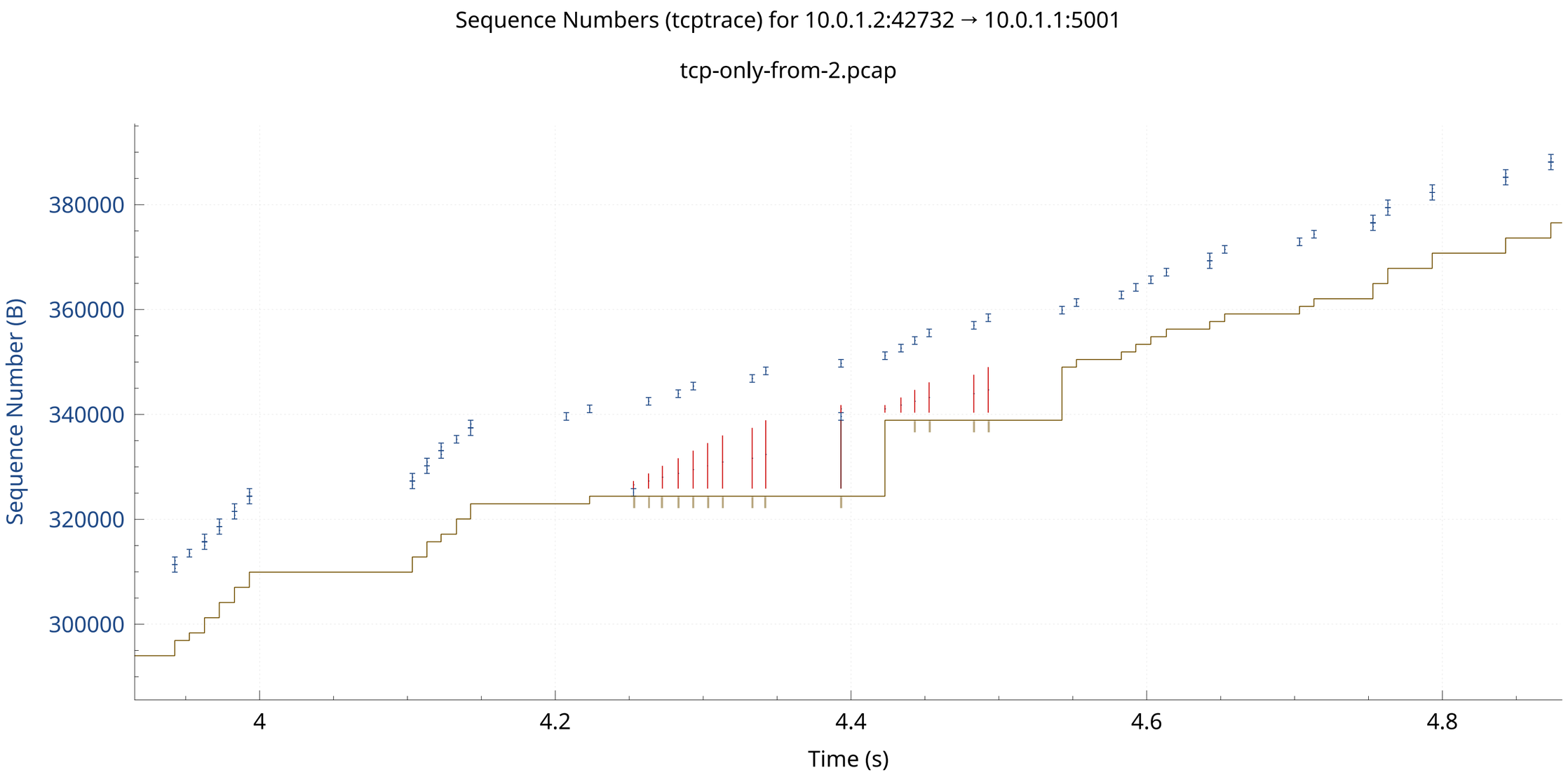


# first data packet




no-operation options used to make TCP header size multiple of 4

# sequence numbers graph



# reading thigs graph

bottom line = last ack number

notches on bottom line = duplicate acks

red lines = selective ACK info

# diff. timing in opposite direction

